

Understanding High-Wage Firms^{*}

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Abstract

I study how firm product and labor market power, along with worker bargaining power, determine wages and welfare. I document new empirical relationships between wages and markups that existing models cannot explain. To address this, I build a model in which firms produce vertically differentiated goods and share profits with workers. The model reveals new challenges in estimating monopsony and bargaining power, and proposes an alternative approach. Quantitatively, differences in product quality are key drivers of wage dispersion. Although markups may further widen wage dispersion through rent-sharing, they ultimately dampen it. Strengthening worker bargaining power improves welfare by counteracting markups.

Keywords: wage inequality, firm heterogeneity, market power, rent-sharing, misallocation

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1 Introduction

The reallocation of economic activity towards highly profitable firms in recent decades has garnered increasing attention due to their impact on labor markets (Autor, Dorn, Katz, Patterson, and Van Reenen, 2020). These firms contribute to wage inequality as they pay a wage premium to identical workers compared to less profitable firms (Van Reenen, 1996; Abowd, Kramarz, and Margolis, 1999; Card, Cardoso, Heining, and Kline, 2018). One line of research quantifies firms’ labor market monopsony power as a driver of profits and determinant of wages (Berger, Herkenhoff, and Mongey, 2022; Lamadon, Mogstad, and Setzler, 2022), but generally abstracts from firm product market monopoly power and worker bargaining power as alternative channels. Another line quantifies product market power as a driver of profits and examines its welfare implications (De Loecker, Eeckhout, and Unger, 2020; Edmond, Midrigan, and Xu, 2023), but largely abstracts from imperfect labor market competition and worker bargaining power, leaving firm-specific wage premia and the welfare implications of bargaining power unaddressed.

How does firm product market power affect wages? Can strengthening worker bargaining power lead to welfare gains? This paper presents a general framework where firms possess both product and labor market power, and workers have bargaining power. The key novel theoretical insight from my model is that the difference between wages and marginal revenue products of labor—the *labor wedge*—can be decomposed into price-cost markups, bargaining power, and standard monopsony markdowns. I derive the implications of this result for how firm market power and worker bargaining power affect wages and labor demand. The model also identifies new challenges in applying existing methods to separately estimate bargaining power and markdowns, offering a new approach for these estimations. In a quantitative version of the model, I show how firm heterogeneity—productivity, product quality, and non-wage amenities—and shocks to them affect firm wage premia. I further formalize the conditions under which strengthening worker power improves welfare. Using detailed French administrative data, which include measures of output prices and hours worked, I estimate markups, markdowns, bargaining power, and firm heterogeneity, and quantify their effects on wages and welfare.

I present five main findings. First, a standard labor market monopsony model, in which differentially productive firms operate in competitive product markets and do not bargain with workers, cannot explain why high-wage firms (i) charge higher output prices and markups, and (ii) pay workers a larger share of their marginal revenue product. Second, a model that accounts for these facts shows that the elasticity of wages to firm-specific shocks depends on the nature of the shock. When products are more differentiated, consumers prioritize quality over price, resulting in lower passthrough of productivity gains and higher passthrough of quality improvements to wages. Both productivity and quality are key drivers of firm wage premia. Third, quantitatively, markup dispersion reduces wage dispersion, but it also lowers aggregate welfare. Although markup

dispersion can amplify wage dispersion through wage bargaining, firms also restrict output and labor demand, dampening wage dispersion. Quantitatively, the latter dominates: equalizing markups leads to a 21% increase in wage dispersion and a 12% welfare gain (in consumption-equivalent terms). Fourth, the combined effect of product and labor market power reduces welfare by about 35% compared to a social planner’s economy, with 80% of this welfare loss attributable to product market power. Fifth, strengthening worker bargaining power can significantly improve welfare by counteracting firm market power. Giving workers full bargaining power closes a third of the welfare gap between the planner’s economy and the baseline economy.

In Section 2, I set up a flexible partial equilibrium framework where product and labor markets are imperfectly competitive. Firms differ in productivity, product quality, and amenities. They face upward-sloping labor supply curves and downward-sloping product demand curves, but at this point the model is agnostic about their specific microfoundation.¹ Prices are set at a markup over marginal costs. Wages are determined through bargaining between workers and firms, taking into account profits generated by imperfect product and labor market competition. Workers at each firm bargain over wages collectively with their employer—in line with a key feature of French wage-setting institutions and a source of tractability for the model. In equilibrium, the firm-level wage is the marginal revenue product of labor multiplied by a labor wedge. A higher labor wedge implies that workers obtain a larger share of their marginal revenue product.

The first main contribution of this paper is to show that the *labor wedge* is driven by price-cost markups, bargaining power, and monopsony markdowns. Prior work that quantifies firm labor market monopsony power largely abstracts from markups and bargaining power (Berger et al., 2022; Lamadon et al., 2022). My model nests monopsony as a special case—when workers have no bargaining power, the labor wedge corresponds to the monopsony markdown. However, when workers have some bargaining power and firms have product market power, workers can capture a share of the profits from markups as wages. This *rent-sharing* channel implies that the labor wedge increases with markups: higher markups imply that workers obtain a larger share of their marginal revenue product as wages. Recent studies that integrate product and labor market power focus on wage-posting models and do not feature bargaining, thus abstracting from this rent-sharing channel (Kroft, Luo, Mogstad, and Setzler, 2024; Deb, Eeckhout, Patel, and Warren, 2024). In those models, markups only reduce labor demand as firms restrict output, thereby reducing wages by moving down the labor supply curve.

One key implication of this result is that worker bargaining power can counteract the effects of firm product and labor market power on labor demand. Without bargaining power, firms exploit their monopsony power by restricting employment and their product market power by limiting output. Worker bargaining power pushes back against firm

¹These may arise for reasons such as search frictions or product/workplace differentiation (Burdett and Mortensen, 1998; Melitz, 2003; Manning, 2011; Gourio and Rudanko, 2014; Card et al., 2018)

product and labor market power by shifting some of the profits to workers, compelling firms to increase production. Moreover, when workers have bargaining power, firms’ product market power counterbalances the impact of their labor market power, as firms increase output when product market rents are appropriated by workers. Higher markups amplifies this effect, as there are more rents for workers to capture. I show that a sufficiently high worker bargaining power can eliminate the labor wedge entirely. Recent work quantifying the macroeconomic impact of markups and markdowns does not incorporate bargaining (Berger et al., 2022; Edmond et al., 2023). Introducing wage bargaining may attenuate the impact of firm market power, and strengthening worker bargaining power may improve welfare. I explore this idea quantitatively in Section 7.

In Section 3, I use my model to highlight new challenges in applying existing approaches to separately estimate monopsony markdowns and bargaining power. Recent advances in the literature show that, using the “production approach”, monopsony markdowns can be directly estimated as the wedge between wages and marginal revenue products of labor (De Loecker and Warzynski, 2012; Yeh, Macaluso, and Hershbein, 2022). However, my model clarifies that this approach only recovers markdowns when workers have no bargaining power. On the other hand, the “rent-sharing” literature measures worker bargaining power by estimating the passthrough of profits to wages, using various instruments such as export demand or innovation-based shocks (Van Reenen, 1996; Kline, Petkova, Williams, and Zidar, 2019; Garin and Silverio, 2024). This approach often relies on wage bargaining models that do not feature firm monopsony power (Card et al., 2018).² In such models, wages depend on firm profits and a reservation wage, usually assumed to be a firm-invariant competitive outside wage or the value of nonemployment that workers receive if they have no bargaining power. Valid instruments for firm profits must then be idiosyncratic and orthogonal to aggregate shocks that affect the reservation wage. In contrast, in my model, workers receive *monopsony wages* if they have no bargaining power. These monopsony wages are unobserved, firm-specific, and endogenous to the same idiosyncratic factors that shift firm profits. Therefore, existing instruments do not meet the exclusion restriction in my model.

The model suggests a new path forward to separately estimating worker bargaining power and monopsony markdowns. Instead of estimating bargaining power using a wage equation as in the rent-sharing literature, my approach uses the labor wedge equation, thereby avoiding the issue of unobserved monopsony wages. This approach relies on the property that labor wedges increase with markups when workers have bargaining power. Given measures of labor wedges and markups, the main challenge in identifying the bargaining power parameter is addressing unobserved monopsony markdowns, which are generally firm-specific as they are endogenous to firm size and amenities. While firm size is observed in the data, amenities are not. Using the model, I propose a theory-

²One exception is Kline et al. (2019), who motivate their analysis with a monopsony model. Card et al. (2018) also explain how a monopsony model could interpret those rent-sharing estimates.

consistent control function approach to address unobserved amenities: firms' wage bill and employment jointly control for differences in amenities. The intuition is that firms offering more desirable amenities can attract more workers at a given wage. This monotonic relationship is the core assumption of my control function approach. Once an estimate for bargaining power is obtained, I then derive the firm-specific monopsony markdown.

Implementing my approach begins with measuring labor wedges and markups. I do this by applying the production approach of [De Loecker and Warzynski \(2012\)](#) and [Yeh et al. \(2022\)](#) to estimate markups and labor wedges. This approach involves estimating production functions using a control function method to account for unobserved firm productivity, following ([Akerberg, Frazer, and Caves, 2015](#)). I then separate labor wedges from markups by noting that labor wedges specifically distort labor demand, while markups distort all input demands. This approach offers the advantage of flexibility, as it does not require the researcher to assume a particular market structure across various product and labor markets, nor a specific wage-setting mechanism.

Section 4 presents the detailed French administrative data I use to estimate markups, markdowns, bargaining power, and firm wage premia. These data are also used to measure the sources of firm heterogeneity in Section 6. The firm balance sheet panel data contains essential variables for production function estimation, such as revenue, materials, and capital. To measure labor inputs and to estimate firm wage premia, I use the matched employer-employee panel data, which has the advantage of including data on hours worked, allowing me to account for differences in hours across firms. I complement these datasets with survey data on firm $\times$ product-level prices for manufacturing firms, which is comprehensive for firms with at least 20 employees or sales exceeding 5 million Euros. The survey data helps address the common challenge of unobserved input and output prices in production function estimation ([De Loecker and Goldberg, 2014](#); [De Loecker, Goldberg, Khandelwal, and Pavcnik, 2016](#)). This is important in the context of markup estimation, as [Bond, Hashemi, Kaplan, and Zoch \(2021\)](#) show that unobserved output prices can bias markup estimates towards 1 (markups of 0%).

Section 5 presents my main empirical findings. I estimate a median labor wedge of 0.62 and a median monopsony markdown of 0.46 for French manufacturing firms. The difference is due to worker bargaining power and markups. I estimate that workers capture about 12% of firm profits, similar to the typical range of bargaining power estimates summarized in [Jäger, Schoefer, Young, and Zweimüller \(2020\)](#). These results suggest that not accounting for worker bargaining power would *understate* firm monopsony power.

Comparing my labor wedge estimates with those from existing studies that use the production approach underscores the importance of accounting for hours worked in measures of labor inputs. My estimates are lower than those of [Yeh et al. \(2022\)](#) and [Mertens \(2020\)](#), who measure labor inputs by employment and report labor wedges upwards of 0.70 for median US and German manufacturing firms. When I measure labor inputs as

employment in the production function, I find median labor wedges of 0.71, significantly narrowing the gap between my original estimates and those from existing literature. This discrepancy reflects an upward bias in the estimated labor output elasticities when hours are omitted, as employment understates the total labor inputs used by the firm and larger firms typically have longer average work hours.

My estimates also reveal two novel empirical relationships between firm wage premia and firm market power. First, high-wage firms tend to charge higher output prices and markups. These patterns persist even within narrow industry categories, and cannot be explained by models that consider only heterogeneous firm productivity, suggesting that differences in product quality drive these patterns. Second, high-wage firms tend to have higher labor wedges, paying a higher share of marginal revenue products of labor as wages. This finding is inconsistent with existing models of monopsony where workers have no bargaining power and firms have no product market power (Manning, 2011; Card et al., 2018). In those models, labor wedges are equal to monopsony markdowns, which often either do not vary across firms or decrease with firm wage premia. My model explains this pattern through positive worker bargaining power, allowing workers to capture some of the product market rents. Consistent with models with endogenous markdowns such as Berger et al. (2022), I also find that larger firms have lower monopsony markdowns, although the gradient of this decline is relatively modest.

In Section 6, I develop a quantitative general equilibrium version of my model to quantify the role of firm market power, worker bargaining power, and the sources of firm heterogeneity—productivity, product quality, and amenities—in shaping wages and welfare. In this model, both product and labor markets are oligopolistic and oligopsonistic, with nested CES preference structures, following the frameworks of Atkeson and Burstein (2008) and Berger et al. (2022). These microfoundations for markups and markdowns are nested within the broader framework presented in Section 2. The model implies that both markups and markdowns are determined by firms’ product and labor market shares. I use the empirical counterparts of these structural relationships to calibrate the underlying preference parameters.

A key advantage of the French administrative data is that it includes output prices and hours worked, which allows me to measure both product quality and non-wage amenities. I measure these based on the model-implied relationship between firms’ sales and wage-bill market shares and their corresponding output prices and (hourly) firm wage premium. The quantified model reveals a strong positive correlation between firm wage premia and both product quality and amenities, suggesting that these factors play an important role in explaining firm wage premia.

The first contribution of the quantitative model is to show how cross-sectional firm heterogeneity and firm-specific shocks affect firm wage premia. In Section 7, the model decomposes the passthrough of a firm-specific shock to wages into three components.

The first is a *direct effect*, capturing the initial impact of the shock on labor demand. A key insight of the model is that the size of this effect depends on the *nature* of the shock. A firm productivity shock generates a larger passthrough than an equally sized product quality shock if product varieties are close substitutes. Conversely, a product quality shock can have a larger effect if products are highly differentiated. The difference arises because sales are more sensitive to prices than quality when products are close substitutes. The second component is a *market power effect*, capturing the impact of endogenous changes in markups and markdowns as firms' market shares change. The third component is a *firm size effect*, capturing the market-level impact of changes in an individual firm's wage and output price.

Simulating a 1% shock to each source of firm heterogeneity, I find that the passthrough elasticity of firm productivity (0.96) is significantly larger than that of product quality (0.27). Most of this difference is due to the direct effect, driven by a relatively high substitutability between product varieties. Nevertheless, quality is at least as important for firm wage premia as productivity: introducing *only quality heterogeneity* to the model generates a 10% larger wage variance than introducing *only productivity heterogeneity*. The empirical literature estimating how firm-level shocks affect wages largely do not distinguish between firm productivity and quality (demand) shocks and typically find estimates below 0.30 (Card et al., 2018; Lamadon et al., 2022).³ My finding suggests that the impact of new innovations on wage inequality may depend on the nature of the innovations, as the passthrough of process (productivity gains) and product (quality improvements or new varieties) innovation to wages can differ.

The second contribution of the quantitative model is to formalize and quantify the welfare implications of worker bargaining power when firms have product and labor market power. Comparing the model's decentralized equilibrium to a welfare-maximizing social planner's equilibrium, I show that markups and markdowns reduce welfare in two ways: (i) their levels act as uniform taxes on aggregate labor demand, and (ii) their dispersion leads to misallocation. Raising worker bargaining power brings the economy closer to the planner's equilibrium, simultaneously correcting the impact of firm market power through (i) and (ii). The reason is that worker power counteracts firm product and labor market power by redistributing profits from firms to workers, compelling firms to increase production and employment. However, even full worker power cannot fully achieve the planner's equilibrium, as markups also distort the demand for non-labor inputs. While economists have long considered that strengthening worker power might counterbalance firm market power, this idea has not been formalized or quantified (Stansbury and Summers, 2020).

Compared to the planner's equilibrium, markups and markdowns jointly reduce welfare by around 35%, measured in consumption-equivalent terms. However, markups are the main source of welfare loss, as their presence alone accounts for four-fifths of this loss.

³Exceptions include Van Reenen (1996) and Kline et al. (2019), who use innovation-based instruments.

An important source of the welfare cost of markups is through markup dispersion, which causes misallocation. This accounts for about 43% of the welfare cost of markups.

Markup dispersion also has implications for wage levels and dispersion. Markups can raise wages through rent-sharing, but also reduce wages by restricting labor demand. In a counterfactual equilibrium in which firms charge uniform markups, average wage levels rise by 21% and the variance of wage premia increase by 21%, accompanied by a 12% welfare gain. Thus, the labor demand channel dominates the rent-sharing channel.

Given the considerable welfare loss attributable to the product and labor market power of French firms, can strengthening worker bargaining power improve welfare? A counterfactual in which worker bargaining power rises from 12% to 20% achieves a similar increase in average wage levels as equalizing markup dispersion. However, it delivers only a quarter of the welfare gains (3%) of equalizing markups (12%). Further increasing worker power to 100% leads to a welfare gain of 13%, which closes about one-third of the welfare gap between the market equilibrium and the planner's equilibrium. The remaining two-thirds reflects the fact that markups distort the demand for non-labor inputs, which cannot be directly addressed by worker bargaining power.

Contributions to related literature. My paper builds on the literature studying the aggregate impact of market power. [De Loecker et al. \(2020\)](#) and [Yeh et al. \(2022\)](#) estimate markups and markdowns for the US economy using the production approach, and [Edmond et al. \(2023\)](#) and [Berger et al. \(2022\)](#) quantify their impact on welfare and other economy-wide aggregates. These papers study product and labor market power separately and do not incorporate wage bargaining. A different set of papers study the wage and welfare impact of labor market concentration using bargaining models, but do not incorporate firm product market power ([Jarosch, Nimczik, and Sorkin, 2024](#); [Azkarate-Askasua and Zerecero, 2024](#)).⁴ [Kroft et al. \(2024\)](#) and [Deb et al. \(2024\)](#) consider models with joint product and labor market power, but do not feature wage bargaining.⁵ My paper contributes to this literature by analyzing the wage and welfare implications of firm product and labor market power, as well as worker bargaining power, in a unified framework. This framework yields novel insights. One key result of my paper is that the labor wedge is determined by markups and bargaining power, in addition to standard monopsony markdowns. This result has the implication that the production approach does not recover markdowns unless workers have no bargaining power, but it also suggests an alternative approach to separately identify bargaining power and markdowns. The model also delivers new quantitative findings, showing that the welfare cost of firm market power primarily originates from product markets, and that raising worker power can significantly offset firm product market power and improve welfare.

⁴A closely related literature studies the effects of worker outside options on wages. See, for example, [Caldwell and Danieli \(2019\)](#) and [Schubert, Stansbury, and Taska \(2019\)](#).

⁵[MacKenzie \(2019\)](#) studies the gains from trade when firms are large in both markets.

My paper also contributes to the literature that studies the role of firm heterogeneity in wage inequality. [Abowd et al. \(1999\)](#) estimate the separate contributions of workers and firms to wage inequality and finds an important role for firms, a finding that has since been widely replicated.⁶ [Bagger, Christensen, and Mortensen \(2014\)](#) estimate firm revenue productivity and quantifies its importance for firm wage premia using a bargaining model, but they do not disentangle productivity from quality or incorporate product market power. [Sorkin \(2018\)](#) shows that amenities account for a substantial share of wage premia, but do not quantify the role of productivity.⁷ [Lamadon et al. \(2022\)](#) show that having both productivity and amenities are crucial for explaining the wage distribution, but do not feature quality, firm-specific markups, or bargaining. My paper shows that product quality is as important a driver of wage premia as productivity. Moreover, markup dispersion may amplify wage dispersion through rent-sharing, but quantitatively dampens it through misallocation.

Finally, my paper contributes to a large body of work providing reduced-form evidence for the passthrough elasticity of firm-specific shocks to wages ([Van Reenen, 1996](#); [Card et al., 2018](#); [Kline et al., 2019](#)).⁸ My model shows that the nature of firm-specific shocks matter for passthrough. A productivity shock can have either a larger or smaller passthrough to wages than a quality shock, depending on the degree of product differentiation. The model also sheds light on the various channels through which a firm-specific shock transmits to wages, for example, through changes in market power or through a firm size effect that affects the wages and prices of competitors.

2 A Flexible Model of Firm and Worker Power

I now set up a model of the labor market in which heterogeneous firms have both labor and product market power, and workers have bargaining power. The model serves several purposes: (i) to structurally interpret regression-based estimates of firm wage premia; (ii) to clarify the role of worker bargaining power in determining labor demand and wages; (iii) to guide how firm market power and worker bargaining power may be estimated from the data (in Section 3). For these purposes, I remain agnostic about the sources of firm labor and product market power in this section. In Section 6, I impose further structure on the model to quantify the implications of firm and worker power for wages and welfare.

⁶See [Card, Heining, and Kline \(2013\)](#), [Song, Price, Guvenen, Bloom, and Von Wachter \(2019\)](#), and [Bonhomme, Lamadon, and Manresa \(2019\)](#).

⁷Other work that structurally evaluate firm productivity and amenities as drivers of firm wage premia include [Postel-Vinay and Robin \(2002\)](#), [Cahuc, Postel-Vinay, and Robin \(2006\)](#), [Barlevy \(2008\)](#), [Manning \(2011\)](#), [Card et al. \(2018\)](#), and [Haanwinckel \(2020\)](#).

⁸See [Faggio, Salvanes, and Van Reenen \(2007\)](#), [Barth, Bryson, Davis, and Freeman \(2016\)](#), [Garin and Silverio \(2024\)](#) and [Chan, Salgado, and Xu \(2021\)](#).

2.1 Model environment

Labor supply. Let Φ_j be the piece-rate wage per efficiency unit of labor paid by firm j . A worker i with efficiency E_i obtains a wage $W_{ij} = E_i \Phi_j$. Taking logs, this wage equation has a log-additive structure reminiscent of the “AKM” regression due to [Abowd et al. \(1999\)](#): $w_{ij} = e_i + \phi_j$, where lowercase letters denote variables in logs. The piece-rate wage (ϕ) is the *firm-specific wage (premium)*.

Efficiency units of labor in the firm can be written as $H_j = \bar{E}_j L_j$, where $\bar{E}_j$ denotes average efficiency and L_j denotes amount of labor. Let the upward-sloping labor supply curve facing each firm be $H_j = \mathcal{H}(\Phi_j, A_j)$. The firm-specific labor supply shifter A_j captures firms’ non-wage amenities and is taken as given by the firm. I assume that the labor supply function is twice differentiable and monotonically increasing in Φ_j and A_j .

Product demand. Let the downward-sloping demand curve for firm j ’s output be $Y_j = \mathcal{G}(P_j, D_j)$. The price charged by firm j is P_j . Firms take as given the demand shifter for its goods D_j , capturing its product quality. I assume that the goods demand function is twice differentiable and monotonically decreasing in P_j and increasing in D_j .

Production technology. Firms operate a general production function with diminishing marginal returns to each input $Y_j = \Omega_j F(K_j, M_j, H_j)$. Ω_j is the Hicks-neutral productivity term, K_j are physical capital, M_j are material inputs, and H_j are units of effective labor. I assume that this production function is twice differentiable. The market for capital and material inputs are perfectly competitive with prices P_k and P_m .

Wage determination. Workers bargain collectively with their employer j . This assumption is consistent with an important feature of wage bargaining in France: annually, firms with at least 50 employees are legally required to bargain with their employees, who are represented by labor union representatives.^{9,10,11} Let Π_j be the firm’s profit. Let workers’ indirect utilities be linear in wages: $V_j = \Phi_j H_j$.

Bargaining is efficient in the sense that workers and firms jointly decide on wages, prices, materials, and capital to maximize total profits, taking into account the product demand curve and labor supply curve.¹² Workers’ bargaining power is κ . In the event that an agreement is not reached between firms and their employees, I assume that firms do not produce and have an outside option of zero profits; workers do not supply labor

⁹See Appendix A for a description of French wage-setting institutions.

¹⁰In Appendix C, I make specific functional form assumptions on revenue functions under a [Stole and Zwiebel \(1996\)](#) bargaining protocol, where firms bargain with individual workers, and show that a similar firm wage premium equation can be obtained.

¹¹See also [Card, Devicienti, and Maida \(2014\)](#), [Card et al. \(2018\)](#), and [Azkarate-Askasua and Zerecero \(2024\)](#), for models of wage bargaining between firms and groups of workers.

¹²When capital is a production input and capital investments are sunk, an important concern is whether workers can holdup their employers and extract rents. However, recent evidence suggests that such holdup problems tend to be small (see [Card et al. \(2014\)](#) and the references therein).

and have an outside option of zero wages.¹³ The worker-firm bargaining problem is:

$$\max_{\Phi_j, P_j, M_j, K_j} \left(\Phi_j H_j \right)^\kappa \left(\Pi_j \right)^{1-\kappa}$$

subject to $H_j = \mathcal{H}(\Phi_j, A_j)$, $Y_j = \mathcal{G}(P_j, D_j)$, and $Y_j = \Omega_j F(K_j, M_j, H_j)$. The firm's profit is $\Pi_j = P_j Y_j - \Phi_j H_j - P_m M_j - P_k K_j$. When workers have no bargaining power ($\kappa = 0$), they have no influence on the firm's input and price/wage-setting decisions.

2.2 How bargaining affects firm wage premia and labor demand

2.2.1 Firm wage premia

Solving the bargaining problem gives the following firm-specific wage (premium):

$$\Phi_j = \underbrace{\kappa \left(\frac{P_j Y_j - P_m M_j - P_k K_j}{H_j} \right)}_{\text{Profits per effective labor}} + (1 - \kappa) \underbrace{\lambda_j MRP H_j}_{\text{Monopsony wage}} \quad (1)$$

The first term represents *profits per effective labor*, also frequently referred to as “quasi-rents” in the rent-sharing literature (Card et al., 2018). The second term is the *monopsony wage*, which is a markdown (λ_j) below the marginal revenue product of effective labor ($MRP H_j$). The monopsony markdown $\lambda_j = \frac{\xi_j}{1+\xi_j}$ is a function of labor supply elasticities $\xi_j = \xi(H_j, A_j)$, whose specific functional form depends on the microfoundation for the labor supply curve.

Equation (1) shows that the firm wage premium is a weighted average of two common wage-setting mechanisms: a pure bargaining outcome and a pure monopsony outcome. When workers have no bargaining power ($\kappa = 0$), firms set wages unilaterally taking into account the labor supply curve. As a result, workers are paid the monopsony wage, as is the case in wage-posting models of the labor market (Berger et al., 2022; Lamadon et al., 2022; Kroft et al., 2024). When workers have some bargaining power ($\kappa \in (0, 1)$), they are able to extract a share of the firm profits. Therefore, bargained wages are a top-up over the monopsony wage. When workers have full bargaining power ($\kappa = 1$), they capture the full profits. Thus, higher worker bargaining power redistributes profits generated by imperfect product and labor market competition from firms to workers. This redistribution of profits also impacts firm labor demand, as I show next. In Section 3, I discuss the implications of equation (1) for existing approaches to estimating worker bargaining power.

¹³This assumption is consistent with the lack of wage response from large increases in unemployment insurance levels (Jäger et al., 2020), and the lack of response of reservation wages to changes in the potential unemployment benefit duration (Le Barbanchon, Rathelot, and Roulet, 2019). This assumption does not alter the results in this section. In Appendix C.1, I derive firm wage premia when firms have positive outside options and discuss the implications for the empirical results presented in Section 5.

2.2.2 Labor demand

To see how markups, markdowns, and worker bargaining power affect firm labor demand, consider a different representation of the firm wage premium equation (1):

$$\Phi_j = \Lambda_j MRP H_j = \Lambda_j \mu_j^{-1} MPH_j \quad (2)$$

where the marginal revenue product of effective labor is $MRPH_j = \mu_j^{-1} MPH_j = \mu_j^{-1} \alpha_{h,j} \frac{P_j Y_j}{H_j}$. Let $MPH_j = \alpha_{h,j} \frac{P_j Y_j}{H_j}$ denote the *value* of the marginal product of labor. $\frac{P_j Y_j}{H_j}$ is the revenue per effective worker, $\alpha_{h,j}$ is the output elasticity with respect to labor inputs, and $\mu_j = \frac{\rho_j}{\rho_j - 1}$ is the *price-cost markup*, which is a function of the price elasticities of demand $\rho_j = \rho(Y_j, D_j)$. The *labor wedge* (Λ_j) is the share of $MRPH_j$ that workers receive as wages.

Markups as negative labor demand shifters. Notice that in equation (2) markups directly drive a wedge between wages and marginal products of labor (MPH_j). Because markups enter inversely, a higher markup represents a negative labor demand shift, capturing the idea that monopolists *reduce output*, and hence labor demand. In a model where firms have both product and labor market power but workers have no bargaining power, all else equal, a markup-induced negative labor demand shift moves firms down the labor supply curve, thus reducing wages (Kroft et al., 2024).

The role of wage bargaining and firm market power in determining the labor wedge. A key insight of the model is that the difference between wages and the marginal revenue product of labor—the labor wedge (Λ_j)—can be expressed as:

$$\Lambda_j = \kappa \left(1 - \frac{\alpha_{m,j} + \alpha_{k,j}}{\mu_j} \right) \frac{\mu_j}{\alpha_{h,j}} + (1 - \kappa) \lambda_j \quad (3)$$

The first term reflects the profits per unit of MRPH (“product market rents” henceforth) captured by workers through bargaining—a *rent-sharing* term. In contrast with the role of markups as negative labor demand shifters, this term increases with markups, as higher markups increase the profits that workers can appropriate. Thus, markups can also act as positive labor demand shifters by raising the labor wedge (Λ_j). This rent-sharing term introduces a new channel through which markups affect labor demand and wages relative to models without worker bargaining power (Kroft et al., 2024).

The second term arises from the monopsony markdown (λ_j), representing firm monopsony power when workers have no bargaining power ($\kappa = 0$). When $\kappa = 0$, the labor wedge is equal to the monopsony markdown $\Lambda_j = \lambda_j$, as workers do not capture any profits. When $\kappa > 0$, the labor wedge exceeds the monopsony markdown. Thus, the labor wedge captures the idea that worker bargaining power can offset firm monopsony power, with pure monopsony as a special case (Lamadon et al., 2022). Additionally, since the labor wedge increases with markups, higher markups make worker bargaining power

more effective at countering firm monopsony power.

Equations (2) and (3) offer several insights on how firm market power and worker bargaining power affect labor demand and the measurement of monopsony markdowns:

1. **LABOR DEMAND IMPLICATIONS.** One implication of equation (3) is that worker bargaining power offsets the effects of firm product and labor market power on labor demand. To illustrate this, I compare equation (3) with benchmarks in the literature on product and labor market power, which do not feature wage bargaining:

$$\begin{aligned}
\text{(B1) Double perfect competition:} & \quad \Phi_j = MPH_j \\
\text{(B2) Imperfect labor market competition:} & \quad \Phi_j = \lambda_j MPH_j \\
\text{(B3) Imperfect product market competition:} & \quad \Phi_j = \mu_j^{-1} MPH_j \\
\text{(B4) Double imperfect competition:} & \quad \Phi_j = \lambda_j \mu_j^{-1} MPH_j
\end{aligned}$$

In B1, firms behave as *pricetakers* and *wagetakers*. In this case, equation (2) converges to a familiar form, where wages are equal to the marginal product of labor. In B2, firms take prices as given but not wages, leading to monopsony markdowns below the marginal product of labor, as in [Berger et al. \(2022\)](#). In B3, firms take wages as given but not prices, so wages equal the marginal product of labor multiplied by the inverse of markups, as in [Edmond et al. \(2023\)](#).¹⁴ In B4, firms possess both product and labor market power, where markups and markdowns jointly act as taxes on labor demand. This highlights the role of “double market power” emphasized by [Kroft et al. \(2024\)](#).

Worker bargaining power governs how close the firm’s optimal labor demand condition is to either the double perfect competition benchmark (B1) or the double imperfect competition benchmark (B4). As equation (3) shows, when firms have product and labor market power, a small positive κ leads labor wedges to be higher than monopsony markdowns, $\Lambda_j > \lambda_j$. In this case, workers are paid closer to their MPH_j , and the firm’s optimal labor demand condition deviates away from the double imperfect competition benchmark (B4) towards the imperfect product market competition benchmark (B3).

As worker bargaining power κ increases, it reaches a threshold $\kappa = \underline{\kappa}_j \in (0, 1]$ where firm j behaves as a *wagetaker*, and labor demand aligns with the imperfect competition benchmark (B3). At this point, wages equal the marginal revenue product of labor, $\Phi_j = MRP_j$, because worker bargaining power fully offsets monopsony power, eliminating markdown distortions. The threshold is:

$$\underline{\kappa}_j \equiv [1 - \lambda_j] / \left[\left(1 - \frac{\alpha_{m,j} + \alpha_{k,j}}{\mu_j} \right) \frac{\mu_j}{\alpha_{h,j}} - \lambda_j \right],$$

which *decreases* with the firm’s price-cost markup. Thus, worker bargaining power is more effective in offsetting monopsony power when price-cost markups are higher. This is

¹⁴In [Edmond et al. \(2023\)](#), in addition to firms being *wagetakers*, the labor supply elasticity a firm faces is infinite, hence, there is no wage dispersion in their model ($\Phi_j = \Phi$).

because higher markups increase the amount of profits that are appropriated by workers, forcing firms to increase production to raise profits.

As worker bargaining power increases further, it reaches a threshold $\kappa = \bar{\kappa}_j \in [\underline{\kappa}_j, 1]$ where firm j behaves *similarly* to a pricetaker in both labor and product markets, converging to the double perfect competition benchmark (B1), where wages equal the marginal product of labor. This threshold is:

$$\bar{\kappa}_j \equiv [\mu_j - \lambda_j] / \left[\left(1 - \frac{\alpha_{m,j} + \alpha_{k,j}}{\mu_j}\right) \frac{\mu_j}{\alpha_{h,j}} - \lambda_j \right],$$

where $\bar{\kappa}_j < 1$ if $\mu_j > \frac{\alpha_{m,j} + \alpha_{k,j}}{1 - \alpha_{h,j}}$, and $\bar{\kappa}_j > \underline{\kappa}_j$ for $\mu_j > 1$. The threshold $\bar{\kappa}_j$ also decreases with markups if $\lambda_j < \frac{\alpha_{m,j} + \alpha_{k,j}}{1 - \alpha_{h,j}}$, implying that higher markups raise the effectiveness of increasing worker bargaining power in achieving B1. Thus, when $\kappa > \underline{\kappa}_j$, labor wedges are eliminated ($\Phi_j = MRP H_j = \mu_j^{-1} MP H_j$), and for $\kappa > \bar{\kappa}_j$, markup wedges are also eliminated ($\Phi_j = MP H_j$). Recent research highlights how markups distort labor demand and reduce welfare (Edmond et al., 2023). Together, equations (2) and (3) suggest that worker bargaining power can help counter the markup-induced distortions, in addition to its role in offsetting monopsony power.

If firms do not have product market power ($\mu_j = 0$), full worker bargaining power ($\kappa = 1$) shifts the firm's labor demand condition from the imperfect labor market competition benchmark (B2) to the double perfect competition benchmark (B1). However, when firms have product market power, even full worker bargaining power cannot achieve B1. This is because worker bargaining power cannot directly address the fact that markups also restrict the firm's demand for capital and materials. The firm's optimal capital and material demand are $P_k = \mu_j^{-1} MP K_j$ and $P_m = \mu_j^{-1} MP M_j$, where the latter terms are the marginal products of capital and materials. This places a limit on the extent to which worker bargaining power can offset the impact of markups on firm production.

2. MEASUREMENT IMPLICATION. Equation (3) also clarifies that the recently developed production approach to markdown estimation (Yeh et al., 2022; Mertens, 2020) recovers monopsony markdowns only if workers do not have bargaining power. This approach measures monopsony markdowns as $\lambda_j = \frac{\alpha_{m,j}}{\alpha_{h,j}} \frac{\Phi_j H_j}{P_m M_j}$, using data on wage bills and material expenditure, as well as estimates of labor and material output elasticities. When $\kappa > 0$, my model implies that this equation measures the labor wedge $\Lambda_j = \frac{\alpha_{m,j}}{\alpha_{h,j}} \frac{\Phi_j H_j}{P_m M_j}$, which is a function of markdowns, markups, and bargaining power.

3 Estimating Firm Wage Premia, Firm Market Power, and Worker Bargaining Power

The model in Section 2 relates firm wage premia (Φ_j) to labor wedges (Λ_j), markups (μ_j), markdowns (λ_j), and worker bargaining power (κ). This section provides a detailed

description of my approach to estimating these objects. Before that, I first provide a broad overview of my approach.

Without committing to a specific product demand or labor supply system or market structure, separately estimating markups from labor wedges is not straightforward. A common approach to measuring markups is the cost share approach, which measures markups using sales to total variable cost ratios. However, a key assumption required by this approach is that all input markets are perfectly competitive, precluding the estimation of labor wedges. To separate markups from labor wedges, I use the production approach to markup estimation of [De Loecker and Warzynski \(2012\)](#), recently extended to quantify labor market power by [Yeh et al. \(2022\)](#). This approach first estimates production functions to obtain output elasticities $(\alpha_{h,j}, \alpha_{k,j}, \alpha_{m,j})$, then separates markups from labor wedges by exploiting the fact that markups distort the demand for all inputs, while labor wedges only distort labor demand. Unlike the cost share approach, the production approach does not require the assumption of competitive labor markets.

As equation (3) shows, labor wedges depend on markups, markdowns, and bargaining power. To understand the contribution of each component to labor wedges, it is necessary to separately estimate markdowns and bargaining power. A central challenge in estimating bargaining power in my model is that realized wages are determined by both profits per worker and *monopsony wages*, which are unobserved, firm-specific, and responds to the same shocks as profits per worker (see equation (1)). This implies that standard instruments for profits per worker, commonly used in the rent-sharing literature, do not satisfy the exclusion restriction in my model ([Card et al., 2018](#)). To address this challenge, I show that using the labor wedge equation (3) instead of the wage equation (1) allows one to reframe the issue of unobserved monopsony wages into one of unobserved amenities. I then propose a theory-consistent control function approach to account for the role of unobserved amenities in driving labor wedges.

3.1 Estimating firm wage premia

A common way of estimating firm wage premia is to estimate firm effects from an AKM regression ([Abowd et al., 1999](#)). Typically, firm effects are assumed to be *fixed over time*. They are then identified from worker mobility between firms. However, as I explain in the next subsection, measuring firms' efficiency units of labor over time requires estimating *time-varying firm wage premia*. Further, a key practical issue in estimating firm effects is the lack of between-firm worker mobility in short panels, which leads to noisy firm effects estimates that upward-bias the variance of firm effects. To address the lack of worker mobility and to allow time-variation in firm wage premia, I implement the k-means classification approach of ([Bonhomme et al., 2019](#)) ("BLM" henceforth).¹⁵

¹⁵I also compare the estimated variance of firm wage premia using the [Bonhomme et al. \(2019\)](#) k-means clustering and [Kline, Saggio, and Solvsten \(2020\)](#) leave-out approaches in Table 2. I find the estimated

I first classify firms into groups using a k-means clustering algorithm, then estimate a version of the AKM regression replacing firm effects with firm-group effects. Specifically, I estimate the following regression:

$$\ln W_{it} = \chi'_{it}\beta + \iota_i + \phi_{g(j(i,t),t)t} + \nu_{it}$$

where i denotes the individual, $j(i, t)$ denotes the firm that employs i at time t , $g(j, t)$ denotes the group of firm j at time t , ι_i are worker fixed effects, $\phi_{g(j(i,t),t)}$ are firm-group effects that vary by t , and χ_{it} includes age polynomials and part-time status. When there are as many firm-groups as there are firms, this regression converges to the AKM regression. The firm-group fixed effects are identified by workers who switch between firm-groups. Relative to the AKM regression, this procedure has the advantage that it substantially increases the number of switchers used to identify firm-group effects, enabling wage premia to be more precisely estimated.¹⁶

To classify firms with similar wage premia into the same group, I group firms based on the similarity of their internal wage distributions. The idea is that, conditional on the log additive wage structure, firms with similar firm effects and worker effects should have similar internal wage distributions. If two firms have internal wage distributions of very similar shapes, but their average wages differ significantly, then they have very different firm effects. If two firms have similar average wages, but the shape of their internal wage distributions differ substantially, they are clustered into different groups. In practice, I apply the clustering algorithm by 2-digit sectors for every overlapping 2-year window, allowing wage premia to vary over time.¹⁷ [Online Appendix B](#) provides more detail on how I cluster firms and addresses the main restrictions underlying the AKM regression.

3.2 Estimating labor wedges and price-cost markups

Estimation approach in theory. Having estimated firm wage premia, I now apply the production approach to separately estimate markups and labor wedges ([De Loecker and Warzynski, 2012](#); [Hershbein, Macaluso, and Yeh, 2018](#)). I start by estimating production functions to obtain output elasticities. Let the production function for each sector s be $y_{jt} = f_s(k_{jt}, m_{jt}, h_{jt}; \beta) + \omega_{jt}$. Lowercase letters represent the natural log counterparts of variables written in uppercase letters, β is the set of production function parameters, and ω_{jt} is the firm's Hicks-neutral productivity. The firm-specific output elasticities with respect to capital, material inputs, and effective labor are: $\alpha_{k,jt} := \frac{\partial y_{jt}}{\partial k_{jt}}$, $\alpha_{m,jt} := \frac{\partial y_{jt}}{\partial m_{jt}}$, and $\alpha_{h,jt} := \frac{\partial y_{jt}}{\partial h_{jt}}$. Consistent with the model, I measure effective labor as $H_{jt} = \bar{E}_{jt}L_{jt}$, where the average efficiency of workers per hour is the ratio of the firm's average wage

variance to be similar between the two approaches.

¹⁶See [Bonhomme, Holzheu, Lamadon, Manresa, Mogstad, and Setzler \(2023\)](#) for a systematic assessment of the importance of clustering firms before estimating firm effects.

¹⁷[Engbom, Moser, and Sauermann \(2022\)](#) and [Chan et al. \(2021\)](#) also estimate time-varying firm effects. They do so within the connected set of firm-years, rather than the connected set of firms.

to the firm wage premium at a given time t , $\bar{E}_{jt} = \frac{\bar{W}_{jt}}{\Phi_{jt}}$, and $\log \bar{E}_{jt}$ is normalized to have a mean of 0 in the cross-section.

Next, I disentangle firms' markups from their labor wedges. I do so using the fact that markups distort the demand for all inputs, while labor wedges only distort labor demand. This has two key requirements: (i) selecting a flexible input in which (ii) firms are pricetakers. Assuming that material inputs satisfy these requirements, markups represent the only distortion to material input demand, and labor wedges are the only distortion to the demand for labor relative to materials. One can then express markups and labor wedges as follows:

$$\mu_{jt} = \alpha_{m,jt} \frac{P_{jt}Y_{jt}}{P_{m,t}M_{jt}} \quad \text{and} \quad \Lambda_{jt} = \frac{\Phi_{jt}H_{jt}}{P_{m,t}M_{jt}} \cdot \frac{\alpha_{m,jt}}{\alpha_{h,jt}} = \frac{\bar{W}_{jt}L_{jt}}{P_{m,t}M_{jt}} \cdot \frac{\alpha_{m,jt}}{\alpha_{h,jt}} \quad (4)$$

Thus, markups and labor wedges can be disentangled given data on the material input expenditure share and the wage-bill-to-material-expenditure ratio, as well as estimates of output elasticities.

Identification challenges. Obtaining estimates of output elasticities requires estimating production functions, which entails a few challenges: (i) unobserved firm productivity, (ii) unobserved output prices, (iii) unobserved input prices. I now address them in turn.

First, firm productivity ω_{jt} is unobserved but plays a crucial role in determining input choices. Since more productive firms demand more inputs, OLS estimates of production function parameters suffer from upward bias—known as transmission bias. To address this issue, I implement the control function approach (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg et al., 2015), which corrects for the endogeneity of input choices by allowing researchers to indirectly observe firm productivity. This is done by inverting the firm's optimal demand function for a fully flexible input. I assume that material inputs are flexible and discuss this assumption in detail in Section 5.4.

Second, researchers often do not observe firms' output prices, which introduces potential bias, as output prices are often correlated with input choices—this is known as output price bias in the estimation of production function parameters (De Loecker and Goldberg, 2014).¹⁸ In practice, the production function is frequently estimated as:

$$p_{jt} + y_{jt} = f_s(k_{jt}, h_{jt}, m_{jt}; \beta) + p_{jt} + \omega_{jt}$$

where $p_{jt} + \omega_{jt}$ is revenue TFP (TFPR) and ω_{jt} is quantity TFP (TFPQ). The potential negative correlation between output prices and input demand introduces a downward bias in the estimated output elasticities. Intuitively, firms that set higher prices tend to sell less output, reducing the demand for inputs. However, if larger firms charge higher prices—due to unobserved quality differences, for instance—output elasticity estimates could be upward-biased. Moreover, as shown by Bond et al. (2021), in the presence

¹⁸When output prices are observed, they are typically specific to certain industries, such as beer brewing (De Loecker and Scott, 2016).

of product market power, this output price bias skews estimated markups towards 1, implying no markups.¹⁹

To address this issue, I measure firm level prices p_{jt} directly using French administrative data on firm-product-year level prices for manufacturing firms and compute firm-year level output y_{jt} as revenue divided by prices.²⁰ I provide further detail on the measurement of output prices in Section 4.

Third, researchers rarely observe firms' input prices directly, so deflated input expenditures are typically used instead of input quantities. This potentially introduces an *input price bias* in the estimated production function parameters (De Loecker and Goldberg, 2014). Conditional on observing firms' output prices, the commonly estimated production function becomes:

$$y_{jt} = f_s(\tilde{k}_{jt}, \tilde{h}_{jt}, \tilde{m}_{jt}; \beta) + B(\mathbf{p}_{\mathbf{x},jt}, \tilde{\mathbf{x}}_{jt}; \beta, \zeta) + \omega_{jt}$$

where $\tilde{\mathbf{x}}_{jt} = \{\tilde{k}_{jt}, \tilde{h}_{jt}, \tilde{m}_{jt}\}$ represents the set of input expenditures (denoted with a tilde) and $\mathbf{p}_{\mathbf{x},jt} = \{p_{k,jt}, \phi_{jt}, p_{m,jt}\}$ the set of input prices. The function $B(\cdot)$ captures the influence of unobserved input prices, with its functional form depending on the functional form of the production function $f(\cdot)$. The parameters ζ must then be estimated to account for unobserved input prices.

Relative to most datasets, the French administrative data includes hours and wages at the worker level, allowing me to measure effective labor h_{jt} . However, capital and material quantities are not observed. The production function I estimate becomes:

$$y_{jt} = f_s(\tilde{k}_{jt}, h_{jt}, \tilde{m}_{jt}; \beta) + B_s(\mathbf{p}_{\mathbf{x},jt}, \tilde{\mathbf{x}}_{jt}, h_{jt}; \beta, \zeta) + \omega_{jt}$$

where the set of input expenditures is now $\tilde{\mathbf{x}}_{jt} = \{\tilde{k}_{jt}, \tilde{m}_{jt}\}$ and the set of unobserved input prices is $\mathbf{p}_{\mathbf{x},jt} = \{p_{k,jt}, p_{m,jt}\}$. Because the prices of material and capital inputs are unobserved, as is the case in most existing datasets, I work under the standard assumption that firms are pricetakers in these input markets (De Loecker and Goldberg, 2014). To the extent that firms in different sectors and locations face different material and capital input prices, I control for sector and location fixed effects in the production function estimation routine.²¹

However, input prices may differ across firms due to differences in input quality. As De Loecker et al. (2016) show, under a large class of consumer demand functions used in International Trade, Industrial Organization, and Macroeconomics, output prices increase monotonically with output quality, which increases monotonically with input

¹⁹Yeh et al. (2022) show that the Bond et al. (2021) critique does not affect the production approach to estimating labor wedges.

²⁰See De Loecker and Goldberg (2014), De Loecker and Syverson (2021), and De Ridder, Grassi, and Morzenti (2021) for systematic discussions of production function and markup estimation with and without output price information.

²¹This can be the case, for example, due to differences in market access by location.

quality. Under the assumption that higher quality inputs come with higher input prices, one can then build a control function for unobserved input prices using output prices. Specifically, let input prices $\mathbf{p}_{\mathbf{x},jt} = \mathbf{p}_{\mathbf{x}}(d_{jt}, \mathbf{G}_j)$ depend on output quality d_{jt} and fixed sector-location characteristics $\mathbf{G}_j$. De Loecker et al. (2016) show that the control function for input prices $\mathbf{p}_{\mathbf{x},jt} = \mathbf{p}_{\mathbf{x}}(p_{jt}, \mathbf{Z}_{jt})$ can be written as a function of output prices p_{jt} and a vector $\mathbf{Z}_{jt}$ containing sector-location fixed effects $\mathbf{G}_j$, export status, and controls for markup heterogeneity such as market shares (more details on markup controls are below).

Relative to De Loecker et al. (2016), I allow labor markets to be imperfectly competitive. This implies that monopsony markdowns will affect output prices through marginal costs. Therefore, for output prices to be valid proxies for input prices, the input price control function needs to account for monopsony markdowns. I do so by including firm wage premia in $\mathbf{Z}_{jt}$ in the input price control function. The function $B(\cdot)$ can then be written as:

$$B_s((p_{jt}, \mathbf{Z}_{jt}) \times \{1, \tilde{\mathbf{x}}_{jt}, h_{jt}\}; \beta, \zeta)$$

which is a function of output prices p_{jt} and the vector of controls $\mathbf{Z}_{jt}$, and their interactions with input expenditures $\tilde{\mathbf{x}}_{jt}$ and effective labor h_{jt} . Since input expenditures $\tilde{\mathbf{x}}_{jt}$ only enter the function $B(\cdot)$ as interaction terms with output prices and other controls $\mathbf{Z}_{jt}$, the production function parameters β are identified. This identification insight from De Loecker et al. (2016) does not hinge on functional form assumptions for $f(\cdot)$.²²

I now construct a control function to address the transmission bias arising from unobserved firm productivity, ω_{jt} . The first step in deriving this control function is to obtain the optimal material input demand function using the first-order conditions for materials:

$$\tilde{m}_{jt} = m_s(\omega_{jt}, \tilde{k}_{jt}, h_{jt}, \mu_{jt}, p_{jt}, \mathbf{Z}_{jt})$$

I invert the material input demand function to obtain the control function for productivity as a function of observed variables:

$$\omega_{jt} = \omega_s(h_{jt}, \tilde{k}_{jt}, \tilde{m}_{jt}, \mu_{jt}, p_{jt}, \mathbf{Z}_{jt}) \quad (5)$$

This relies on the following assumptions:

Scalar unobservability. *The firm's idiosyncratic Hicks-neutral productivity ω_{jt} is the only unobservable variable determining firm material input demand.*

Monotonicity. *Conditional on the variables in the control function, material input demand is monotonically increasing in ω_{jt} .*

The control function (5) points to an important identification challenge when product markets are imperfectly competitive: it includes prices and markups $\mu_{jt} = \mu(P_{jt}, D_{jt})$, which are typically unobserved. As Bond et al. (2021) explain, when prices and markups

²²I refer interested readers to De Loecker et al. (2016) for derivations of this identification result. The reason that input expenditures do not enter $B(\cdot)$ as lone variables is that the control function for input prices is built only from the consumer demand side.

are unobserved, the material input demand function generally cannot be inverted (violation of the scalar unobservable assumption), unless all variation in prices and markups are only driven by unobserved TFP ω_{jt} . In the French data, I observe output prices. If firm heterogeneity in markups is driven by ω_{jt} or regional and sectoral differences product market competition, these are controlled for in the control function. However, differences in idiosyncratic demand (D_{jt}) uncorrelated with TFP could still drive markup variation. Therefore, I additionally include controls for markup heterogeneity. Informed by oligopolistic competition trade models such as [Edmond, Midrigan, and Xu \(2015\)](#), I include export status and market shares as additional controls. Informed by models of customer capital ([Gourio and Rudanko, 2014](#)), which predict that firms accumulate customers over time, I also include firm age. Finally, to capture the potentially nonlinear relationship between markups and output prices, I include a third-order polynomial of output prices. The key assumption here is that these additional controls sufficiently capture variation in markups uncorrelated with TFP.

Estimation. I estimate production functions for each 2-digit manufacturing sector. To allow output elasticities to vary across firms and over time, I estimate translog production functions:²³

$$y_{jt} = \beta_{k,s}k_{jt} + \beta_{h,s}h_{jt} + \beta_{m,s}m_{jt} + \beta_{kk,s}k_{jt}^2 + \beta_{hh,s}h_{jt}^2 + \beta_{mm,s}m_{jt}^2 \\ + \beta_{kh,s}k_{jt}h_{jt} + \beta_{km,s}k_{jt}m_{jt} + \beta_{hm,s}h_{jt}m_{jt} + \beta_{khs,s}k_{jt}h_{jt}m_{jt} + \omega_{jt}$$

I maintain the assumption that production functions are the same across firms within 2-digit sectors and are constant over time. A simpler alternative is to estimate a sector-specific Cobb-Douglas function: $y = \beta_{k,s}k + \beta_{h,s}h + \beta_{m,s}m + \omega$. However, this restricts the output elasticities to a constant, which attributes all variation in α_m to markups and all variation in α_h/α_m to labor wedges. In Section 5.4, I provide suggestive evidence that the output elasticities vary systematically with firm characteristics.

The choice of production functions also have implications for the functional form of $B(\cdot)$. Under Cobb-Douglas, $B_s(\mathbf{p}_x(p_{jt}, \mathbf{Z}_{jt}); \beta)$, so output prices p_{jt} do not interact with input expenditures $\tilde{\mathbf{x}}_{jt}$. Under translog, $B_s(\mathbf{p}_x(p_{jt}, \mathbf{Z}_{jt}), \tilde{\mathbf{x}}_{jt}, h_{jt}; \beta, \zeta)$, as described before. It is also worth noting that the production function parameters β are separately identified from parameters of the function $B(\cdot)$, because input expenditures and effective labor only appear in $B(\cdot)$ as interaction terms with output prices.²⁴

I estimate the production functions following the two-step GMM approach described in [Akerberg et al. \(2015\)](#). In the first-stage, I combine the control function with the production function and estimate the following by OLS:

²³In this paper, I focus on gross output production functions. I refer interested readers to [Rubens \(2021\)](#) for a method of estimating markups and markdowns when production functions are Leontief.

²⁴For more details on production function and markup estimation under unobserved input prices, I refer the interested reader to [De Loecker et al. \(2016\)](#).

$$y_{jt} = \Psi_s(\tilde{k}_{jt}, h_{jt}, \tilde{m}_{jt}, \mu_{jt}, p_{jt}, \mathbf{Z}_{jt}; \beta, \zeta) + \epsilon_{jt} \quad (6)$$

The vector $\mathbf{Z}_{jt}$ contains 5-digit sector-location fixed effects, year effects, export status, market shares, and firm age. This step estimates and removes the residual term ϵ_{jt} , capturing measurement error and productivity shocks that are unobserved by the firm and are therefore orthogonal to input choices.²⁵

In the second stage, I estimate the production function parameters β and input-price-related parameters ζ by forming moment conditions. Firm productivity ω_{jt} can be written as a function of the parameters to be estimated $\{\beta, \zeta\}$:

$$\omega_{jt}(\beta, \zeta) = \hat{\Psi}_{jt} - f_s(\tilde{k}_{jt}, h_{jt}, \tilde{m}_{jt}; \beta) - B_s(\mathbf{p}_x(p_{jt}, \mathbf{Z}_{jt}), \tilde{\mathbf{x}}_{jt}, h_{jt}; \beta, \zeta)$$

Specify law of motion for the log of Hicks-neutral productivity as:

$$\omega_{jt} = g_s(\omega_{jt-1}) + \varpi_{jt} \quad (7)$$

where $g_s(\cdot)$ is a flexible function and ϖ_{jt} is a productivity shock.²⁶ Combining equation (6) and the law of motion for productivity (7), I obtain productivity shocks ϖ_{jt} as a function of the parameters of interest:

$$\varpi_{jt}(\beta, \zeta) = \omega_{jt}(\beta, \zeta) - g_s(\omega_{jt-1}(\beta, \zeta))$$

I then form the following moment conditions:

$$E[\varpi_{jt}(\beta, \zeta) \mathbf{X}_{jt}] = \mathbf{0}$$

where $\mathbf{X}_{jt}$ includes current and lagged capital, lagged effective labor, lagged material inputs, lags of the squared factor inputs, lagged interaction terms between the factor inputs, lagged output prices, lagged market shares, lagged export status, lagged firm age, lagged wage premia, and the interaction terms between lagged output prices and lagged factor inputs. Under the timing assumption on the productivity process, these lagged variables are orthogonal to current productivity shocks. In addition, capital inputs are assumed to be dynamic and pre-determined, therefore firms' current capital input demand are also orthogonal to current productivity shocks.

As discussed above, the availability of output price data for French manufacturing firms enables me to estimate production functions that address (i) output price bias, (ii) input price bias, and (iii) potential violations of the scalar unobservable assumption due to the presence of markups and output prices as material demand shifters. In Section 5.4 (with further details in Appendix B.6), I compare the markup and labor wedge estimates across various specifications that address some or all of points (i)-(iii).²⁷

²⁵I approximate $\Psi(\cdot)$ with a third-order polynomial using each variable except dummy variables.

²⁶I let $g_s(\cdot)$ be linear. I approximate $B(\cdot)$ with a third-order polynomial using each variable except dummy variables. Dummy variables enter linearly.

²⁷See also [De Ridder et al. \(2021\)](#) and [Yeh et al. \(2022\)](#) for systematic assessments of how unobserved output prices and markups in the control function can affect markup and markdown estimates.

The control function approach requires selecting an input which is flexible and whose prices are taken as given by firms. In estimating production functions, I have assumed that material inputs satisfy these two conditions. In Section 5.4, I compare materials to electricity consumption (from an annual survey of French manufacturers) as an alternative for the flexible input. Although electricity consumption is a natural alternative candidate—electricity consumption is plausibly flexible and its production is typically concentrated within a few producers—I show that electricity consumption in the survey data is significantly noisier than the data on materials, which is from administrative sources. This limits electricity’s usefulness as a proxy variable and as a markup measurement device in practice. Additionally, I show in the same section that it is still possible to recover the correct markup and labor wedge measures when firms have monopsony power in materials, if production functions are estimated using material expenditure instead of material quantities.

Finally, it is important to acknowledge a key challenge in identifying the material output elasticity, as highlighted in the production function estimation literature. [Gandhi, Navarro, and Rivers \(2020\)](#) show that, under perfectly competitive product markets, the only source of variation that identifies the material output elasticity is time-series variation in material prices. However, [De Ridder et al. \(2021\)](#) show that with imperfectly competitive product markets—a key feature of my model—variation in output prices and productivity shocks to firms’ competitors generate additional variation in firms’ material input demand, which aids the estimation of the material output elasticity. To further address this identification concern, in Section 5.4, I implement an alternative approach suggested by [Flynn, Gandhi, and Traina \(2019\)](#), which imposes constant returns to scale on the production function, instead of relying on output prices and shocks to competitors as additional sources of variation. I then compare the corresponding markup and labor wedge estimates to my baseline estimates.

3.3 Estimating workers’ bargaining power and wage markdowns

Recall that the estimated labor wedges (Λ) are not generally equal to monopsony markdowns (λ). Disentangling the two requires an estimate of worker bargaining power (κ). I first discuss the identification challenges that often arise when estimating κ and standard approaches taken to address them in existing literature. I then show that an additional identification challenge arises in the context of my model which cannot be immediately addressed using established approaches. To make progress on estimating κ , I then propose an alternative approach motivated by my model.

To illustrate the identification challenges that frequently arise when estimating κ , it is useful to first derive a wage equation that motivates the analyses in the “rent-sharing” literature ([Card et al., 2018](#)). This literature estimates the elasticity of wages with respect to measured profits per worker. As in Section 2, wages W_{ij} have a worker-

specific component E_i and a firm-specific component Φ_j , $W_{ij} = E_i\Phi_j$. Suppose that a worker i 's utility at firm j is $V_{ij} = W_{ij}A_j$. Let $W_i^o = E_i\Phi^o$ be the alternative wage, which is commonly modelled as the utility value of nonemployment or a competitively determined wage in an outside labor market with an amenity value normalized to one. The only difference of the current set up compared to my model in Section 2 is that now firms *do not* have monopsony power.²⁸ The efficient bargaining problem becomes:

$$\max_{\Phi_j, P_j, H_j, M_j, K_j} \left((\Phi_j A_j - \Phi^o) H_j \right)^\kappa \left(P_j Y_j - \Phi_j H_j - P_m M_j - P_k K_j \right)^{1-\kappa}$$

subject to $Y_j = \mathcal{G}(P_j, D_j)$ and $Y_j = \Omega_j F(K_j, M_j, H_j)$. This delivers the wage equation.²⁹

$$\Phi_j = \kappa Q_j^{rent} + (1 - \kappa) \Phi_j^r \quad (1')$$

where $\Phi_j^r \equiv \frac{\Phi^o}{A_j}$ is the reservation wage and $Q_j^{rent} \equiv \frac{P_j Y_j - P_m M_j - P_k K_j}{H_j}$ are profits per worker, often referred to as “quasi-rents” in the rent-sharing literature.

Common identification challenges and existing solutions. For the following reasons, an OLS regression of firm-level average wages W_j on quasi-rents Q_j^{rent} will give an upward-biased estimate of κ :

- IC1. *Unobserved worker heterogeneity.* The positive correlation between W_j and Q_j^{rent} may simply reflect differences in unobserved worker ability E_i .
- IC2. *Common shocks.* An aggregate productivity shock will raise firm-level Q_j^{rent} but also the outside wage Φ^o .
- IC3. *Firm-specific non-wage amenities.* The positive correlation between W_j and Q_j^{rent} may reflect differences in unobserved firm amenities, since low-amenity firms must pay higher wages and also employ fewer workers.

To address IC1, the literature makes use of matched worker-firm panel data by estimating rent-sharing elasticities for workers who do not switch employers (“stayers”). To address IC2 and IC3, the literature proposes a set of instruments for Q_j^{rent} and makes the case that they are: (i) relevant, (ii) idiosyncratic (addressing IC2), and (iii) orthogonal to firm-specific amenities (addressing IC3).³⁰

A new identification challenge. Estimating κ using my main wage equation (1) would lead to the same endogeneity concerns. My approach to addressing IC1 follows the

²⁸This is a common feature, for example, in labor market models with search frictions and wage bargaining (Diamond, 1982; Mortensen, 1982; Pissarides, 1985).

²⁹Card et al. (2018) propose a monopsony model that does feature firm-specific labor supply curves and generates the same wage equation. However, in their model, there is no wage bargaining. This has two implications. First, κ is instead a function of labor supply preference parameters. Second, and more importantly, the rest of this section shows that without bargaining the realized wage is the monopsony wage—the monopsony wage is thus observed—which is key for the validity of existing instruments.

³⁰See Card et al. (2018) for a survey of this literature. Recent work used idiosyncratic export demand shocks or patent shocks as instruments (Garin and Silverio, 2024; Kline et al., 2019).

literature by using worker-firm panel data. However, a comparison of the wage equation (1') with the more general wage equation (1) shows why instrumental variables that satisfy the conditions listed above will not deliver an estimate of κ in the context of my model. Recall that equation (1) is:

$$\Phi_j = \kappa Q_j^{rent} + (1 - \kappa) \Phi_j^{monop}$$

where $\Phi_j^{monop} = \lambda_j \mu_j^{-1} \alpha_j^h \frac{P_j Y_j}{H_j}$ is the monopsony wage. This comparison highlights the key difference between the two wage equations: the alternative wage in my model is the monopsony wage. This difference stems from the fact that firms have monopsony power in my model, while they typically do not in the bargaining models traditionally employed in the rent-sharing literature (Card et al., 2018).

This presents a new challenge in identifying κ : any idiosyncratic shock that shifts Q_j^{rent} will also affect the monopsony wage Φ_j^{monop} at the same time, implying that the exclusion restriction cannot be satisfied. This is because both Q_j^{rent} and Φ_j^{monop} are determined by firm productivity, product quality, and amenities. Thus, applying existing instruments to estimate worker bargaining power will yield biased estimates in the context of my model.

An alternative approach to estimating worker bargaining power. In light of the above discussion, I propose an alternative theory-consistent approach to estimate κ that avoids the presence of unobserved monopsony wages. Instead of using the wage equation (1), my approach exploits the structural relationship between price-cost markups and the labor wedges given by equation (3). Recall that the labor wedges are:

$$\Lambda_{jt} = \kappa \tilde{\mu}_{jt} + (1 - \kappa) \lambda_{jt}$$

where $\tilde{\mu}_{jt} \equiv \left(1 - \frac{P_{m,t} M_{jt} + P_{k,t} K_{jt}}{P_{jt} Y_{jt}}\right) \frac{\mu_{jt}}{\alpha_{h,jt}} = \left(\frac{P_{jt} Y_{jt} - P_{m,t} M_{jt} - P_{k,t} K_{jt}}{P_{m,t} M_{jt}}\right) \frac{\alpha_{m,jt}}{\alpha_{h,jt}}$ are product market rents. Suppose that labor wedges (Λ_{jt}), markups (μ_{jt}), markdowns (λ_{jt}), and output elasticities ($\alpha_{h,jt}, \alpha_{m,jt}$) are observed. Then, conditional on markdowns, variation in product market rents identify worker bargaining power. The intuition is that when workers have bargaining power, they are able to appropriate some of the rents generated by markups.

Although the production function estimation step provides estimates of Λ_{jt} and $\tilde{\mu}_{jt}$, at this stage markdowns (λ_{jt}) remain unobserved. Recall from Section 2 that markdowns may vary endogenously depending on firm size and amenities $\lambda_{jt} = \lambda(H_{jt}, A_{jt})$. This implies an endogeneity issue, because markups and markdowns will be correlated. However, because firm size is observed, using the labor wedge equation (3) in place of the wage equation (1) turns the issue of unobserved monopsony wages Φ_{jt}^{monop} into an issue of unobserved amenities A_{jt} . This is similar to an identification challenge (IC3) faced in the rent-sharing literature when using the wage equation (1') to estimate κ .

Before addressing unobserved amenities, it is worth noting that one advantage of using the labor wedge equation (3) instead of the wage equations ((1) or (1')) is that unobserved amenities *do not necessarily* lead to an identification challenge in estimating κ . Even

when markdowns depend on firm size, if amenities and the disutility of supplying labor (or wages) are *multiplicatively separable* in worker preferences, markdowns do not directly depend on amenities. To show this, consider the following example of a representative household who maximizes utility by choosing consumption (C) and labor supply to each firm (H_j):

$$\max_{C, H_j} C - \sum_j A_j^{-1} \mathcal{L}(H_j) \quad \text{s.t.} \quad \sum_j \Phi_j H_j = C$$

where the function $\mathcal{L}(\cdot)$ represents the disutility of supplying labor. It satisfies $\mathcal{L}' > 0$, and $\mathcal{L}'' > 0$. This function features variable elasticity of substitution between jobs (firms) and nests constant elasticity of substitution as a special case. Thus, it allows labor supply elasticities, and hence, monopsony markdowns, to vary endogenously across firms, as in [Berger et al. \(2022\)](#). The inverse labor supply function and labor supply elasticities are:

$$\Phi_j = A_j^{-1} \mathcal{L}'(H_j) \quad \text{and} \quad \xi_j = \frac{\mathcal{L}'(H_j)}{\mathcal{L}''(H_j) H_j}$$

Thus, under multiplicative separability, amenities are proportional labor supply shifters and do not directly affect labor supply elasticities—firm size is a sufficient statistic for monopsony markdowns. In this case, unobserved amenities do not pose a direct threat to the identification of κ when using the labor wedge equation (3). This multiplicative separability is a common modeling assumption for amenities ([Card et al., 2018](#); [Lamadon et al., 2022](#)).³¹

However, in general, amenities may not be multiplicatively separable from wages. For this more general case, I propose a control function approach to address unobserved variation in amenities.³² My approach relies on the following assumption:

Assumption. The firm-specific labor supply function $H_{jt} = \mathcal{H}(\Phi_{jt}, A_{jt})$ is monotonically increasing in the value of its amenities A_{jt} , conditional on Φ_{jt} .

Under this assumption, the control function for amenities can be written as $A_{jt} = \mathcal{A}(H_{jt}, \Phi_{jt} H_{jt})$.³³ The estimating equation becomes:

$$\hat{\Lambda}_{jt} = \kappa \hat{\mu}_{jt} + (1 - \kappa) \lambda(H_{jt}, \mathcal{H}^{-1}(H_{jt}, \Phi_{jt} H_{jt})) + \tilde{\epsilon}_{jt} \quad (3')$$

where $\tilde{\epsilon}$ is a residual and variables with a hat denote estimates from previous steps. The idea behind the amenity control function is that variation in unobserved amenities can be

³¹Analogously for product markets, it is also a common modeling approach for quality or demand shifters, see [Hottman, Redding, and Weinstein \(2016\)](#) and [Eslava, Haltiwanger, and Urdaneta \(2023\)](#) for example.

³²Here, I focus on the case where workers have a null outside option $\Phi^o = 0$, consistent with empirical findings from Austria and France that wages and reservation wages are insensitive to changes in unemployment benefit policies ([Le Barbanchon et al., 2019](#); [Jäger et al., 2020](#)). In Appendix C.1, I estimate κ when $\Phi^o > 0$. I find similar estimates for worker bargaining power.

³³This expression for the control function is consistent with recent models of imperfect labor market competition such as [Berger et al. \(2022\)](#) and [Azkarate-Askasua and Zerecero \(2024\)](#), where wage-bill or employment market shares are sufficient controls for markdown variation. An equivalent way of writing the control function is in terms of effective labor and firm wage premia $A_{jt} = \mathcal{A}(H_{jt}, \Phi_{jt})$.

controlled for jointly by employment and wage bills. The intuition is that amenities shift labor supply, which breaks the direct correspondence between wages and employment in the labor supply curve. The bargaining weight κ is then identified from variation in markups, conditional on firm size in terms of employment and wage bills.

I implement equation (3') as an OLS regression of the estimated labor wedges ($\hat{\Lambda}_{jt}$) on estimated product market rents ($\hat{\mu}_{jt}$) and an approximation of the markdown function $\lambda(\cdot)$. I approximate $\lambda(\cdot)$ with a 4th-order polynomial in (effective) labor and wage bills. In my baseline implementation, I include sector $\times$ year fixed effects to absorb sector-specific time trends and firm fixed effects to absorb any unmodelled time invariant firm characteristics that may determine the labor wedges. Finally, I compute monopsony markdowns using the estimated bargaining parameter: $\hat{\lambda}_{jt} = \frac{1}{1-\hat{\kappa}}(\hat{\Lambda}_{jt} - \hat{\kappa}\hat{\mu}_{jt} - \hat{\epsilon}_{jt})$.

Although my approach to estimating worker bargaining power flexibly accommodates the presence of monopsony markdowns, there are important limitations to note. First, both the measurement of labor wedges and product market rents requires first estimating output elasticities ($\alpha_{h,jt}, \alpha_{m,jt}$), which are subject to the challenges of estimating flexible production functions discussed in Section 3.2. Because the estimated output elasticities enter the measurement of Λ_{jt} and μ_{jt} multiplicatively, mismeasurement in the output elasticities will then lead to a mechanical upward bias in the estimated κ . I refer to this as *multiplication bias* and address it in Section 5.4. Second, my analysis has so far assumed that labor is a flexible input. To the extent that labor adjustment costs are present, they will be captured by my measure of labor wedges and bias the estimated κ . In Section 5.4, I infer the extent to which labor adjustment costs affect my estimates and to incorporate it into my estimation approach.

3.4 Discussion

While the estimation of the model described in Section 2 allows for richer firm heterogeneity and features wage bargaining relative to existing labor market monopsony models, there are certain environments in which my analysis does not immediately extend to.

First, the presence of downward-sloping product demand curves implies that firms face diminishing returns to labor. In models where labor markets are characterized by search frictions and on-the-job search (e.g. [Burdett and Mortensen \(1998\)](#)), diminishing returns to labor can result in an equilibrium where all firms pay the same wage.^{34,35} My estimates of firm wage premia and labor wedges do not readily extend to these settings.

Second, my structural framework and estimation approach allows for within-firm wage

³⁴An example is when workers and firms bargain over wages and hiring costs per worker are constant. In this environment, firms will hire until marginal products of labor equal hiring costs. Since marginal products are equalized, so are wages.

³⁵Recent developments in modeling firm dynamics with diminishing returns to labor, search frictions, and on-the-job search are [Bilal, Engbom, Mongey, and Violante \(2022\)](#) and [Elsby and Gottfries \(2021\)](#).

dispersion due to differences in worker skills, but does not extend to settings in which workers have different outside options or wage contracts within firms. An important example is the sequential auctions mechanism (Postel-Vinay and Robin, 2002), where wages within firms can differ between two identical workers because one worker received an outside offer that induces the incumbent employer to engage in wage competition. Further, incumbent workers and new hires who are otherwise identical may be paid differently. In such cases, equation (2) is misspecified. In Appendix B, I show that my empirical findings are robust to considering only hiring wages following Di Addario, Kline, Saggio, and Solvsten (2020), allowing incumbents to be paid differently.

Third, implicit in the efficiency units specification of the production function, I assume that worker types are perfect substitutes, although average worker efficiency and firm productivity are complements. This restrictive assumption implies that the model abstracts from worker-firm sorting based on production complementarities (Eeckhout and Kircher, 2011). In return, this assumption (i) delivers a close mapping between AKM regressions and the structural wage equation; and (ii) keeps the production function estimation procedure computationally feasible—introducing multiple worker types exponentially increases the number of parameters to be estimated when estimating flexible production functions. When workers are imperfect substitutes, the log-additive AKM regression is misspecified—an interaction term between the worker and firm effect needs to be present (see, for example, Lamadon et al. (2022)). I address this issue in two ways. First, I estimate an AKM regression augmented with this interaction term, following Bonhomme et al. (2019), but find a limited role for it. Second, in Appendix E, I provide an extension that distinguishes between two skills. I find that the results of this extension are similar to those reported in Section 5.

4 Data Description

4.1 Administrative datasets from France

Implementing the estimation approach described above requires three types of datasets. Firm wage premia are estimated using matched employer-employee data, which track workers over time and across different firms. Labor wedges and markups are estimated using firm balance sheet panel data and a survey of firm-product level output prices. While these types of datasets have become increasingly accessible, they are often not jointly available. Therefore, I use data from France.

Firm balance sheet data. My source for firm balance sheet information is the *Fichier approché des résultats d’Esane* (FARE) dataset, available from 2008 to 2019. FARE is compiled by the fiscal authority of France, *Direction Générale des Finances Publiques* (DGFIP), from compulsory filings of firm annual accounting information. The dataset

contains balance sheet information for all firms in France. From these datasets, I obtain information on variables such as sales, employment, material input and capital expenditure. I provide details on measurement in [Appendix A](#).

Output price data. To obtain output prices at the firm level, I use the *Enquête Annuelle de Production* (EAP), available from 2009 to 2019. These are survey data compiled by the national statistical institute of France, *Institut National de la Statistique et des Études Économiques* (INSEE). The dataset contains firm-product-level sales and output for all manufacturing firms with at least 20 employees or with sales exceeding 5 million Euros, and a representative sample of manufacturing firms with less than 20 employees.

More specifically, for each year, firm, and ten-digit product code combination, the survey reports the unit of account (e.g., kilograms, meters, and pieces), total quantity sold, and total revenue. I define a product as the combination of a ten-digit product code and a unit of account, treating firms that report different units of account for the same product code as producing a different product. I drop firm-product combinations for which either quantity or revenue is missing.

COMPUTING FIRM-LEVEL OUTPUT PRICES. Let g denote a product, G_{jt} the set of products sold by firm j in year t , and J_{gt} the set of firms that sell product g in year t . I compute the firm-level output price as:

$$P_{jt} = \sum_{g \in G_{jt}} \mathcal{W}_{jgt} \frac{P_{jgt}}{\bar{P}_{gt}}, \text{ with } \mathcal{W}_{jgt} = \frac{P_{jgt} Y_{jgt}}{\sum_{g \in G_{jt}} P_{jgt} Y_{jgt}}, \bar{P}_{gt} = \sum_{j' \in J_{gt}} \left(\frac{P_{j'gt} Y_{j'gt}}{\sum_{j'' \in J_{gt}} P_{j''gt} Y_{j''gt}} \right) P_{j'gt}$$

That is, I first compute firm-product prices (P_{jgt}) for each year by dividing firm-product revenues by the corresponding quantities. I then normalize the firm-product price measure by dividing it by the sales-weighted average price of the particular product across all firms in a given year ($\bar{P}_{gt}$). The firm-specific output price (P_{jt}) is then computed as the sales-weighted ($\mathcal{W}_{jgt}$) average of the price index ($\frac{P_{jgt}}{\bar{P}_{gt}}$) across all products sold by a given firm. This procedure follows [De Ridder et al. \(2021\)](#). It is worth noting that my paper does not have a theory of multiproduct firms to construct theory-motivated firm-specific price aggregators. Additionally, it is not straightforward to compare output across firms when products are differentiated. These price measurement issues are the subject of ongoing research, discussed in detail by [De Loecker and Syverson \(2021\)](#).

Matched employer-employee data. I also use annual French administrative data on employed workers, available from 1995 to 2018, under the umbrella *Déclarations Annuelles de Données Sociales* (DADS). The DADS datasets are compiled by INSEE from compulsory reports of employee information to the French authorities. They contain information at the worker level, such as age, gender, earnings, hours, and occupational category. One advantage of the DADS is that work hours are observed, allowing researchers to construct hourly wages. This addresses concerns that variation in earnings simply reflect variation in hours worked. Data on hours also allows researchers to ac-

count for differences in hours across firms, which are not captured by employment, when measuring firm-level labor inputs. They also include employer identifiers, called SIREN, which enables linking with firm balance sheet data. One disadvantage is that information about workers' education is not available.

The first DADS dataset is the DADS-Panel, which provides information on all employed workers in the private sector born in October in a panel structure.³⁶ Because workers are followed over time and their employer identifiers are observed, I use this dataset to estimate firm wage premia. The second DADS dataset is the DADS-Postes, which contains information on all existing jobs in France. Unlike the DADS-Panel, this is not a proper panel dataset. It is organized in an overlapping structure—each observation appears in the dataset under the same identifier for at most two periods. Therefore, this dataset cannot be used to estimate firm wage premia directly. Instead, to maximize the number of firms for which firm wage premia are estimated using the DADS-Panel, I first use the DADS-Postes to k-means cluster firms into groups of similar firms following the procedure described in the previous section. This approach has the advantage that firm wage premia can be estimated for firms that exist in the firm balance sheet data but not in the DADS-Panel because they do not have an employee who is born in October.

4.2 Estimation sample

I restrict firm-level observations from the FARE balance sheet data to manufacturing firms whose output prices are observed in the EAP. I include only firms with at least 5 employees. I harmonize all industry codes to the latest available version (Nomenclature d'activités Française, NAF rév. 2) and exclude 2-digit sectors with fewer than 300 observations.

For both of the DADS datasets, I focus on workers between the age of 16 to 65, who hold either a part-time or full-time job principal job (jobs in which workers are paid for at least 30 days of work and at least 120 hours of work that year). I use only years 2009-2016, since INSEE reports that wages are recorded with errors in 2017 and 2018. I keep workers in almost all 1-digit occupational categories, except farm workers. The included occupational categories are top management, senior executives and technical professions, middle management, non-supervisory white-collar workers, and blue-collar workers. Occupation codes are harmonized and updated to the latest version (PCE-ESE 2003). Workers whose wages fall outside 3 standard deviations of the mean are excluded.

Firm wage premia in the wage regression are only identified for the sets of firms connected by worker mobility. I focus on the largest connected set of firms. In practice, due to the clustering of firms into groups using the DADS-Postes, my analysis pertains to the largest connected set of firm-groups, of which very few firms are not a part. This

³⁶Only October-workers born in even years are observed prior to 2002.

group consists of 158,163,180 people-year observations, an average of 7,908,159 per year. After clustering firms into groups, I link the DADS-Postes and DADS-Panel via the firm identifier to allocate each firm-year observation a firm-group identifier and construct the estimation sample for firm wage premia. I estimate firm wage premia on this sample. Figure 24 in Appendix D shows that larger firms pay higher wage premia.

Table 1: Summary statistics of observed firm characteristics.

	Sales (’000€)	Employment	Sales per hour (’000€)	Prices	Prices (2-digit)	Prices (5-digit)
10 th percentile	814	8	0.05	-1.75	-1.37	-0.92
25 th percentile	1,502	12	0.06	-0.37	-0.41	-0.36
Median	3,528	24	0.09	0.28	0.16	0.07
Mean	16,920	76	0.13	0.00	0.00	0.00
75 th percentile	10,352	53	0.14	0.53	0.53	0.42
90 th percentile	34,268	144	0.22	0.94	0.98	0.91
Std. deviation	46,283	289	0.27	1.02	0.95	0.86

This table reports the summary statistics for firm-level sales, employment, sales per hour, and output prices. Sales and sales per hour are denoted in 2009 prices. Output prices are in logs, normalized to zero at the mean, and winsorized by 5% on either side of the distribution. The last two columns, with ‘2-digit’ and ‘5-digit’ in parentheses, display variation in log output prices within 2-digit and 5-digit sector classifications.

After estimating firm wage premia, I collapse the dataset to the firm level and link it to the FARE-EAP balance sheet and output price data to construct the estimation sample. I implement the production function estimation routine on this sample. There are 126,836 firm-year observations in total and an average of 16,501 firms per year in this sample. The summary statistics in Table 1 show that there is significant price dispersion within narrowly defined sectors. Figure 25 in Appendix D further shows that this price dispersion is positively correlated with firm size.

5 The Characteristics of High and Low-Wage Firms

5.1 The distribution of firm wage premia and firm market power

In this section, I report the distribution of estimated firm wage premia and firm heterogeneity. I show that the median French manufacturing firm charges high markups and pays wages below the marginal revenue product of labor.

Firm wage premia. Table 2 reports statistics about firm wage premia. The variance is modest, accounting for 4.6% of wage dispersion, similar to the numbers for the United States, Sweden, Austria, Norway, and Italy from Bonhomme et al. (2023). Nevertheless, the dispersion of firm wage premia is a quantitatively important deviation from the law of one wage. Column 2 in Table 2 shows that a firm at the 90th percentile of the firm

wage premium distribution pays a given worker a wage that is on average 30% more than a firm at the 10th percentile. This difference amounts to approximately 4 Euros per hour or 25% of the hourly wage of the median French worker in 2016. The interquartile range is 15%, similar in magnitude to the typical estimate for the costs of job displacement (Schmieder, Von Wachter, and Heining, 2018). This gap is also approximately three times as large as the gender wage gap in France. Among manufacturing firms, the 90-10 wage premium difference is slightly smaller at 23%.

Table 2: Dispersion of firm wage premia.

	BLM		AKM		KSS	
	MN	Overall	MN	Overall	MN	Overall
$\frac{Var(\phi)}{Var(w)}$	4.6%	6.9%	9.2%	15.9%	4.3%	7.2%
$Var(\phi)$	0.008	0.011	0.016	0.026	0.006	0.012
90-10 ratio	1.23	1.30	1.33	1.43	-	-
75-25 ratio	1.10	1.15	1.16	1.20	-	-
90-50 ratio	1.11	1.14	1.14	1.21	-	-
50-10 ratio	1.11	1.14	1.17	1.19	-	-
# firms	15,934	369,091	13,289	343,470	13,289	343,470
# firm-groups	418	4,156	13,289	343,470	13,289	343,470
# workers	1,128,966	10,737,479	123,952	1,829,455	123,952	1,829,455

This table reports the summary statistics for firm wage premia in 2016. Columns labeled ‘MN’ are estimates for manufacturing firms in my estimation sample. Columns labeled ‘Overall’ are estimates for all firms in the private sector. The first and second columns report statistics for firm effects estimated at the firm-group level following BLM. The columns under AKM report the firm effects without grouping firms. The columns under KSS report firm effects corrected for limited mobility bias following the Kline et al. (2020) leave-out approach.

Markups and labor wedges. The first row of Table 3 reports that markups (μ) are heterogeneous across firms. The median markup among French manufacturers is 1.32. De Loecker and Warzynski (2012) estimate markups using Slovenian manufacturing data and find median markups between 1.10 and 1.28. De Loecker et al. (2020) find markups at the 75th percentile between 1.30 and 1.60 in 2016 in the US economy, while my estimates for French manufacturing is 1.60 in 2016. Edmond et al. (2023) report an interquartile range for markups of 1.31-0.97=0.34. The interquartile range for my estimates is 0.43.

The second row of Table 3 describes the distribution of labor wedges (Λ), which captures the wage-setting power of employers relative to their employees. Most French manufacturers appear to have significant wage-setting power. France has one of the highest national minimum wages.³⁷ Nevertheless, half of the firms in my sample pay less than two-thirds of the marginal revenue product of labor as wages. I also find substantial dispersion of labor wedges across firms. Firms at the 75th percentile of the labor wedge

³⁷In 2016, about 15% of French workers are earning at or close to the minimum wage. In manufacturing, the corresponding number is 10%. See Appendix A for a description of French wage-setting institutions.

distribution pay a wage that is 77% of the marginal revenue product of labor. At the 25th percentile, workers obtain around half of their marginal revenue productivity.

The set of direct estimates of labor wedges is small. I compare my estimates to those of [Yeh et al. \(2022\)](#) and [Mertens \(2020\)](#), whose estimates are methodologically the closest to mine.³⁸ They estimate that the median US and German manufacturing firm pay 0.73 and 0.68 of the marginal revenue product of labor as wages. For the median French manufacturing firm the corresponding number is 0.62.

Table 3: Summary statistics for measured markups and labor wedges.

	Mean	Median	25 th Pct	75 th Pct	Var	Var (i)	Var (ii)
Markups (μ)	1.37	1.32	1.17	1.60	0.07	0.05	0.01
Labor wedge (Λ)	0.63	0.62	0.52	0.77	0.12	0.08	0.03
Number of firms	15,934						

This table reports the summary statistics in 2016 for the measured price-cost markups and labor wedges. They are obtained by implementing my baseline approach: three-input translog production function taking into account input prices, output prices, and markup heterogeneity. Variances are reported for the log of the corresponding variable. The column Var (i) reports the variances corrected for measurement error following [Krueger and Summers \(1988\)](#) and [Kline et al. \(2020\)](#), while the column Var (ii) reports the variances between firm-groups. Markups and labor wedges are winsorized by 2%.

The role of hours and output price data. What explains the difference between my labor wedge measures for France and those for the US and Germany? Like [Yeh et al. \(2022\)](#) and [Mertens \(2020\)](#), I estimate gross output translog production functions to measure labor wedges. However, my implementation differs in several ways. First, the availability of output price data for French manufacturing firms makes it possible to: (i) measure output in terms of quantities instead of revenues, addressing potential output price bias; (ii) build a control function for unobserved input quality variation, addressing potential input price bias ([De Loecker et al., 2016](#)); and, (iii) control for output price and markups in the productivity control function, which would otherwise violate the scalar unobservable assumption because prices and markups appear when goods markets are imperfectly competitive. Second, the availability of matched employer-employee data with information on worker-level hours worked allows me to measure labor inputs in three ways: employment, total hours, or total effective hours (my baseline measure). Measuring labor inputs as total hours accounts for the contribution of hours to firm output, while measuring labor as total effective hours further accounts for the contribution of differences in worker efficiency across firms.

To see whether differences in methodology or measurement drive the differences in

³⁸However, the interpretation of these labor wedges differ between my paper and those in [Yeh et al. \(2022\)](#) and [Mertens \(2020\)](#). In the aforementioned papers, labor wedges are interpreted as monopsony markdowns, corresponding to the case where workers do not have bargaining power ($\kappa = 0$) in my framework in Section 2. Allowing $\kappa > 0$ also helps rationalize that about 7% of firms have a labor wedge greater than 1. Indeed, firms with $\Lambda > 1$ tend to have much higher markups.

labor wedge measures, I compare the level and dispersion of labor wedges under different specifications and measures of output and labor. This comparison is presented in Table 19 in Appendix B.6. Rows (1), (2), and (3) show the labor wedges I obtain when I estimate production functions using a “basic” specification that *does not*: (i) address output price bias; (ii) address input price bias; (iii) control for markups and prices in the productivity control function. The only difference between the rows is how labor inputs are measured.

My main finding is that the choice of labor input measure significantly affects both the level and dispersion of measured labor wedges. Using total hours as the labor measure results in lower labor wedges with considerably smaller dispersion than when labor is measured as employment. When I measure labor as total effective hours, the median labor wedge is 0.62, which increases to 0.64 when using total hours, and further rises to 0.71 when using employment. This difference in labor input measurement accounts for a large part of the difference between my labor wedge estimates and those of Yeh et al. (2022) and Mertens (2020). Moreover, when labor is measured as employment, the 90-10 difference in labor wedges is 0.68; when I use effective hours instead, it is 0.48—a 20 percentage point difference.

Similarly, both the level and dispersion of measured markups fall when I measure labor inputs as total hours instead of employment. When measured as employment, the median markup is 1.37 and 90-10 difference is 0.99. Instead, when measured as total hours, the median markup is 1.34 and 90-10 difference is 0.93. When measured as total effective hours, the corresponding numbers fall further to 1.31 and 0.88.

These findings suggest that using employment instead of hours or effective hours would understate the *level* of firm labor market power, and overstate their *dispersion*. This matters because recent models of labor and product market power highlight that the level of labor wedges or markups act as a uniform tax on labor demand for all firms, while the dispersion leads to misallocation of labor across firms (Berger et al., 2022; Edmond et al., 2023). Since the availability of data on hours depends on the country, these findings also suggest that researchers should interpret the estimated moments of labor wedges and markups with this issue in mind. In Appendix B.6, I discuss two reasons for why using employment in place of total hours leads to upward-biased measures of labor wedges: (i) omitted variable bias (average hours); and (ii) mismeasurement of output elasticities, even with known production function parameters.

The availability of output price data enables the estimation of richer specifications that account for output and input price biases, as well as the presence of prices and markups in the productivity control function. A key question is whether these richer specifications produce significantly different markup and labor wedge estimates compared to a basic specification that does not address these biases. At first glance, comparing the basic specification (row (3) in Table 19) to the specification that addresses each of these biases (row (9)) yields very similar markup and labor wedge estimates, both in terms

of levels and dispersion. However, a closer comparison of rows (3) and (4) shows that addressing only the output price bias leads to slightly lower and less dispersed markups, along with much higher and more dispersed labor wedges. Comparing rows (4) and (5) indicates that accounting for markups in the control function does not significantly alter the estimates. In contrast, comparing rows (4) and (6) shows that addressing the input price bias, in addition to the output price bias, returns markup and labor wedge estimates that closely resemble those from the basic specification. These results suggest that the output and input price biases roughly offset each other. As shown in Appendix B.6, this offsetting effect can occur when firms differ in output quality. Consequently, addressing only one source of bias but not the other would yield more biased markup and labor wedge estimates.

5.2 How prices, markups, and labor wedges vary across high-wage and low-wage firms

I now compare how output prices and market power differ between high-wage and low-wage firms. I provide new empirical evidence demonstrating that high-wage firms tend to charge higher output prices and markups, while also paying a larger share of the marginal revenue product of labor as wages. These empirical patterns cannot be readily explained by existing monopsony models of the labor market where firms do not have product market power and workers do not have bargaining power.

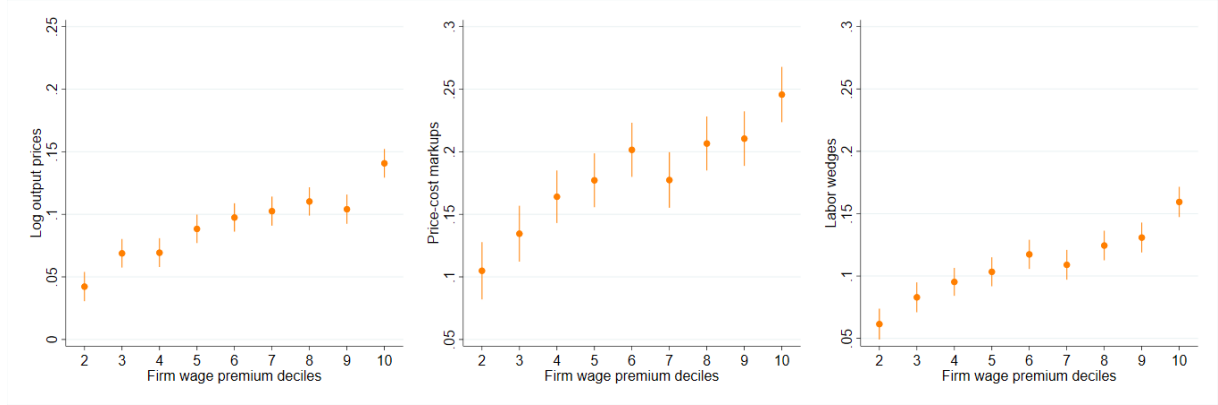


Figure 1: Prices, markups, and labor wedges by high-wage status, conditional on TFPQ.

Notes: This figure shows how log output prices, price-cost markups, and labor wedges vary by deciles of firm wage premia relative to firms in the first decile, controlling for TFPQ and 5-digit sector $\times$ year fixed effects. Decile 10 represents high-wage firms. Confidence intervals at the 95% level are plotted.

To document how output prices, markups, and labor wedges vary across high-wage and low-wage firms, I separately regress log output prices (p_{jt}), markups (μ_{jt}), and labor wedges (Λ_{jt}) on deciles of firm wage premia (ϕ_{jt}), controlling for 5-digit sector $\times$ year fixed effects, and TFPQ. The estimated coefficients are shown in Figure 1. When I do

not control for differences in TFPQ, I find similar patterns (see Figure 26). Figure 27 also shows similar patterns when I compare by firm size (sales) instead of firm wage premia.

Figure 1 shows that high-wage firms tend to charge higher output prices and higher markups. According to theory, more productive firms typically charge higher markups (Atkeson and Burstein, 2008) and pay higher wages (Card et al., 2018), suggesting that productivity differences between firms could explain the positive relationship between markups and wage premia. However, if productivity were the sole driver of wage premia and markups, then one would expect that high-wage firms charge lower output prices, contrary to the data. Indeed, Figure 1 shows that the positive correlation between output prices, markups, and wage premia remains after conditioning on firm productivity. These empirical patterns suggest that differences in product quality is a driver wage premia.

Figure 1 also shows that high-wage firms pay a larger share of the marginal revenue product of labor as wages (higher labor wedges). This empirical pattern is difficult to reconcile with a purely monopsonistic labor market model where workers lack bargaining power. In such environments, the estimated labor wedge (Λ) would coincide with the monopsony wage markdown (λ), which is often constant or decreasing with wages (Burdett and Mortensen, 1998; Berger et al., 2022; Lamadon et al., 2022).³⁹ In such models, high-wage firms would typically pay a smaller fraction of the marginal revenue product of labor as wages. However, as shown in equation (3), when workers have bargaining power ($\kappa > 0$), the labor wedge becomes positively correlated with markups, which can explain the observed positive relationship between wages and labor wedges.

5.3 Worker bargaining power & firm monopsony markdowns

Worker bargaining power. My estimates suggest that worker bargaining power is relatively low. Table 4 shows that, across various specifications, workers obtain around 12% of firm profits.

Column (1) presents the pooled OLS estimate of bargaining power, yielding an estimate of approximately 0.12. Under the following conditions (also mentioned in Section 3.3), this is an unbiased estimate of κ : (i) monopsony markdowns are homogenous across firms ($\lambda_{jt} = \lambda_t$); (ii) the labor wedges do not reflect labor adjustment costs; and (iii) no mismeasurement of labor wedges and product market rents. In Section 5.4, I show that violations of assumptions (ii) and (iii) can lead to upward-biased estimates of κ , and I assess the robustness of these estimates when addressing such concerns. Column (2) additionally includes firm fixed effects, and also returns an estimate of around 0.12. Although this specification continues to rely on conditions (ii) and (iii), it relaxes condition (i), allowing for firm-specific monopsony markdowns ($\lambda_{jt} = \lambda_j$) as long as these markdowns do not exhibit firm-specific trends over time.

³⁹The oligopsonistic labor market model in Berger et al. (2022) allows markdowns to increase with wages if jobs across labor markets are closer substitutes than jobs within the same labor market.

Columns (3) and (4) implement the control function approach described in Section 3.3 to address unobserved amenities. These specifications accommodate the case where amenities are proportional labor supply shifters and do not directly determine markdowns (see the discussion multiplicative separability in Section 3.3), but also the more general case where amenities may directly affect markdowns. I continue to find a bargaining power estimate of around 0.12, suggesting that amenities do not have a significant direct impact on monopsony markdowns.

Finally, columns (5) and (6) further instrument measured product market rents using their lags. As outlined in Section 3.3 and further discussed in detail in Section 5.4, measurement error in labor wedges and product market rents will be positively correlated since they depend on the same variables (e.g. the estimated material output elasticities). This correlation could mechanically bias the bargaining power estimates upward. Under a classical measurement error assumption, using lagged product market rents as instruments corrects this bias. These specifications return estimates between 0.11 and 0.15. These estimates align with those from the rent-sharing literature, which typically finds values between 0.05 and 0.15 (Card et al., 2018; Jäger et al., 2020). However, it is important to note that studies using external, innovation-based instruments tend to find larger rent-sharing elasticities, around 0.30 (Van Reenen, 1996; Kline et al., 2019).

Table 4: Estimated worker bargaining power.

	Labor wedge (Λ_{jt})					
	(1)	(2)	(3)	(4)	(5)	(6)
Product market rents ($\tilde{\mu}_{jt}$)	0.124 (0.005)	0.125 (0.005)	0.122 (0.006)	0.124 (0.005)	0.113 (0.007)	0.143 (0.012)
Sector×year fixed effects	✓	✓	✓	✓	✓	✓
Firm fixed effects		✓		✓		✓
Control for $\lambda(\cdot)$			✓	✓	✓	✓
IV for measurement error					✓	✓
Observations	126,836	122,239	126,836	122,239	99,238	95,666

This table reports the estimated workers' bargaining power parameter. Even columns controls for firm fixed effects. Columns (3) through (6) controls for differences in monopsony markdowns reflecting differences in amenities. Columns (5) and (6) instruments measured product market rent $\tilde{\mu}_{jt}$ with its lag to address classical measurement error. Bootstrapped standard errors are reported in parentheses.

At this stage, it is important to note two caveats that I address in detail in Section 5.4. First, the inclusion of firm fixed effects does not adequately address the presence of labor adjustment costs, which are unobserved, time-varying, likely correlated with product market rents, and directly determine labor wedges. In Appendix B.8, I find that labor adjustment costs do not appear to be the main driver of labor wedges, and that an estimating equation that accommodates convex labor adjustment costs yields a very similar estimate of κ .

Second, when material or labor output elasticities are not well-identified, any biases that arise in their estimation will be serially correlated over time, mechanically causing an upward bias in $\hat{\kappa}$. This is a concern because identifying material output elasticities in gross output production functions is a well-known challenge (Gandhi et al., 2020). In this case, using lagged product market rents as instruments does not adequately address the mechanical upward bias. In Appendix B.7, I propose an alternative estimating equation that avoids the need to first estimate output elasticities and find a bargaining power of around 0.13 using a specification analogous to column (4) in Table 4.

How different are the bargaining power estimates across sectors? Table 28 in Appendix D reports these estimates by sector. The sector-specific bargaining power are estimated using my baseline regression specification (following column (4) in Table 4), which includes year fixed effects, firm fixed effects, and a control function for monopsony markdowns. In all 2-digit French manufacturing sectors, I find that worker bargaining power is below 0.2. However, the estimates range from 0.07 in the Paper and Publishing sector, to 0.19 in the Apparel sector. The sectors with the lowest worker bargaining power are sectors producing paper products, rubber and plastic products, and transportation equipment. The sectors with the highest worker bargaining power include sectors producing apparel and leather products, and the sector “repair and installation of machinery”.⁴⁰

Monopsony markdowns. I compute the implied monopsony markdowns taking the bargaining power estimate from specification (4) in Table 4 as my baseline estimate. Table 5 reports the estimated monopsony markdowns. At the median, the markdown is 46% of the marginal revenue product of labor, implying significant monopsony power. The corresponding firm-specific labor supply elasticities are 0.56, 0.85, and 1.33 at the 25th, 50th, and 75th percentiles. These estimates are similar to those found for the US labor market using the Burdett-Mortensen model by Webber (2015), who reports firm-specific labor supply elasticities of 0.44, 0.75, and 1.13 at the same percentiles. In an oligopsonistic setting, Berger et al. (2022) find labor supply elasticities ranging from 0.76 to 3.74 in the US. Meanwhile, a meta-analysis by Sokolova and Sorensen (2020), covering over 700 studies, finds a median labor supply elasticity of around 1.10. My median markdown-implied labor supply elasticity (0.85) is somewhat lower than this benchmark.

There is a significant difference between the median labor wedge (0.62) and median monopsony markdown (0.46), which is accounted for by bargaining power and markups. The production approach to markdown estimation only delivers estimates of monopsony markdowns when workers have no bargaining power. Otherwise, the estimates reflect labor wedges instead. As a result, production-based markdown estimates that do not

⁴⁰The French statistical institute, INSEE, describes the sector “repair and installation of machinery” as responsible for “the specialized repair of metalwork, machinery and equipment and other items produced in the manufacturing industry with a view to restoring them to working condition”. This includes installation and assembly of medical equipment, photographic, cinematographic and optical equipment, and installation and assembly of motors and other electrical machinery.

account for bargaining power will understate the true extent of firm monopsony power.

How do monopsony markdowns differ across firms and over time? Figure 2 shows how markdowns vary by deciles of firm wage premia and employment. I find that monopsony markdowns increase with firm wage premia, although this gradient is quite flat—the differences between deciles are under 3 percentage points. When I compare markdowns by firm size, I find that larger firms have lower markdowns—larger firms have more monopsony power, although this gradient is also relatively flat. The negative relationship between markdowns and firm size is consistent with the predictions of monopsonistic or oligopsonistic labor market models in which markdowns vary endogenously by firm size, such as Gouin-Bonenfant (2022), Berger et al. (2022), and Jarosch et al. (2024). Overall, there are no significant trends observed, though markdowns show a slight increase over the sample period, suggesting a slight increase in labor market competition.

Table 5: The distribution of monopsony markdowns.

Summary statistics	Mean	Median	25 th Pct	75 th Pct	Var	Var (i)	Var (ii)
Markdowns (λ_{jt})	0.45	0.46	0.36	0.57	0.04	0.03	0.03

This table reports the summary statistics in 2016 for the estimated monopsony markdowns. Markdowns are computed using the estimated bargaining parameter κ from specification (4) in Table 4. The column Var (i) reports the variances corrected for measurement error following Krueger and Summers (1988) and Kline et al. (2020), while the column Var (ii) reports the variances for firm-groups. Each variable is winsorized by 2%.

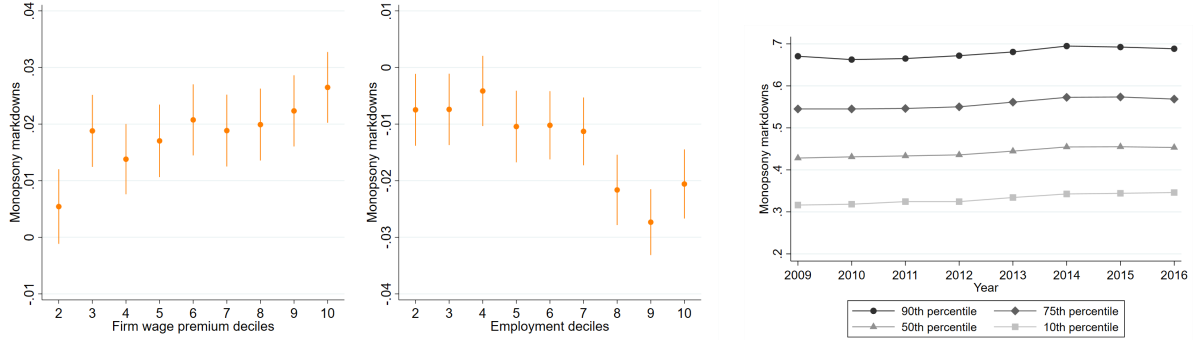


Figure 2: Monopsony markdowns by high-wage status, firm size, and year.

Notes: The first and second figures shows how monopsony markdowns vary by deciles of firm wage premia and employment relative to firms in the first decile, controlling for 2-digit sector $\times$ year fixed effects. Decile 10 represents high-wage firms or large firms. Confidence intervals are at the 95% level. The third figure presents the evolution of monopsony markdowns over the sample time frame (2009-2016).

Since the monopsony markdowns are measured after estimating worker bargaining power, their accuracy depends on the extent to which it is possible to obtain unbiased and precise estimates of κ . In the next subsection, I discuss and address a number of concerns that may affect the accuracy of my κ estimates. Nevertheless, it is instructive to compare the inferred markdowns under different values of κ . I make this comparison in Appendix B.7. Table 22 shows that the implied level of markdowns falls and the

dispersion (interquartile range) increases as I increase the value of κ . Figure 18 shows that for low values of κ , markdowns are increasing with firm wage premia. This positive correlation eventually turns negative as κ increases. Figure 18 also shows that markdowns are generally decreasing with firm size, and this gradient becomes steeper as κ increases.

5.4 Robustness and further discussions

In this section, I provide robustness analyses and additional discussions of the assumptions underlying the measurement of markups and labor wedges, as well as the estimation of worker bargaining power and monopsony markdowns.

5.4.1 Assumptions underlying markup and labor wedge measurement

The choice of production functions. In Section 3.2, I estimate translog production functions to obtain measures of markups and labor wedges that account for firm heterogeneity in output elasticities. A simpler alternative is the Cobb-Douglas production function, in which output elasticities are assumed to be homogenous across firms (within a given sector). Under Cobb-Douglas (CD henceforth), markups and labor wedges are proportional to the sales-to-material-expenditure ratio ($\mu_{jt}^{CD} = \alpha_m \frac{P_{jt}Y_{jt}}{P_{m,t}M_{jt}}$) and the wage-bill-to-material-expenditure ratio ($\Lambda_{jt}^{CD} = \frac{\alpha_m}{\alpha_h} \frac{\Phi_{jt}H_{jt}}{P_{m,t}M_{jt}}$).⁴¹ Researchers can then infer how markups and labor wedges vary depending on firm characteristics without having to first estimate production functions, avoiding the identification challenges that come with this task. However, to the extent that output elasticities vary systematically across firms, CD production functions may lead to incorrect conclusions about the how markups and labor wedges vary across firms.

In Appendix B.3, I compare the markups and labor wedges measured under the assumption of CD and translog production functions. I show that my baseline estimates of translog output elasticities are strongly correlated with the input cost shares (expenditure on an input as a share of total variable costs), which are readily available in the data and correspond directly to output elasticities under certain assumptions—constant returns to scale and perfectly competitive input markets. My estimates suggest that material (labor) output elasticities increase (decrease) with firm size and are stable over time. The translog-implied markups also increase with firm size, consistent with existing evidence showing more muted passthrough of cost shocks to prices among larger firms (Amiti, Itskhoki, and Konings, 2014). By contrast, CD-implied markups decrease with firm size, which is inconsistent with existing evidence on cost passthrough, as well as existing models of imperfect product market competition, such as Edmond et al. (2023).

The choice of flexible inputs. The production function approach to markup estimation

⁴¹This assumes that materials and labor are flexible inputs, and that material prices are competitively determined.

requires selecting a flexible input to measure markups with. When implementing this approach, I assume that materials are flexible inputs, as is standard in the literature (De Loecker and Warzynski, 2012). However, another natural candidate for the flexible input is electricity consumption. Relatedly, electricity prices may be less subject to potential monopsony power by buyer firms, compared to materials. However, after analyzing firm-level electricity consumption data from the French Survey of Energy Consumption in Industry (EACEI), I find that the measurement error in electricity consumption is likely much larger than in materials. Combined with the small share of electricity expenditures in revenue, this leads to highly dispersed and unreliable markup estimates.

The EACEI survey data provides information on energy consumption and purchase for French manufacturing firms from 1983 to 2021. The survey includes approximately 5,000 firms per year. For each year and plant combination, I observe the total amount of electricity consumed (in megawatt hours, MWh) and total expenditure on electricity (in Euros). I aggregate the data to the firm-year level. I drop firms with missing or negative values of electricity consumption and expenditure. I then link this survey to my estimation sample (EAP-FARE 2009-2016). However, this significantly reduces the sample size in my estimation sample to 26,171.

Table 6: The cost and revenue shares of electricity and materials.

	Materials			Electricity		
	Mean	Median	Variance	Mean	Median	Variance
Cost share	0.68	0.72	0.03	0.03	0.01	0.00
Revenue share	0.42	0.42	0.03	0.01	0.01	0.00
$\log \frac{\text{Revenue}}{\text{Input expenditure}}$	0.99	0.86	0.39	4.85	4.85	1.18

This table reports the cost and revenue shares of electricity. In this table, costs are measured as the sum of wage bills, material expenditure, capital investments, and electricity expenditure. The variances are across firms within 2-digit sectors. The sample consists of firms that are surveyed in the EACEI energy consumption data and the EAP-FARE output price and balance sheet data.

The survey reveals that electricity expenditure is a small share of total costs and revenue among French manufacturing firms. Table 6 shows that the cost share and revenue share of electricity is about 1% at the median French firm in this sample. In comparison, the cost and revenue shares of material inputs are many times larger, at 72% and 42%.

This small revenue share of electricity presents a challenge for using it as a flexible input to measure markups. Even a small amount of measurement error in electricity expenditure can cause large variability in the resulting markup estimates, since markups are calculated as the output elasticity of the flexible input multiplied by the inverse of its revenue share.

Table 6 shows that the revenue-to-electricity-expenditure ratio (in logs) is much more

dispersed than the revenue-to-material expenditure ratio. Moreover, the last row of Table 7 shows that electricity consumption growth is much more volatile than materials growth. When examining the comovement of electricity consumption with other factor inputs, I find that materials exhibit a stronger comovement with capital and labor compared to electricity consumption (see the third and fourth rows of Table 7). Taken together, these descriptive evidence is suggestive of larger measurement error in electricity expenditure.

Table 7: Firm-level comovement of factor inputs.

	%Δ Capital	%Δ Labor	%Δ Materials	%Δ Electricity
%Δ Capital	1.00			
%Δ Labor	0.13	1.00		
%Δ Materials	0.12	0.11	1.00	
%Δ Electricity	0.03	0.06	0.07	1.00
Variance	0.03	0.04	0.10	0.18

This table reports the Pearson correlation coefficients between the firm-level growth rates of capital, effective labor, materials, and electricity consumption.

Price-taking firms in the markets for material inputs. The production approach to markup estimation requires firms to be pricetakers in the market for the flexible input used to measure markups. In Section 3, I assume that firms have no monopsony power over material prices. If this assumption is violated, then measured markups may reflect both actual markups and monopsony markdowns for materials:

$$\mu_{jt} = \lambda_{m,jt} \alpha_{m,jt} \frac{P_{jt} Y_{jt}}{P_{m,jt} M_{jt}}$$

where $\lambda_{m,jt} = \frac{\epsilon_{m,jt}}{1+\epsilon_{m,jt}}$ is the monopsony markdown for materials, which is a function of the material supply elasticity $\epsilon_{m,jt}$. Similarly, the labor wedge measure becomes:

$$\Lambda_{jt} = \lambda_{m,jt} \frac{\alpha_{m,jt}}{\alpha_{h,jt}} \frac{\Phi_{jt} H_{jt}}{P_{m,jt} M_{jt}}$$

For markups and labor wedges to be measured without capturing firm monopsony power in materials, one would need to estimate the composite elasticity $\tilde{\alpha}_{m,jt} \equiv \lambda_{m,jt} \alpha_{m,jt}$, a product of the material output elasticity and material markdowns. In Appendix B.4, I show that the elasticity of output with respect to material expenditure is equal to this composite elasticity. In the absence of material price data, as is typically the case in practice, production function estimation recovers the output elasticity with respect to material expenditure. Thus, it is still possible to obtain correct measures of markups and labor wedges when firms have monopsony power in material input markets.⁴² However, without material prices, one cannot separately identify the material output elasticity from material markdowns.

⁴²If firms are pricetakers in the market for materials, then the elasticity of output with respect to material expenditure and material quantity are the same.

Identification of the material elasticity. In a framework with perfectly competitive output and input markets, [Gandhi et al. \(2020\)](#) show that under standard assumptions of the control function approach, the only source of variation left that helps identify the material output elasticity is time-series variation in material prices (provided one is willing to assume that production function parameters are constant over time). In such cases, identification relies heavily on persistence in material prices to generate persistence in material input choices.

However, [De Ridder et al. \(2021\)](#) show that when product markets are *imperfectly competitive*—a key feature of my model—variation in output prices and shocks to firms’ competitors generate additional variation in firms’ material input demand, which aids the estimation of the material output elasticity. Additionally, as discussed in the “*Choice of production functions*”, my baseline estimates of output elasticities are strongly correlated with input cost shares. The translog-implied markups also increase with firm size, in line with both existing evidence on the passthrough of cost shocks to prices and existing theoretical models of imperfect product market competition.

To further explore the stability of my output elasticity estimates, I implement the approach of [Flynn et al. \(2019\)](#), who show that imposing a constant returns to scale (CRS) restriction on the production function aids identification of the material output elasticity. Using this approach, I find output elasticity, markup, and labor wedge estimates that are stable over time and vary by firm size in a similar manner as my baseline estimates. One difference between my baseline estimates and those obtained under CRS is that the material elasticities are slightly smaller, and labor elasticities slightly larger, under CRS. This translates into a higher *level* of measured markups and lower level of measured labor wedges under CRS.

Finally, I also implement the dynamic panel approach in [Blundell and Bond \(1998\)](#), which avoids the need to build control functions. In practice, however, this approach returns unreasonably low estimates of output elasticities and returns to scale. This could be driven by the first-differencing of input and output data, which amplifies the effect of measurement error and may lead to significant attenuation bias. This issue is also discussed in [De Loecker and Syverson \(2021\)](#) and [Yeh et al. \(2022\)](#).

5.4.2 Assumptions underlying bargaining power and markdown estimation

Endogeneity: multiplication bias. In Section 3.3, I propose using the labor wedge equation (3) to estimate worker bargaining power, instead of the wage equation (1). This approach avoids the issue of unobserved monopsony wages. However, it could introduce other endogeneity issues. One concern is that the measure of labor wedges Λ_{jt} and product market rents $\tilde{\mu}_{jt}$ rely on the same variables—material expenditure and material/labor output elasticities—so measurement error or estimation biases in these variables will be positively correlated and mechanically bias $\hat{\kappa}$ upwards. If there are measurement error in

the factor inputs, then the estimated output elasticities will also be measured with error since they are a function of those inputs. Additionally, given the challenges of identifying the output elasticities of flexible inputs (Bond and Söderbom, 2005; Gandhi et al., 2020), any biases that arise in their estimation will introduce a mechanical upward bias in $\hat{\kappa}$.

To illustrate the upward bias on $\hat{\kappa}$, consider the following example. Assume that the true markdown $\lambda_{jt} = \lambda$ is constant and homogenous. Let ϵ_{jt}^{bias} be the source of bias (such as measurement error or estimation biases), such that $\frac{\hat{\alpha}_{m,jt}}{\hat{\alpha}_{h,jt}} = \frac{\alpha_{m,jt}}{\alpha_{h,jt}} \epsilon_{jt}^{bias}$. Let $\tilde{\epsilon}_{jt}$ be a random error uncorrelated with $\hat{\mu}_{jt}$. The estimating equation becomes:

$$\underbrace{\frac{\Phi_{jt}H_{jt}}{P_{m,t}M_{jt}} \frac{\hat{\alpha}_{jt}^m}{\hat{\alpha}_{jt}^h}}_{\hat{\Lambda}_{jt}} = \underbrace{\kappa \left(\frac{P_{jt}Y_{jt} - P_{m,t}M_{jt} - P_{k,t}K_{jt}}{P_{m,t}M_{jt}} \right)}_{\hat{\mu}_{jt}} \frac{\hat{\alpha}_{jt}^m}{\hat{\alpha}_{jt}^h} + (1 - \kappa)\lambda + \tilde{\epsilon}_{jt}$$

Then, an OLS regression of $\hat{\Lambda}_{jt}$ on $\hat{\mu}_{jt}$ will give an upward-biased estimate of κ :

$$\hat{\kappa}^{OLS} = \kappa + (1 - \kappa) \frac{CV(\lambda \epsilon_{jt}^{bias}, \tilde{\mu}_{jt} \epsilon_{jt}^{bias})}{V(\tilde{\mu}_{jt} \epsilon_{jt}^{bias})} \geq \kappa, \text{ where } \kappa \in [0, 1]$$

The smaller the true κ , the greater the upward bias. The bias will also be present if material expenditure $P_{m,t}M_{jt}$ is measured with error. This “multiplication bias” is reminiscent of the “division bias” problem highlighted by Borjas (1980) in research estimating intensive margin labor supply elasticities.

In Appendix B.7, I address this concern through several approaches, depending on the source of the bias. First, under the classical measurement error assumption, I estimate the potential measurement error in $(\hat{\alpha}_m/\hat{\alpha}_h)$ and simulate the implied extent of bias on $\hat{\kappa}$. While this source of the multiplication bias inflates $\hat{\kappa}$ relative to κ , the difference is small: $\hat{\kappa}$ is no more than 0.005 higher than κ . Second, I follow the well-established practice of using instrumental variables to address measurement error. Specifically, assuming classical measurement error, I instrument product market rents using their lags, and report the estimates in Table 4 (columns 5 and 6). This produces a $\hat{\kappa}$ of 0.14, similar to my baseline estimate of 0.12.

However, if the source of the multiplication bias is that the output elasticities are not well-identified—an important concern given the challenges of doing so discussed in Bond and Söderbom (2005) and Gandhi et al. (2020)—then the lags-as-instruments approach does not address the multiplication bias. This is because in this case ϵ_{jt}^{bias} will be serially correlated. While this concern is partially mitigated by the robustness of the baseline output elasticity estimates discussed in “*The Choice of production functions*” and “*Identification of the material elasticity*”, to address this concern more directly, I propose an alternative estimating equation that avoids having to first estimate the output elasticities:

$$\frac{\Phi_{jt}H_{jt}}{P_{m,t}M_{jt}} = \kappa \left(\frac{P_{jt}Y_{jt} - P_{m,t}M_{jt} - P_{k,t}K_{jt}}{P_{m,t}M_{jt}} \right) + (1 - \kappa)\lambda(H_{jt}, \mathcal{H}^{-1}(H_{jt}, \Phi_{jt}H_{jt})) \frac{\alpha_h(K_{jt}, M_{jt}, H_{jt})}{\alpha_m(K_{jt}, M_{jt}, H_{jt})}$$

Although estimating κ using this alternative equation avoids the need to first estimate output elasticities, it requires approximating a more complex, potentially highly nonlinear term ($\lambda(\cdot)\alpha_m(\cdot)/\alpha_h(\cdot)$). This alternative equation may still yield upward-biased estimates of κ , as measurement error in material expenditure and wage bills could still lead to multiplication bias. When I estimate κ using this alternative equation, I therefore also instrument the right-hand-side variables with their lags under a classical measurement error assumption. Using this specification, my most conservative estimate of κ of 0.11.

Finally, to provide a sense of how the implied monopsony markdowns depend on the value of κ , I derive an upper bound for κ and compare the implied λ for values of κ within this bound. The lower bound for κ is 0, in which case the labor wedge is the monopsony markdown, $\Lambda = \lambda$. I show that, for a given value of the labor wedge (Λ) and product market rent ($\tilde{\mu}$), the higher κ is, the lower the implied λ becomes. I then compute the upper bound for κ (denoted as κ_{max}) as the value of κ consistent with a full monopsony markdown ($\lambda = 0$): $\Lambda = \kappa_{max}\tilde{\mu}$, so the labor wedge is driven solely by workers' ability to capture product market rents. I find an upper bound of approximately 0.27. Within this range of κ , I find that larger firms have more monopsony power (lower λ).

Endogeneity: labor adjustment costs. My approach to estimating worker bargaining power and firm monopsony markdowns assumes that firms do not face labor adjustment costs. When this assumption does not hold, the labor wedge (Λ) captures not only rent-sharing and monopsony markdowns, but also a wedge due to labor adjustment costs. In Appendix B.8, I assess the likely magnitude of labor adjustment costs and adapt my bargaining power estimation approach to account for it. My findings suggest that labor adjustment costs do not materially affect my estimates of worker bargaining power.

To illustrate how labor adjustment costs might drive my bargaining power and monopsony markdown estimates, I set up a version of my model in which firms face convex labor adjustment costs, but charge homogenous markups and markdowns. I assume that firms differ only in productivity, and that productivity is time invariant—a deterministic model. The convex adjustment cost function is motivated by the fact that firm employment growth rates are well-approximated by a normal distribution with a mean of 0. In this model, the labor wedge differs across firms only due to labor adjustment costs: $\Lambda_{jt} = \frac{\kappa\mu + (1-\kappa)\lambda}{\Gamma_{jt}}$, where $\Gamma_{jt} \geq 1$ represents the adjustment cost wedge.

The model offers several predictions. First, it suggests that, all else equal, as firms age, their labor wedges increase, as labor adjustment costs become less relevant when firms converge to their optimal size ($\Gamma_{jt} \rightarrow 1$). Second, all else equal, firms with a higher growth rate face larger adjustment costs, resulting in a higher Γ and lower labor wedge Λ . Third,

the presence of unobserved labor adjustment costs will lead to upward-biased estimates of worker bargaining power, and measured monopsony markdowns will overstate firm monopsony power ($\hat{\lambda}$ will be lower than λ).

I test the first two predictions by examining labor wedges within firms over time. When I track estimated labor wedges as firms age, I observe a flat profile. Similarly, the correlation between labor wedges and firm employment growth rates is relatively weak. These patterns suggest that labor adjustment costs are unlikely to be the primary factor driving labor wedges.

To further evaluate whether labor adjustment costs impact my estimates of worker bargaining power, I modify the estimation approach to account for these costs. Under quadratic labor adjustment costs, the adjustment wedge (Γ) depends on both current and expected future employment growth rates. While data on current growth rates is available, expectations of future growth rates are not. I address this in two ways: (i) using realized future growth rates as a proxy for expectations, and (ii) instrumenting future growth rates with current state variables, including capital stock, employment, and productivity. In both cases, the estimated κ remains nearly identical to my baseline estimate of 0.12.

Extension: heterogeneous skills. My model in Section 2 does not feature skills. However, firm monopsony power and worker bargaining power may differ by skill type. Additionally, differences in skill compositions across firms may affect my measured labor wedges and hence estimated bargaining power. In Appendix E, I provide an extension that distinguishes between two skill types and allows skill compositions to differ between firms. I estimate skill-specific labor wedges, bargaining power, and markdowns in this extension. My main takeaway is the following: my baseline estimates which do not allow for skill heterogeneity masks interesting differences in bargaining power (higher for high-skill workers) and monopsony markdowns (lower for high-skill workers) between skills, but captures the firm-level average markdown and labor wedge well.

However, it is worth noting that this extension comes with a few caveats. First, it requires a clear definition of “skill”. Second, firm wage premia are typically estimated using worker mobility between firms, but estimating firm-specific wage premia for different skill types exacerbates the limited mobility bias, as mobility is often low in matched employer-employee data. This can lead to imprecise wage premia estimates, and the finer the skill classification, the more difficult this issue becomes. Limiting the analysis to two skill groups mitigates this issue. Third, incorporating skills into the model involves selecting a production function that generates firm-level differences in skill composition and is feasible to estimate. This limits the type of complementarity or substitutability that the model can accommodate.

6 A Quantifiable Model of Firm and Worker Power

The previous section provides empirical estimates and descriptive evidence on firm market power, worker bargaining power, output prices, and firm wage premia. However, it leaves several key questions unanswered: How important are markups in determining wages? What are the underlying sources of firm heterogeneity that shape the observed distributions of wages and market power? And, can strengthening worker bargaining power lead to welfare improvements?

Answering these questions requires specifying the sources of product and labor market power in Section 2. I assume that product varieties and workplace amenities are both horizontally and vertically differentiated, leading to firm market power in these markets. The specific functional form that I adopt for consumption and labor supply preferences are both nested-CES. I assume that product and labor market structures are oligopolistic and oligopsonistic, giving rise to firm-specific markups and markdowns that depend on their market shares (Atkeson and Burstein, 2008; Berger et al., 2022). As in Section 2, three key sources of heterogeneity shape the firm size distribution: productivity, quality, and amenities.

I use the model for three purposes. First, I measure and show how product quality and workplace amenities vary across firms.⁴³ Second, I use it to quantify the channels through which firm heterogeneity affects wage premia. Third, I formalize and quantify the role of worker bargaining power in determining social efficiency.

6.1 Environment

There is a continuum of product markets $g \in [0, G]$ and local labor markets $s \in [0, S]$. Each product market and local labor market contains an exogenously given finite number of firms n_g and n_s . Each firm j belongs to one product market and one local labor market.

Labor supply. The representative household maximizes utility by choosing the amount of numeraire final goods to consume C and effective labor to supply to each firm H_{js} :

$$U = \max_{\{C, H_{js}\}} C - \frac{H^{1+\varphi}}{1+\varphi}$$

subject to the budget constraint $C = \Phi H + \Pi$, where Φ is the aggregate wage index and Π are aggregate profits. Jobs across and within markets are imperfect substitutes. Aggregate labor supply H is a composite of market-level labor supply H_s , with a constant elasticity of substitution ν , governing the degree of horizontal job differentiation across markets. Within each market, the market-level labor supply is a composite of firm-

⁴³Appendix C.5 shows that CES-implied measures of quality and amenities are closely aligned with those implied by variable elasticity of substitution (VES) preferences, which nests CES as a special case.

specific labor supply with a constant elasticity of substitution η , governing the degree of horizontal job differentiation within markets. The aggregate and market-level labor supply aggregators are:

$$H = \left[\int_0^S \tilde{A}_s^{-\frac{1}{\nu}} H_s^{\frac{\nu+1}{\nu}} ds \right]^{\frac{\nu}{\nu+1}} \quad \text{and} \quad H_s = \left[\sum_{j=1}^{n_s} \tilde{A}_{js}^{-\frac{1}{\eta}} H_{js}^{\frac{\eta+1}{\eta}} \right]^{\frac{\eta}{\eta+1}}$$

where $\tilde{A}_s$ are market-level labor supply shifters such that $\int_s \tilde{A}_s ds = 1$, and $\tilde{A}_{js}$ are firm-specific labor supply shifters such that $\sum_{j=1}^{n_s} \tilde{A}_{js} = 1$. When $\eta \geq \nu$, jobs within markets are closer substitutes than jobs across markets. When $\eta \rightarrow \infty$ ($\nu \rightarrow \infty$), there is no job differentiation within (between) markets. The firm-specific labor supply curve is:

$$H_{js} = \tilde{A}_{js} \tilde{A}_s \Phi_{js}^\eta \Phi_s^{\nu-\eta} \Phi^{-\nu} H \quad (8)$$

I refer to $A_{js} \equiv \tilde{A}_{js} \tilde{A}_s$ as amenities. This provides a microfoundation for the labor supply curve in Section 2, $H_{js} = \mathcal{H}_s(\Phi_{js}, A_{js})$, where the subscript s on the function $\mathcal{H}_s(\cdot)$ accommodates the endogenous market-wide and economy-wide aggregates.

Product demand. The competitive final good producer produces good Y by combining the output of firms in each product market. The CES goods aggregator across markets and within markets are:

$$Y = \left[\int_0^G \tilde{D}_g^{\frac{1}{\theta}} Y_g^{\frac{\theta-1}{\theta}} dg \right]^{\frac{\theta}{\theta-1}} \quad \text{and} \quad Y_s = \left[\sum_{j=1}^{n_g} \tilde{D}_{jg}^{\frac{1}{\sigma}} Y_{jg}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

where $\tilde{D}_g$ are market-level demand shifters such that $\int_g \tilde{D}_g dg = 1$, and $\tilde{D}_{jg}$ are firm-specific demand shifters such that $\sum_{j=1}^{n_g} \tilde{D}_{jg} = 1$. The within-market and between-market elasticities of substitution between goods are σ and θ . When $\sigma \geq \theta$, goods within a market are more substitutable than goods across markets. When $\sigma \rightarrow \infty$ ($\theta \rightarrow \infty$), there is no product differentiation within (between) markets. Firms face the following inverse goods demand curve:

$$Y_{jg} = \tilde{D}_{jg} \tilde{D}_g P_{jg}^{-\sigma} P_g^{\sigma-\theta} Y \quad (9)$$

where I refer to $D_{jg} \equiv \tilde{D}_{jg} \tilde{D}_g$ as product quality. This provides a microfoundation for the product demand curve in Section 2, $Y_{jg} = \mathcal{G}_g(P_{jg}, D_{jg})$, where the subscript g on the function $\mathcal{G}_g(\cdot)$ accommodates the endogenous market-wide and economy-wide aggregates.

Resource constraint. Final goods can be consumed by the representative household or used as factor inputs—capital and material—such that $Y = C + K + M$, where $K = \int_g \sum_j^{n_g} K_{jg} dg$ and $M = \int_g \sum_j^{n_g} e^{\tau_{jg}} M_{jg} dg$ are aggregate capital and material use.

Wage bargaining. Wage-setting is as described in Section 2. Workers and firms jointly decide wages, output, capital, and materials to maximize the following Nash product:

$$\max_{\Phi_{js}, Y_{jg}, K_j, M_j} \left(\Phi_{js} H_{js} \right)^\kappa \left(\Pi_j \right)^{1-\kappa}$$

subject to the labor supply curve (8), product demand curve (9), and production function $Y_j = \Omega_j K_j^{\alpha_k} M_j^{\alpha_m} H_j^{\alpha_h}$. Profits are $\Pi_j = P_{jg} Y_{jg} - \Phi_{js} H_{js} - P_k K_j - P_{m,j} M_j$. Since final goods are used as capital and material inputs, we have $P_k = 1$ and $P_{m,j} = e^{\tau_j}$, where τ_j denotes heterogeneity in material input prices due to differences in material quality.⁴⁴

Decentralized market equilibrium. Given a set of firms that each belong to a product market g and a local labor market s , an equilibrium is: (i) An aggregate output Y , consumption C , labor supply H , price index $P = 1$, wage index Φ , capital inputs K , materials M ; (ii) A sequence of market-level aggregates: output Y_g , labor supply H_s , price index P_g , wage index Φ_s ; (iii) A sequence of firm-level prices and allocations: output Y_j , labor H_j , capital inputs K_j , materials M_j , output prices P_j , wages Φ_j that maximize household utility, the worker-firm Nash product, and clear all goods and labor markets.

6.2 The Social Planner's Equilibrium

I now define the social planner's equilibrium to derive the implications of worker bargaining power for social efficiency. The planner chooses capital, materials, and labor at each firm to maximize household utility, subject to the same preferences, production technologies, and aggregate resource constraint as the market economy. The planner maximizes:

$$\max_{H_j, K_j, M_j} C - \frac{H^{1+\varphi}}{1+\varphi}$$

subject to:

$$\begin{aligned} C + K + M &= Y, & Y &= \left[\int_0^1 \tilde{D}_g^{\frac{1}{\theta}} Y_g^{\frac{\theta-1}{\theta}} dg \right]^{\frac{\theta}{\theta-1}}, & Y_g &= \left(\sum_{j \in g} \tilde{D}_{jg}^{\frac{1}{\sigma}} Y_{jg}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \\ K &= \int_g \sum_{j \in g} K_{jg} dg, & M &= \int_g \sum_{j \in g} M_{jg} dg, & Y_j &= \Omega_j K_j^{\alpha_k} M_j^{\alpha_m} H_j^{\alpha_h} \\ H &= \left[\int_0^1 \tilde{A}_s^{-\frac{1}{\nu}} H_s^{\frac{\nu+1}{\nu}} ds \right]^{\frac{\nu}{1+\nu}}, & H_s &= \left(\sum_j \tilde{A}_{js}^{-\frac{1}{\eta}} H_{js}^{\frac{\eta+1}{\eta}} \right)^{\frac{\eta}{\eta+1}} \end{aligned}$$

⁴⁴Material price heterogeneity helps the model to match the empirical firm size distribution. There is broad evidence that material prices vary across firms due to differences in quality. For example, the literature on input trade liberalization shows that firms use inputs of different quality and that tariff reductions lead some firms to import higher quality inputs, raising their productivity and size (Kugler and Verhoogen, 2012; Bas and Strauss-Kahn, 2015; Fieger, Eslava, and Xu, 2018). Material quality heterogeneity can be accommodated by my model; the estimation approach in Section 3 accounts for material input quality heterogeneity by using output price data, following De Loecker et al. (2016). For the same reason, my model can also accommodate capital quality heterogeneity, but I do not model this explicitly. However, not all sources of firm heterogeneity can be accommodated by my model. For example, my model cannot accommodate heterogeneous monopsony power in material and capital markets due to the absence of input price data. More detailed discussions are available in Section 3.

The planner's aggregate and firm-level labor allocations are characterized by:

$$H^\varphi = \underbrace{\alpha_h \frac{Y}{H}}_{MPL} \quad \text{and} \quad H^\varphi \left(\frac{H_{js}}{H_s} \right)^{\frac{1}{\eta}} \left(\frac{H_s}{H} \right)^{\frac{1}{\nu}} = \underbrace{\alpha_h \frac{Y_{jg}}{H_{js}}}_{MPL_{js}} \tilde{D}_g^{\frac{1}{\theta}} \tilde{D}_{jg}^{\frac{1}{\sigma}} \left(\frac{Y_{jg}}{Y_g} \right)^{-\frac{1}{\sigma}} \left(\frac{Y_g}{Y} \right)^{-\frac{1}{\theta}}$$

Table 8: Labor allocation in the market and social planner's equilibrium.

	Market equilibrium	Planner's equilibrium
Aggregate labor demand	$\Phi = H^\varphi = \frac{\Lambda}{\mu} \Theta \alpha_h \frac{Y}{H}$	$H^\varphi = \alpha_h \frac{Y}{H}$
Firm-level labor demand	$\Phi_{js} = \Phi \tilde{A}_{js}^{-\frac{1}{\eta}} \tilde{A}_s^{-\frac{1}{\nu}} \left(\frac{H_{js}}{H_s} \right)^{\frac{1}{\eta}} \left(\frac{H_s}{H} \right)^{\frac{1}{\nu}}$ $= \frac{\Lambda_{js}}{\mu_{jg}} \alpha_h \frac{P_{jg} Y_{jg}}{H_{js}}$	$H^\varphi \tilde{A}_{js}^{-\frac{1}{\eta}} \tilde{A}_s^{-\frac{1}{\nu}} \left(\frac{H_{js}}{H_s} \right)^{\frac{1}{\eta}} \left(\frac{H_s}{H} \right)^{\frac{1}{\nu}}$ $= \alpha_h \frac{Y_{jg}}{H_{js}} \tilde{D}_g^{\frac{1}{\theta}} \tilde{D}_{jg}^{\frac{1}{\sigma}} \left(\frac{Y_{jg}}{Y_g} \right)^{-\frac{1}{\sigma}} \left(\frac{Y_g}{Y} \right)^{-\frac{1}{\theta}}$
Firm-level capital demand	$P_k = 1 = \mu_{jg}^{-1} \alpha_k \frac{P_{jg} Y_{jg}}{K_{jg}}$	$1 = \alpha_k \frac{Y_{jg}}{K_{jg}} \tilde{D}_g^{\frac{1}{\theta}} \tilde{D}_{jg}^{\frac{1}{\sigma}} \left(\frac{Y_{jg}}{Y_g} \right)^{-\frac{1}{\sigma}} \left(\frac{Y_g}{Y} \right)^{-\frac{1}{\theta}}$
Firm-level material demand	$P_{m,jg} = e^{\tau_{jg}} = \mu_{jg}^{-1} \alpha_m \frac{P_{jg} Y_{jg}}{M_{jg}}$	$e^{\tau_{jg}} = \alpha_m \frac{Y_{jg}}{M_{jg}} \tilde{D}_g^{\frac{1}{\theta}} \tilde{D}_{jg}^{\frac{1}{\sigma}} \left(\frac{Y_{jg}}{Y_g} \right)^{-\frac{1}{\sigma}} \left(\frac{Y_g}{Y} \right)^{-\frac{1}{\theta}}$

Firm market power and social efficiency. Table 8 compares the aggregate and firm-level input demand conditions in the market economy to that of the social planner. Comparing the aggregate labor demand conditions in the first row highlights the two channels through which firm market power distorts the market equilibrium away from the planner's equilibrium.

The first channel through which firm market power affects aggregate labor demand is as a *uniform tax* on firm labor demand, Λ/μ . This component is due to the *level* of firm labor and product market power and is less than 1. The wedge $\Lambda \equiv \kappa \tilde{\mu} + (1 - \kappa) \lambda$ (where $\tilde{\mu} \equiv (1 - \frac{\alpha_k + \alpha_m}{\mu}) \frac{\mu}{\alpha_h}$) is defined as the common labor wedge when price-cost markups and monopsony markdowns are constant across firms: $\mu \equiv \frac{\sigma}{\sigma-1}$ and $\lambda = \frac{\eta}{1+\eta}$.

The second channel is through the *misallocation* of factor inputs across firms. Let Θ represent a misallocation wedge on aggregate labor demand:

$$\Theta \equiv \int_g \left[\sum_{j \in g} \left(\frac{\Lambda_{jg}/\Lambda}{\mu_{jg}/\mu} \right) \frac{P_{jg} Y_{jg}}{P_g Y_g} \right] \frac{P_g Y_g}{PY} dg$$

where $\Theta < 1$ when larger firms have lower labor wedges and/or charge higher markups, and $\Theta = 1$ when firms have the same markups and markdowns. Thus, the *dispersion* of markups and markdowns also reduce aggregate labor demand.

If firms behave as pricetakers and wagetakers, and firms and workers do not bargain over wages, the market equilibrium coincides with the social planner's equilibrium. To show this, the last three rows of Table 8 compare the firm-level input decisions in the market economy to the planner's choice. The planner chooses the allocation of labor across firms such that the marginal product of labor intersects the firm-specific labor sup-

ply curve. Similarly, the planner chooses capital and materials such that their marginal revenue products are equalized across firms.

Can strengthening worker bargaining power improve welfare? I start by showing that worker bargaining power can counterbalance firm market power as a uniform tax on aggregate labor demand, bridging the gap between the market equilibrium and the planner’s equilibrium. To see this, suppose for now that markups and monopsony markdowns are constant across firms ($\mu_j = \mu$ and $\lambda_j = \lambda$), so there is no misallocation ($\Theta = 1$); markups and markdowns only act as a uniform tax on labor demand (Λ/μ). When workers have no bargaining power, the tax is $\Lambda/\mu = \lambda/\mu < 1$. With sufficiently high bargaining power, $\kappa = \bar{\kappa} \equiv [\mu - \lambda]/[\tilde{\mu} - \lambda] \in [0, 1]$, the tax is $\Lambda/\mu = 1$. In this case, the firm-level and aggregate labor demand conditions coincide with those of the planner’s (see the first two rows of Table 8). Indeed, as discussed in Section 2.2.2, the presence of markups makes bargaining power more effective in eliminating the uniform tax (Λ/μ)— $\bar{\kappa}$ is decreasing in markups—as the presence of markups raise the amount of profits that workers can extract.

However, there are limits to the extent that raising worker bargaining power can achieve the planner’s equilibrium. This is because the markup also acts as a uniform tax on capital and material demand, which cannot be directly addressed by worker bargaining power (see the last two rows of Table 8).

Besides addressing firm market power as a uniform tax, worker bargaining power can also offset the impact of labor misallocation on welfare, which is induced by the dispersion of markups and labor wedges. To see this, consider the following example. Continue to suppose for now that markups are constant across firms ($\mu_j = \mu$), but monopsony markdowns differ across firms ($\lambda_j \neq \lambda_{j'}$, where $j \neq j'$), leading to labor misallocation. Then, the labor wedge becomes $\Lambda_j = \kappa\tilde{\mu} + (1 - \kappa)\lambda_j$. This shows that when workers have full bargaining power ($\kappa = 1$), they appropriate all of the firm profits, so there is no longer any dispersion in the labor wedge. Therefore, the misallocation wedge becomes $\Theta = 1$, eliminating the role of markdown dispersion in causing labor misallocation.

More generally, worker bargaining power can also mitigate the impact of markup dispersion on labor misallocation. To see this, I now reintroduce markup dispersion. Suppose that larger firms charge higher markups and have lower monopsony markdowns (more monopsony power)— μ_j and λ_j are negatively correlated—so labor is misallocated ($\Theta < 1$). When workers have no bargaining power, this also means that markups and labor wedges (Λ_j) are negatively correlated. In this case, firm labor demand features a “double market power” tax, as highlighted in Kroft et al. (2024) (see Section 2.2.2). Indeed, the stronger the negative correlation between markups and labor wedges, the larger the dispersion of the joint wedge $\frac{\Lambda_{jg}/\Lambda}{\mu_{jg}/\mu}$, and so labor becomes even more misallocated (an even lower Θ). As worker bargaining power increases, this negative correlation between markups and labor wedges *weakens*, and the misallocation wedge Θ increases towards 1.

That is, labor misallocation is dampened by higher worker bargaining power.

6.3 Calibrating the model

I now describe my calibration approach. The calibrated parameter values are reported in Table 9. I first calibrate parameters that govern markups and markdowns $(\theta, \sigma, \nu, \eta)$, then measure product quality (D_{jg}) and amenities (A_{js}) .

Parameters related to markups. The relationship between markups and product market shares in this model can be written as:

$$\rho_{jg}^{-1} \equiv \frac{\mu_{jg} - 1}{\mu_{jg}} = \frac{1}{\sigma} + \left(\frac{1}{\theta} - \frac{1}{\sigma} \right) \frac{P_{jg} Y_{jg}}{\sum_{j'g}^{n_g} P_{j'g} Y_{j'g}}$$

To calibrate the within-sector elasticity of substitution between varieties (σ) , I first measure product market shares as the share of sales within 5-digit sectors. I then run a pooled OLS regression of the inverse price elasticity of demand $(\hat{\rho}_{jg}^{-1})$ —computed using the estimated markups—on market shares, controlling for market $\times$ year fixed effects. This gives an estimate for the passthrough of market shares to markups $\widehat{\left(\frac{1}{\theta} - \frac{1}{\sigma} \right)}$. I compute an economy-wide σ as the median of the inverse of $\hat{\rho}_{jg}^{-1} - \widehat{\left(\frac{1}{\theta} - \frac{1}{\sigma} \right)} \frac{P_{jg} Y_{jg}}{\sum_{j'g}^{n_g} P_{j'g} Y_{j'g}}$. To calibrate the between-sector elasticity of substitution (θ) , I compute the inverse of $\widehat{\left(\frac{1}{\theta} - \frac{1}{\sigma} \right)} + \frac{1}{\sigma}$ using the calibrated σ . Appendix C.4 discusses the measurement of market shares and presents values of σ by 2-digit sectors.

Parameters related to markdowns. The relationship between oligopsonistic markdowns and wage bill shares in this model can be written as:

$$\xi_{js} \equiv \frac{\lambda_{js}}{1 - \lambda_{js}} = \eta + (\nu - \eta) \frac{\Phi_{js} H_{js}}{\sum_{j's}^{n_s} \Phi_{j's} H_{j's}}$$

Analogous to the calibration of the demand elasticities, I run a pooled OLS regression of the firm-specific labor supply elasticity $(\hat{\xi}_{js}^{-1})$ —computed using the estimated monopsony markdowns—on labor market shares, controlling for market $\times$ year fixed effects. I measure labor market shares as the share of wage bills within 5-digit sectors $\times$ commuting-zone pairs. This gives a market share passthrough estimate $\widehat{(\nu - \eta)}$. Next, I compute an economy wide η as the median of $\hat{\xi}_{js}^{-1} - \widehat{(\nu - \eta)} \frac{\Phi_{js} H_{js}}{\sum_{j's}^{n_s} \Phi_{j's} H_{j's}}$. To obtain ν , I measure $\widehat{(\nu - \eta)} + \eta$ using the calibrated value of η . In Appendix C.4, I find similar parameter values for η and ν when I widen the definition of a local labor market to a 2-digit sector $\times$ commuting-zone.

Calibrating heterogeneous productivity, quality, and amenities. Firm productivity (TFPQ) is estimated directly from the data in Section 3. Given the parameters governing product demand and labor supply curves, I back out firm heterogeneity in product quality and amenities. The product and labor market shares of a given firm are:

$$\frac{P_{jg}Y_{jg}}{P_gY_g} = D_{jg} \left(\frac{P_{jg}}{P_g} \right)^{1-\sigma} \quad \text{and} \quad \frac{\Phi_{js}H_{js}}{\Phi_sH_s} = A_{js} \left(\frac{\Phi_{js}}{\Phi_s} \right)^{1+\eta}$$

Using these equations to measure quality and amenities requires measuring firm-level output prices and (hourly) wage premia. A key advantage of the French administrative data over many existing datasets (e.g., the US LEHD and Census data) is that it allows one to do so, instead of relying on earnings data, for example.

I compute D_{jg} such that $\sum_j^{n_g} D_{jg} = 1$ and A_{js} such that $\sum_j^{n_s} A_{js} = 1$. Similarly, the sectoral product demand and labor supply shifters (D_g and A_s) are measured using the relative size of sectors: $\frac{P_gY_g}{PY} = D_g \left(\frac{P_g}{P} \right)^{1-\theta}$ and $\frac{\Phi_sH_s}{\Phi H} = A_s \left(\frac{\Phi_s}{\Phi} \right)^{1+\nu}$. Appendix C.5 compares the quality and amenity measures under non-CES preferences that nest the CES as a special case, finding that CES-implied measures are closely aligned with non-CES-implied measures.

Table 9: Calibrated parameters.

Parameter		Value	Source
Frisch labor supply elasticity	φ	0.25	Chetty (2012)
Between-market e.o.s. (labor)	ν	0.74	Calibrated
Within-market e.o.s. (labor)	η	0.87	Calibrated
Between-market e.o.s. (product)	θ	1.53	Calibrated
Within-market e.o.s. (product)	σ	4.53	Calibrated
Worker bargaining power	κ	0.12	Estimated (see Section 3.3)
Number of firms within goods markets	n_g	[279, 182, 357]	FARE-EAP sample
Number of firms within labor markets	n_s	[5, 2, 10]	FARE-EAP sample
Labor elasticity of output	α_h	0.45	Average of estimates
Material elasticity of output	α_m	0.49	Average of estimates
Capital elasticity of output	α_k	0.06	$1 - \alpha_h - \alpha_m$
Log TFPQ	ω_j	[0.00, -0.08, 0.88]	Production function estimation
Log product quality	d_j	[0.00, 0.13, 3.27]	Sales shares
Log non-wage amenities	a_j	[0.00, -0.04, 0.82]	Wage bill shares
Log material price heterogeneity	τ_j	[0.00, 0.00, 0.47]	Firm size distribution (eff. labor)

Notes: This table reports the calibrated values of the model parameters. The abbreviation “e.o.s.” means “elasticity of substitution”. The numbers in the brackets represent the mean, median, and standard deviation, respectively. TFPQ, quality, amenities, and material price heterogeneity have been de-measured.

Other parameters. The parameter value for the Frisch labor supply elasticity is obtained from Chetty (2012). I calibrate workers’ bargaining power to 0.12 using my estimates in Table 4 from specification (4). The number of firms within each 5-digit sector is obtained from my FARE-EAP sample. The production function parameters are calibrated as the average of my production function estimates. Finally, material price heterogeneity τ_j is calibrated to match the firm size distribution (in efficiency units of labor). Combined with the calibrated values of non-wage amenities, the model exactly reproduces the empirical firm wage premium distribution.

Model fit. The nested-CES product demand (labor supply) structure imposes a specific relationship between markups (markdowns) and market shares. The first panel of Figure

3 shows how the estimated markups and the calibrated nested-CES markups vary by sales market share quantiles. The orange line shows that the estimated markups increase particularly steeply with market shares among the top two quantiles. The calibrated nested-CES markups are generally able to capture this pattern, particularly the markups of firms in the top quantile of market shares.

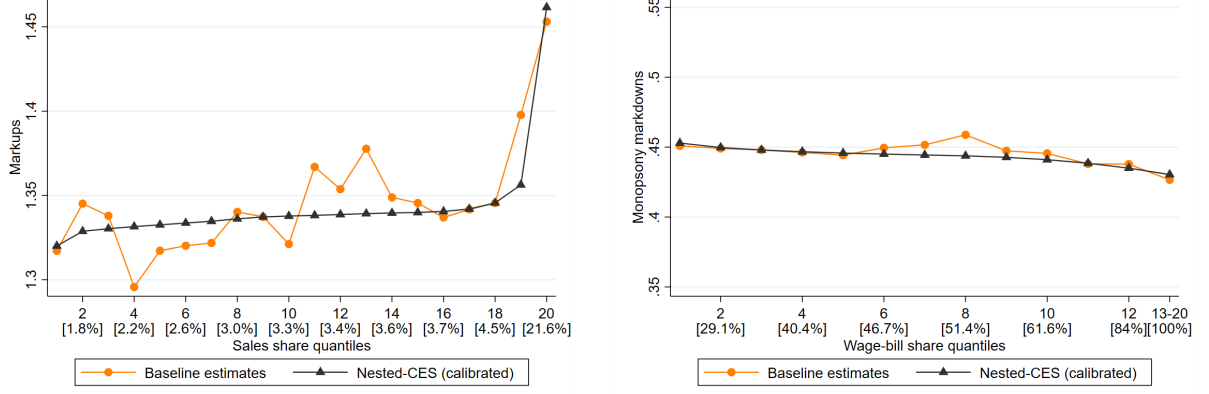


Figure 3: Nested-CES markups and monopsony markdowns.

Notes: This figure shows how estimated markups (markdowns) and nested-CES markups (markdowns) vary by sales (wage-bill) market share. Markup estimates, markdown estimates, and market share measures are residualized by the market $\times$ year fixed effects. Firms in the top 5% of the market share distribution are in bin 20. The numbers in square brackets are the median market share within each bin.

The second panel of Figure 3 shows how the estimated monopsony markdowns and the calibrated nested-CES markdowns vary by wage-bill market share quantiles. Both the estimated markdowns and the calibrated markdowns are relatively flat across the market share quantiles.

7 Quantifying the Wage and Welfare Implications of Firm and Worker Power

In this subsection I use the calibrated model for three main exercises. First, I present how the measured product quality and amenities vary across high-wage and low-wage firms. Second, I decompose the passthrough of firm heterogeneity to wage premia into various channels and quantify their contributions. Third, I quantify the importance of firm market power and worker bargaining power for wages and welfare.

7.1 Product quality and amenities among high-wage firms

Analogously to Section 5, I compare (the logs of) product quality and amenities across deciles of firm wage premia, controlling for 5-digit sector $\times$ year fixed effects. Figure 4 shows that the difference between high-wage and low-wage firms in quality and amenities

are larger than the difference in productivity (TFPQ). While firms in the top decile of the wage premium distribution are on average 13% more productive than firms at the bottom decile, they have over a 100% greater quality. That is, for a given price, firms in the top decile can sell over 100% more goods than firms in the bottom decile. Amenities also increase more steeply with firm wage premia than productivity. Similarly, for a given wage premium, firms in the top decile can employ about 15% more labor than firms in the bottom decile. That high-wage firms have more desirable amenities is consistent with the findings of [Lamadon et al. \(2022\)](#) for US firms.

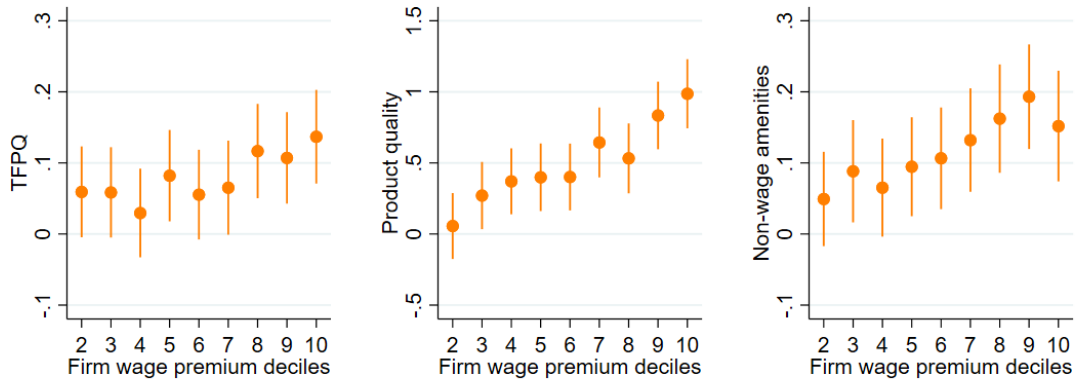


Figure 4: Measured firm heterogeneity by high-wage status.

Notes: This figure shows how measured TFPQ, product quality, and non-wage amenities vary by deciles of firm wage premia relative to firms in the first decile (low-wage firms), controlling for 5-digit sector $\times$ year fixed effects. All variables are in logs. Decile 10 represents high-wage firms. Confidence intervals at the 95% level are plotted.

7.2 On the passthrough of firm heterogeneity to wages

What is the relative importance of the three sources of firm heterogeneity—productivity, quality, and amenities—for the cross-sectional dispersion of firm wage premia? And, what determines the size of the passthrough of firm-specific shocks to wages? I find that the degree of product differentiation governs the extent of productivity passthrough relative to product quality passthrough. When there is little product differentiation, the passthrough of productivity is larger than the passthrough of quality. Further, firm size dampens the passthrough of productivity, leading to a smaller passthrough among high-wage firms. Overall, I find that product quality is as important as productivity in explaining firm wage premia.

To see how firm heterogeneity affects firm wage premia, I write wage premia (in *logs*)

from the model of Section 6 as:

$$\begin{aligned}
\phi_j = & \underbrace{\frac{(\sigma - 1)}{1 + \eta + \alpha_h(\sigma - 1)}\omega_j + \frac{1}{1 + \eta + \alpha_h(\sigma - 1)}(d_{jg} - a_{js}) - \frac{\alpha_m(\sigma - 1)}{1 + \eta + \alpha_h(\sigma - 1)}\tau_j}_{\text{direct effect}} \\
& + \underbrace{\frac{1 + \alpha_h(\sigma - 1)}{1 + \eta + \alpha_h(\sigma - 1)} \overbrace{\log \Lambda_j}^{\text{labor wedge}} - \frac{\sigma}{1 + \eta + \alpha_h(\sigma - 1)} \overbrace{\log \mu_j}^{\text{variable markup}}}_{\text{variable market power effect}} + \underbrace{\bar{\alpha}}_{\text{constant}} \\
& + \underbrace{\frac{\sigma - \theta}{1 + \eta + \alpha_h(\sigma - 1)}p_g + \frac{\eta - \nu}{1 + \eta + \alpha_h(\sigma - 1)}\phi_s}_{\text{size effect}} \\
& + \underbrace{\frac{1}{1 + \eta + \alpha_h(\sigma - 1)}\log\left(\frac{Y}{\Phi^{-\nu}H}\right)}_{\text{general equilibrium effect}}
\end{aligned} \tag{10}$$

where $\bar{\alpha}$ is a constant.

The first line of equation (10) shows the *direct effect* of firm heterogeneity on wage premia. The direct effect is present even when firms are atomistic and markups and markdowns are constant. A higher labor output elasticity α_h reduces the passthrough of productivity, product quality, and amenities, since a smaller increase in employment can now achieve a given increase in output. When jobs within markets are closer substitutes (higher η), the passthrough of firm heterogeneity becomes smaller, because a smaller wage increase is sufficient to achieve a given increase in employment.

The substitutability of product varieties (σ) affects the passthrough of productivity and product quality in opposite directions. When σ is higher, productivity passthrough increases, while quality passthrough decreases. Intuitively, with higher σ , product varieties become closer substitutes, making consumers more price-sensitive. This magnifies the impact of productivity heterogeneity on labor demand and wages, making productivity a more significant determinant of wage premia in high- σ environments.

The second line shows that endogenous changes in labor wedges and markups affect wage premia. I refer to this term the *variable market power effect*. A higher markup can raise wage premia through *rent-sharing*—by raising the labor wedge. However, it can also reduce wage premia by inducing the firm to *restrict output* and labor demand. Additionally, labor wedges also depend on monopsony markdowns, and lower markdowns reduce wages and labor demand.

The third line shows that firm wage premia depend on sectoral price and wage indices. I refer to this term as the *size effect* since individual firms can only affect the sectoral price and wage index when they are large relative to the product and labor markets they belong (they are not atomistic). They can also indirectly affect the price and wage index by influencing their competitor's price and wage-setting decision. For example, if a large firm reduces prices, it lowers the market-level price index, forcing its competitors to also reduce prices. Since $\sigma > \theta$, a lower sectoral price index induces firms to produce and

hire less, reducing wages. Similarly, given that $\eta > \theta$, a higher sectoral wage index forces firms to pay higher wages.

The final term reflects the general equilibrium effects on wage levels, which impact all firms equally and thus do not affect the distribution of firm wage premia. The numerator, aggregate output Y , captures the effect of aggregate product demand on wages, while the denominator, $\Phi^{-\nu}H$, represents an endogenous aggregate labor supply shifter (see equation (8)). Together, these general equilibrium effects capture the role of markups and markdowns in distorting aggregate labor demand (see the market versus social planner comparison in the first row of Table 8).

Passthrough of productivity, product quality, and amenities. I now present quantitative findings that point to firms' price-setting power as an important determinant of passthrough. The passthrough is defined as the elasticity of firm wage premia with respect to a firm-specific shock. To compute the passthrough of a shock, I randomly select one firm from each sector and assign it a 1% positive shock, while holding general equilibrium aggregates constant. I then compute the average elasticity of wages to the shock across shocked firms. Panel (A) of Table 10 shows the passthrough of a 1% productivity shock, product quality shock, and amenity shock.⁴⁵

The passthrough of a productivity shock is 0.96, almost four times larger than the passthrough of a product quality shock, as the last column of Panel (A) shows. To see why, columns one through five breaks down the total passthrough into various channels. The *direct* channel accounts for most of the difference between productivity and product quality passthrough. As equation (10) shows, the direct component of productivity passthrough is increasing in the within-market goods substitutability σ , while for product quality passthrough it is decreasing in σ . Given the relatively high calibrated σ of 4.53, the direct effect is larger for productivity passthrough. However, for sufficiently low σ , the passthrough of product quality can be higher than the passthrough of productivity, as Table 29 in Appendix C shows. On the other hand, as η increases, the passthrough of each shock becomes smaller.

The *variable market power* channel also plays a role in the passthrough of productivity shocks. The first row of columns two and three in Table 10 shows how variable markups and markdowns contribute to the productivity passthrough. A higher productivity leads to higher markups, leading to an increase in rent-sharing (higher labor wedges) but also a stronger incentive to restrict output. Overall, the output restriction effect dominates, implying that increased markups dampen the passthrough of productivity shocks.

The *size* channel further dampens the passthrough of productivity shocks. A positive productivity shock reduces a firm's output price, which directly reduces the sectoral

⁴⁵These passthrough simulations show how wage premia change in response to shocks at the firm-level, but they do not show how wages at the worker-level would respond. The latter will require accounting for job mobility, as workers may be induced to switch firms, e.g. due to a negative shock. Chan et al. (2021) show that job mobility is an important determinant of passthrough in response to negative shocks.

price index if the firm is large relative to the sector. Moreover, when σ is higher so consumers are more price-sensitive, the firm's competitors reduce prices more aggressively in response to the price reduction of the shocked firm, further reducing the price index. Given the calibrated σ , the reduction in sectoral output prices offsets the shocked firm's wage increase by -0.06. Being a large firm in the labor market also means that its wage directly affects the sectoral wage index and induces competitors to raise wages. Given the relatively low substitutability of jobs within sectors η , the response of the sectoral wage index to a firm-specific productivity shock is muted and does not affect the productivity passthrough.

Table 10: Passthrough of firm heterogeneity to firm wage premia.

Panel (A)	Channels					Total passthrough
	Direct	Var. market power	Size			
		$\Delta \log \Lambda$	$-\Delta \log \mu$	Δp_s	$\Delta \phi_s$	
Δ Productivity	1.05	0.02	-0.05	-0.06	0.01	0.96
Δ Product quality	0.29	0.01	-0.01	-0.02	0.00	0.27
Δ Amenity	-0.29	0.00	-0.01	-0.01	0.00	-0.32

Panel (B)	Firm wage premium distribution		
	10 th pct.	50 th pct.	90 th pct.
Δ Productivity	1.05	1.04	0.72
Δ Product quality	0.29	0.29	0.21
Δ Amenity	-0.29	-0.30	-0.37

This table reports the passthrough of firm heterogeneity to firm wage premia. The passthrough measures are with respect to a positive 1% shock. Panel (A) decomposes the total passthrough into different effects. The first column shows the *direct effect* of a shock on wage premia, which is present even when firms are atomistic (see equation (10)). The second and third columns show the effects coming from variable labor wedges and markups, capturing *rent-sharing* and *output restriction* motives. The fourth and fifth columns present the effects coming from firms being large relative to the market size. The final column presents the total passthrough, which is the sum of columns 1 through 5. Panel (B) looks at the passthrough at different percentiles of the firm wage premium distribution.

The total passthrough of an amenity shock is larger than that of a product quality shock. Table 10 shows that the direct effect of an amenity shock is the mirror image of a product quality shock (-0.29 compared to 0.29). However, while variable markups and sectoral price indices dampen the passthrough of positive product quality shocks, they amplify the passthrough of positive amenity shocks. This is because an increase in the value of amenities allows the firm to hire workers at a lower wage, effectively acting as a negative shock to production costs and expanding the size of the firm.

Given the above discussion on the role of markups and sectoral price indices in determining the wage passthrough, a shock of a given size should have a different passthrough at different points in the wage premium distribution. Panel (B) of Table 10 shows that productivity and quality passthrough is substantially smaller for firms in the 90th per-

centile, while the amenity passthrough is larger. In comparison, firms in the 10th and 50th percentiles have a passthrough that is well-approximated by the direct effect only.

7.3 Decomposing the dispersion of firm wage premia

I now decompose firm wage premia and present two main findings. First, product quality is the main driver of wage premium dispersion, closely followed by productivity, and then by amenities. Second, if markups were constant instead of variable, wage premium dispersion increases substantially, driven by high-wage firms.

Table 11: Importance of firm heterogeneity for firm wage premia.

Panel (A)	Counterfactual		
	Vary only TFP	Vary only quality	Vary only amenity
$\frac{V(\phi^{\text{counterfactual}})}{V(\phi^{\text{benchmark}})}$	152.51	168.11	35.47

Panel (B)	Counterfactual		
	Constant markups	Constant markdowns	Both constant
$\frac{V(\phi^{\text{counterfactual}})}{V(\phi^{\text{benchmark}})}$	1.21	1.01	1.22
Δ 99-90 ratio (p.p.)	0.04	0.00	0.04
Δ 90-50 ratio (p.p.)	0.01	0.00	0.01
Δ 50-10 ratio (p.p.)	0.00	0.00	0.00

This table reports the variance of firm wage premia under different counterfactual experiments. $\phi^{\text{benchmark}}$ refers to firm wage premia in the benchmark model, which has heterogeneity in productivity, product quality, amenities, and material price heterogeneity, as well as variable markups and markdowns. Note that the variance of $\phi^{\text{benchmark}}$ is the same as the variance of ϕ^{data} . Panel (A) compares the wage premium distribution when there is only one dimension of heterogeneity to that of the benchmark model. Panel (B) compares the counterfactual when markups (μ) and/or markdowns (λ) are constant to the benchmark model.

Which sources of firm heterogeneity are most important for firm wage premia?

I introduce *one* source of firm heterogeneity at a time into the model. I then compare the generated variance of wage premia to that of the benchmark economy, which exactly reproduces the wage premium distribution in the data. Panel (A) of Table 11 shows the contribution of productivity, product quality, and amenities to wage premia, in descending order of importance. When only productivity varies, the model generates a variance of wage premia that is 153 times that of the full model. The corresponding numbers for product quality and amenities are 168 times and 35 times. Product quality is thus a larger contributor to wage premium dispersion than productivity and amenities. Nevertheless, productivity and amenities play an important role in driving firm wage premia.

How important are markups and markdowns for firm wage premia? To go from variable markups to constant markups, I derive and implement subsidy schemes that induce firms to choose the same markups, building on Edmond et al. (2015). I do

the same for markdowns. Let $T_y(Y_{jg})$ and $T_h(H_{js})$ be output and labor subsidies to firm j . The subsidies are financed by a lump sum tax on the household. Firm gross profits, which includes the subsidies, are $\Pi_j^{gross} = P_{jg}Y_{jg} - \Phi_{js}H_{js} - P_kK_j - P_{m,j}M_j + T_y(Y_{jg}) + T_h(H_{js})$. In Appendix C.3, I show that output subsidies that satisfy the following condition removes *dispersion* in markups and implements a specific *level* of markups:

$$T'_y(Y_{jg}) = \left(\frac{1}{\rho_{jg}} - \frac{1}{\bar{\sigma}} \right) P_{jg}$$

Any given level of markups can be implemented by an appropriate choice of the policy parameter $\bar{\sigma}$. For example, to implement constant CES markups $\mu = \frac{\sigma}{\sigma-1}$, we can set $\bar{\sigma} = \sigma$. Similarly, labor subsidies that satisfy the following condition removes *dispersion* in markdowns and implements a specific *level* of markdowns:

$$(1 - \kappa)T'_h(H_{js}) = (1 - \kappa) \left(\frac{1}{\xi_{js}} - \frac{1}{\bar{\eta}} \right) \Phi_{js} - \kappa \lambda_{js}^{-1} \frac{T_y(Y_{jg}) + T_h(H_{js})}{H_{js}}$$

any given level of markdown can be implemented by an appropriate choice of the policy parameter $\bar{\eta}$, where $\bar{\eta} = \eta$ implements the CES markdown $\lambda = \frac{\eta}{1+\eta}$. The second term on the right-hand side of the equality shows that the labor subsidies schedule takes into account the fact that worker bargaining power partly offsets the distortionary effects of markups and monopsony markdowns on labor demand. The social planner's allocation, derived in Section 6.2, can be implemented by setting both $\bar{\sigma}$ and $\bar{\eta}$ to infinity. In this case, the subsidies lead firms to behave as pricetakers in the product and labor market.

I now discuss the effects of markups on firm wage premia. Panel (B) of Table 11 shows how the variance of firm wage premia changes when markups and markdowns are constant. When firms charge constant markups, wage premia are 21% more dispersed. The increase in wage dispersion comes primarily from the top end of the wage premium distribution: the 99-90 ratio increases by 4 percentage points (p.p.) while the 50-10 ratio does not change. This is because markups at high-wage firms fall the most, leading to higher output and labor demand at these firms. The fall in markups also leads to a decline in profits shared with workers, which tends to reduce wages. Overall, the output restriction effect dominates the rent-sharing effect, resulting in higher wages.

The increase in wage dispersion is only slightly larger when markups *and* markdowns are both equalized, compared to when *only* markups are equalized. In the latter, the expansion of high-wage firms also increases their labor market share, allowing them to mark down wages more, which partially erases the wage gains from lower markups. However, given the similarity of the calibrated within-market and across-market substitutability of jobs (η and ν), the increase in firms' labor market power is small.

7.4 The impact of worker bargaining power and firm market power on wages and welfare

In Section 6.2, I compare the market equilibrium to the social planner's equilibrium and find that worker bargaining power simultaneously (i) transfers profits from firms to workers, (ii) reduces the effects of markups and monopsony markdowns as a uniform tax on labor demand, and (iii) reduces the misallocation effects of markup and markdown dispersion. In my quantitative simulations below, I find that raising worker bargaining power can lead to substantial welfare gains by offsetting the impact of markups and markdowns on labor demand. However, even with full worker bargaining power, removing markup dispersion still raises welfare significantly, because bargaining power does not eliminate the distortionary effect of markups on the demand for non-labor inputs.

Wages and welfare. I measure welfare in consumption-equivalent terms. Let household utility be $U(C, H)$ in the baseline economy and $U(C^*, H^*)$ in a counterfactual economy. The consumption-equivalent welfare gain in percentages ($\hat{c}$) is measured as $U(C^*, H^*) = U((1 + \hat{c})C, H)$. Table 12 presents my findings.

Table 12: Effects of worker bargaining power and firm product and labor market power on wages and welfare.

	Counterfactuals								
	(1) $\mu_j = 1$	(2) $\lambda_j = 1$	(3) $\mu_j = 1,$ $\lambda_j = 1$	(4) $\mu_j = \bar{\mu}$	(5) $\lambda_j = \bar{\lambda}$	(6) $\mu_j = \bar{\mu},$ $\lambda_j = \bar{\lambda}$	(7) $\kappa = 0.2$	(8) $\kappa = 0.5$	(9) $\kappa = 1.0$
$\% \Delta V(\phi_j)$	21	1	22	21	1	22	2	7	12
$\% \Delta \Phi$	84	81	266	21	1	22	20	93	215
$\hat{c} \times 100$	28	9	35	12	0	12	3	10	13

This table shows the percentage changes in the variance of firm wage premia, the aggregate wage index, and aggregate welfare under various counterfactual simulations. Welfare changes $\hat{c}$ are measured in consumption-equivalent terms. Simulations involving constant markups set markups to $\mu = \frac{\sigma}{\sigma-1}$. Simulations involving constant monopsony markdowns set markdowns to $\lambda = \frac{\eta}{1+\eta}$.

To see how far raising worker bargaining power can go in raising welfare, I start by computing the welfare gain of implementing the social planner's equilibrium, which removes both markups and markdowns entirely. The third column shows that removing both markups and markdowns leads to a 22% increase in the dispersion of firm wage premia, a 266% increase in average wage levels, and a 35% welfare gain. A comparison between the first column and the third column shows that about 80% of the welfare gain comes from removing markups.

Comparing the first column to the fourth column shows that about 40% of the welfare cost of markups comes from misallocation. Moving from variable to constant CES markups ($\mu = \frac{\sigma}{\sigma-1}$) delivers a welfare gain of 12%. In this counterfactual economy, both average wages and the variance of firm wage premia are 21% higher. On the other hand, equalizing monopsony markdowns does not materially raise welfare (column 5), although

the *level* of markdowns is a significant source of welfare loss (column 2).

Can raising worker bargaining power (κ) lead to large welfare gains when firms have product and labor market power? Column 7 shows that raising κ from 0.12 to 0.20 generates an average wage gain similar to that of equalizing markups (column 4). However, there are two notable differences. First, the increase in κ leads to a much smaller increase in wage premium dispersion than equalizing markups. This suggests that the increased worker bargaining power mainly counteracts the role of market power as a uniform tax, and only slightly reduces markup-dispersion-induced misallocation. Second, raising worker bargaining power yields a smaller welfare gain (3%) for the *same increase in average wages* compared to markup equalization (12%). This is because the increase in κ only offsets the impact of markups on labor demand, but does not offset the impact of markups on the demand for non-labor inputs.

For the same reason, giving workers full bargaining power does not restore the planner's equilibrium. Full worker bargaining power forces firms to be wagetakers, removing the distortionary effects of markups and markdowns on labor demand. This raises welfare by 13% (column 9), which is around 40% of the welfare gains from implementing the planner's equilibrium (column 3).

Interaction effects of worker bargaining power and firm price-cost markups.

Panel (A) of Table 13 compares the effects of removing markup dispersion under different levels of workers' bargaining power. When bargaining power is low ($\kappa = 0.12$), removing markup dispersion leads to a 12% welfare gain. When workers have full bargaining power, the welfare gain falls to 8%.

Table 13: Interaction effects of bargaining power and product market competition.

Panel (A)	Counterfactual: constant markups		
Bargaining power (κ)	$\% \Delta V(\phi_j)$	$\% \Delta \Phi$	$\hat{c} \times 100$
Low ($\kappa = 0.12$)	21	21	12
Medium ($\kappa = 0.5$)	13	12	10
Full ($\kappa = 1.0$)	8	8	8

Panel (B)	Counterfactual: raise bargaining power (0.12 to 0.20)		
Markups & markdowns are ...	$\% \Delta V(\phi_j)$	$\% \Delta \Phi$	$\hat{c} \times 100$
Both variable	2	20	3
Both constant	0	16	2

This table shows the percentage changes in the aggregate wage index, variance of firm wage premia, and welfare under two counterfactual exercises. Panel (A) equalizes markups across firms and compares its effects under different levels of worker bargaining power. Markdowns remain variable in these exercises. Panel (B) raises worker bargaining power by 8 p.p. above the baseline value.

Why do the effects of equalizing markups depend on the level of worker bargaining power? Equalizing markups removes the misallocation effect of variable markups. When bargaining power is low, bargaining does little to affect the allocation of labor, thus the

market allocation of labor is far from the social planner’s allocation to begin with. When bargaining power is high, bargaining does more to improve the allocation of workers to firms, forcing high-markup firms to share more rents and employ more workers. Therefore, the market allocation of labor is already closer to the social planner’s allocation, resulting in smaller welfare gains from removing markup dispersion.

Similarly, panel (B) shows that increasing worker bargaining power from a baseline of 0.12 to 0.20 achieves a higher welfare gain when markups and markdowns are variable. However, a comparison with the case of constant markups and markdowns shows that the main source of welfare gains from higher bargaining power comes from redistributing profits from firms to workers, mainly offsetting the role of markups and markdowns as a uniform tax on labor demand.

8 Conclusion

This paper builds a structural model in which firms have product and labor market power, and workers have bargaining power, to study how they determine the wages that firms pay and the accompanying welfare implications. The model provides new insights on several key questions: What determines the gap between wages and marginal revenue products of labor? What role does market power in product and labor markets play in shaping wages and welfare? How can worker bargaining power be estimated, and can increasing it improve welfare? Additionally, what are the sources of firm heterogeneity driving wage differences between firms, and how do firm-specific shocks affect wages?

A central theoretical insight of the model is that the labor wedge between wages and marginal revenue products of labor depends not only on firm monopsony markdowns, but also on price-cost markups and worker bargaining power. This means that markups can either amplify firm wage premia through rent-sharing or dampen wage dispersion by suppressing labor demand. The model also shows that worker bargaining power can improve welfare by mitigating the distortions caused by firm market power. Moreover, it highlights the challenges of separately estimating worker bargaining power and firm monopsony power using existing approaches, and offers an alternative approach for doing so. Applying this approach on detailed French administrative data, I find novel empirical relationships that cannot be explained by existing labor market monopsony models. My model explains these patterns through the sharing of markup-induced profits between firms and workers, where product quality plays a central role in driving those markups.

A quantified version of the model delivers several key findings. First, the degree of product differentiation determines the size of the passthrough of productivity or product quality shocks to wages. Product quality heterogeneity is as important as productivity heterogeneity in explaining the cross-sectional dispersion of firm wage premia. Second, markup dispersion dampens firm wage premium dispersion, as the labor demand effect

is quantitatively stronger than the rent-sharing effect. Third, raising worker bargaining power can significantly improve welfare. However, even full worker bargaining power cannot fully close the gap between the market equilibrium and a welfare-maximizing social planner’s equilibrium, as bargaining power cannot directly address the distortionary effects of markups on the demand for non-labor inputs.

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Appendix (for online publication)

Understanding High-Wage Firms

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A Appendix: Data, Institutions, and Measurement

A.1 Wage determination in France

Wages in France are mainly determined at three levels of aggregation – the national level, the industry level, and the firm level. At the national level, the French government sets the national minimum wage. At the industry level, labor unions and employers' organizations negotiate industry wage floors. At the firm level, wage bargaining occurs between an individual firm and labor union representatives, representing the collective group of employees at the firm. This allows firms to depart upwards of the national minimum wage or industry wage floor.

The national minimum wage as of 2016 is 9.67 Euros per hour. In the same year, approximately 15% of workers in the French economy are minimum wage workers, defined as workers earning at, or at most 5% above, the minimum wage. In the manufacturing sector, on which the analysis in the paper is based, approximately 10% of workers are minimum wage workers.

Collective wage bargaining between individual firms and their employees is prevalent. The 1982 Auroux Laws require firms in which a labor union representative is present to bargain over wages with the union annually. Among firms with 50 employees or more, the presence of at least one union representative is a binding legal requirement.⁴⁶ According to the French Ministry of Labor, among firms with 20 to 49 employees, 34% had at least one labor union representative in 2010 (Naouas and Romans, 2014). Among firms with 11 to 19 employees, the corresponding number is 22%.

The vast majority of union-employer bargaining occurred at the firm level, rather than at the establishment level. Only 9% of multi-establishment firms negotiated wages with their employees at the establishment level. Among workers employed by firms with at least 20 employees, 70% of them are covered by firm level collective bargaining agreements. These collective bargaining agreements extend to all workers within the firm, regardless of whether the worker holds a union membership (Fougere, Gautier, and Roux, 2016).

⁴⁶See Garicano, Van Reenen, and Lelarge (2016) for further details about the specific restrictions faced by firms with at least 50 employees.

A.2 Data and measurement

To estimate production functions using FARE-EAP-DADS (2009-2016) firm balance sheet, output price, and matched employer-employee data from France, I measure the key variables in the following way:

- Sales revenue (PY): measured by the variable CATOTAL in FICUS, and REDI_R310 in FARE.
- Efficiency units of labor ($H = \bar{E}L$): the DADS provides the number of hours worked for each worker under NBHEUR, which enables the researcher to measure total hours (L) at a given firm. The average efficiency of workers ($\bar{E}$) is then measured as the difference between the unconditional mean wage and the firm wage premium, according to the theory.
- Capital (K): measured as total fixed physical assets under variable names IMMO-COR in FICUS, and IMMO_CORP in FARE.
- Materials (M): the French balance sheet data provides a breakdown of material inputs into three components – materials purchased to be used as inputs in production (ACHAMPR in FICUS, REDI_212 in FARE), goods purchased to be resold (ACHAMAR in FICUS, REDI_210 in FARE), and purchase of services (details provided next). I correct for changes in inventory for materials to be used in production (using VARSTMP in FICUS, REDI_213 in FARE) and for goods purchased to be resold (VARSTMA in FICUS, REDI_211 in FARE). I measure M as the sum of these variables, except services.
- Services (O): measured as AUTACHA in FICUS, and REDI_214 in FARE. These variables include the costs of outsourcing and advertising.
- Hourly wages (W): measured by dividing BRUT by NBHEUR in DADS.
- Output prices (P): PRODFRA defines the ten-digit product codes, C_UNITE_VAR gives the quantity and revenue indicator, and VAL_REF gives the values in quantity and revenue terms. These variables are obtained from EAP.
- Market shares: measured within 5-digit sectors.

A.3 Summary statistics

Employees		
Sample size		
People-years	9,233,319	
Firm-years	126,836	
Average number of workers per year	1,154,215	
Average number of firms per year	16,501	
Wage distribution		
Mean log Wage	2.53	
Variance log wage	0.19	
Fraction between-firms	0.44	
Efficiency Units & Firm Premium		
Variance $\bar{e}$	0.03	
Variance ϕ	0.007	
Correlation $(\bar{e}, \phi)$	0.27	
Employers		
	Mean	Variance
Log production value ('000)	8.31	2.02
Log employment	3.29	1.36
Log capital stock ('000)	7.06	3.17
Log material inputs ('000)	7.22	2.72

Table 14: Summary statistics: manufacturing sector employers and employees (2009-2016).

Table 15: Summary statistics for 2-digit French manufacturing sectors.

Sector	# Observations	Sales share	Employment share	Average ϕ	Price-cost markups	Labor wedges
Textile	4,850	1.9%	2.3%	2.79	1.47	0.75
Apparel	3,157	1.6%	2.1%	2.76	1.35	0.73
Leather	1,433	0.5%	0.8%	2.74	1.40	0.92
Wood products (except furniture)	6,859	2.2%	2.8%	2.75	1.25	0.59
Paper and publishing	5,257	5.6%	4.8%	2.82	1.46	0.67
Recorded media	7,817	1.8%	2.6%	2.79	1.78	0.88
Chemicals	6,277	11.4%	7.8%	2.84	1.30	0.48
Pharmaceutical	380	2.5%	1.5%	2.94	1.60	0.16
Rubber & plastics	13,333	12.1%	13.3%	2.81	1.34	0.63
Non-metallic minerals	9,295	7.8%	7.7%	2.79	1.33	0.57
Basic metals	3,343	8.5%	6.0%	2.81	1.29	0.55
Fabricated metals (except machinery)	22,845	9.4%	11.9%	2.80	1.43	0.64
Computers, electronic, & optical	4,460	6.7%	7.1%	2.85	1.33	0.60
Electrical equipment	5,416	3.4%	3.8%	2.83	1.30	0.59
Machinery & equipment	12,785	7.5%	8.3%	2.83	1.15	0.46
Motor vehicles	4,189	7.5%	6.4%	2.83	1.17	0.57
Other transport equipment	948	2.9%	2.4%	2.85	1.20	0.45
Furniture	7,037	2.1%	3.1%	2.78	1.47	0.76
Other manufacturing	4,032	2.4%	2.7%	2.80	1.45	0.70
Repair & installation of machinery	4,238	2.2%	2.5%	2.79	1.37	0.57
Total	126,836	100%	100%	-	-	-

This table reports the summary statistics for manufacturing sectors in my sample (2009-2016). The last two columns report the average price-cost markup and labor wedges in each sector.

B Appendix: Estimation

B.1 K-means clustering of firms into groups

Specifically, let $g(j) \in \{1, 2, \dots, G\}$ denote the cluster of firm j , and G the total number of clusters. The k-means algorithm finds the partition of firms such that the following objective function is minimized:

$$\min_{g(1), \dots, g(J), H(1), \dots, H(G)} \sum_{j=1}^J N_j \int \left(\hat{F}_j(\ln W_{ij}) - H_{g(j)}(\ln W_{ij}) \right)^2 d\gamma(\ln W_{ij})$$

where $H(g)$ denotes the firm-group level cumulative distribution function for log wages at group g , $\hat{F}_j$ is the empirical CDF of log wages at firm j , and N_j is the employment size of firm j . The total number of groups G is the choice of the researcher. I choose sector-specific G such that the variance of log wages between firm-groups captures at least 95% of the total between-firm variance. This choice is motivated by the following consideration: having a coarse classification of firms into fewer groups leads to many more workers who switch between firm-groups, which substantially improves the precision of firm wage premium estimates. However, this comes at the cost of potentially averaging away considerable amounts of multidimensional firm heterogeneity within firm-groups.

B.2 Robustness of AKM restrictions: conditional exogenous mobility and log-additivity

AKM regressions rely on the assumption that worker mobility is as good as random conditional on observed worker characteristics, worker fixed effects, and firm fixed effects. Formally, $E(\nu_{it} | \chi_{it}, \iota_i, \phi_{g(j(i,t),t)t}) = 0$. This assumption rules out worker mobility based on wage realizations due to the residual component of wages. In addition, AKM regressions impose log additivity of the worker and firm components of wages. If these assumptions are reasonable approximations, then one should observe systematic worker mobility up and down the firm wage quartiles. Moreover, workers should experience approximately symmetric wage changes as they move along the firm wage quartiles, given the log additive regression specification. On the other hand, in structural models of worker-firm sorting based on comparative advantage ([Eeckhout and Kircher, 2011](#)), worker mobility is based on the match-specific component of wages, which is captured by the residual component of wages in the AKM regression. In this class of models the AKM regression is misspecified in the sense that the wage gains depend on value of the particular worker-firm match, for example, if highly skilled workers have a comparative advantage in high productivity firms. In the event-study exercise show in [Figure 5](#), I compare the changes in mean log wages for workers who move between firms in different quartiles of coworker pay, following

Card et al. (2018). Figure 5 shows that workers who move up firm quartiles experience a wage gain similar in magnitude to the wage loss of workers who move down firm quartiles.

An alternative way to assess the AKM regression specification is to compare the changes in residual wages to changes in firm effects, following Sorkin (2018). This is similar to the above method. I run the following regression among all employer-to-employer transitions:

$$w_{it}^r - w_{it-1}^r = \alpha_0 + \alpha_1 (\phi_{g(j(i,t))} - \phi_{g(j(i,t-1))}) + \epsilon_{it} \quad \forall (i, t), g(j(i, t)) \neq g(j(i, t-1))$$

where $w_{it}^r = w_{it} - x_{it}'\hat{\beta}$ denotes residualized wages and $\phi_{g(j(i,t))}$ are the firm-group fixed effects. If the AKM regression is not mis-specified, the estimated coefficient $\hat{\alpha}_1$ will equal 1. I find $\hat{\alpha}_1 = 0.857$, with a standard error of 0.007. To see this visually, Figure 6 plots the changes in residual wages and the changes in firm fixed effects in 100 bins of changes in firm fixed effects. In models of assortative matching based on comparative advantage (Lopes de Melo, 2018), worker mobility is strongly driven the residual component of the AKM regression, implying that AKM regressions are mis-specified. As Sorkin (2018) shows, these models predict that worker mobility entails a wage gain, regardless of the direction of worker mobility in terms of the estimated firm effects, as workers move to firms at which they have a comparative advantage: there is a V-shape around zero changes in firm effects. The patterns of wage changes upon changes in firm fixed effects shown in Figure 6 do not resemble a V-shape around zero.

Another way to assess the log additivity of the worker and firm components of wages is to group worker and firm fixed effects into 10 deciles each, generating 100 worker-firm fixed effect deciles, then plot the mean estimated residuals within each worker-firm fixed effect decile. If the firm wage premium depends strongly on the worker's unobserved ability type, log additivity would be severely violated, and one should observe that the estimated residuals systematically varies across worker-firm fixed effect deciles. Figures 7 and 8 show that the mean estimated residuals are approximately zero across worker-firm fixed effect deciles, with the exception of the very top deciles of high-wage workers who are employed at low-wage firms at the very bottom deciles.

As a further robustness check, I follow Bonhomme et al. (2019) and run the BLM regression with worker-firm interactions, but with only 20 firm groups and 6 worker groups to maintain computational tractability. Moving from an additive to an interacted regression model gives a gain in R^2 of 0.01.

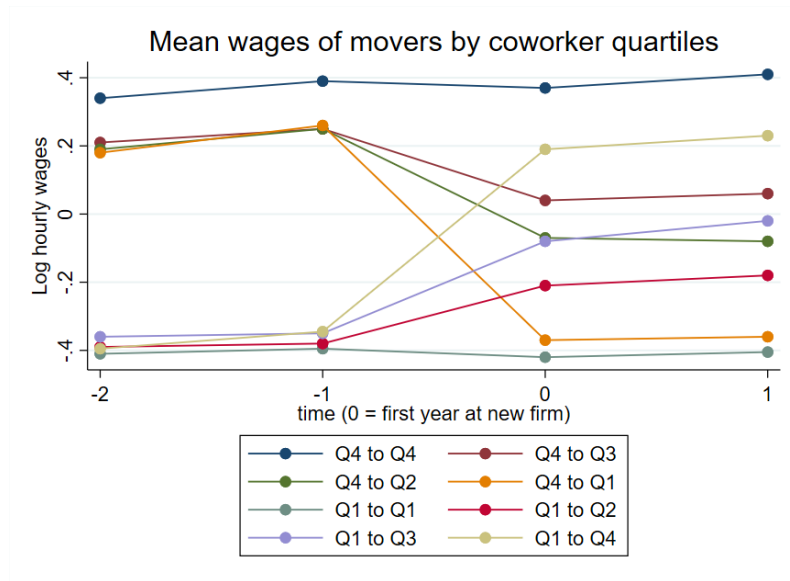


Figure 5: Worker mobility and wage changes by quartiles of coworker effects (2009-2016).

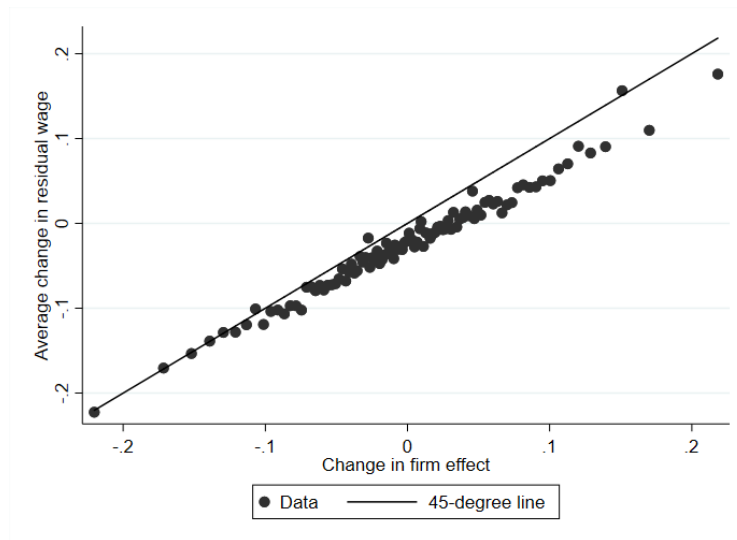


Figure 6: Average wage changes from worker mobility by declines of changes in firm premia (2009-2016).

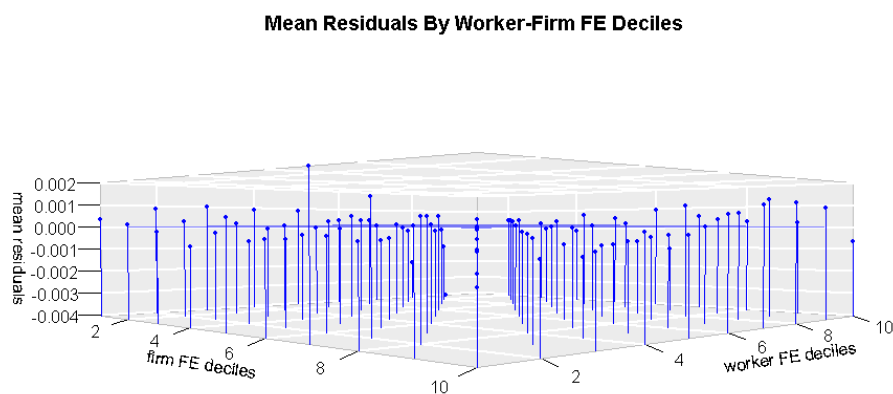


Figure 7: Mean estimated residuals by worker-firm deciles (2014)

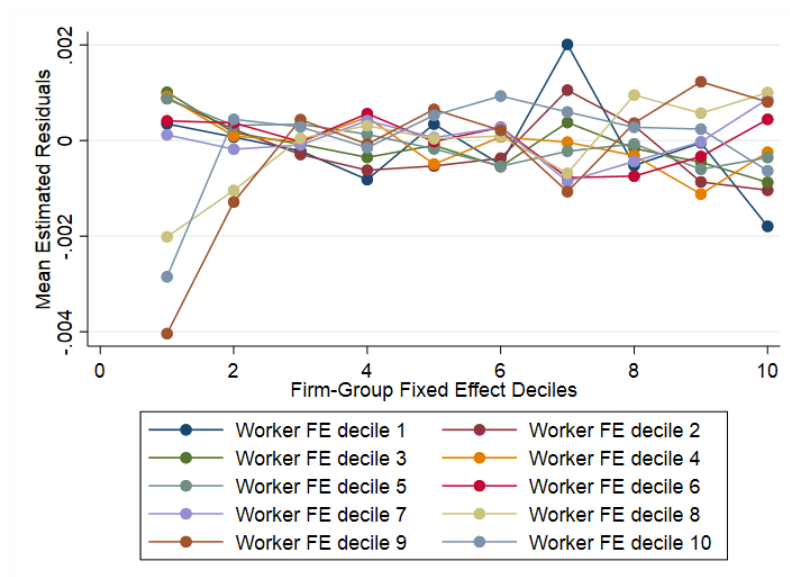


Figure 8: Mean estimated residuals by worker-firm deciles (2016)

B.3 The choice of production functions

In Section 3, I estimate translog production functions to obtain firm-specific output elasticities, which are used to measure markups and labor wedges. The Cobb-Douglas production function is a simpler alternative that allows one to document how markups and labor wedges vary across firms without having to estimate production functions. The advantage is that one can sidestep the challenges of estimating gross output production functions (Gandhi et al., 2020). However, this requires assuming that output elasticities do not vary across firms. In this section, I provide suggestive evidence that output elasticities vary systematically across firms, and show that they can lead to a spurious negative correlation between markups and firm size. I then show that Cobb-Douglas-implied markups decline with firm size, which is inconsistent with a body of evidence showing smaller price passthrough of cost shocks among larger firms (Amiti et al., 2014). I show that my estimates of translog output elasticities vary systematically across firms and are stable over time, leading to markup estimates that increase with firm size and are stable over time. Moreover, the estimated translog output elasticities closely track the input cost shares, which can be readily computed in the data and directly correspond to output elasticities under certain conditions.

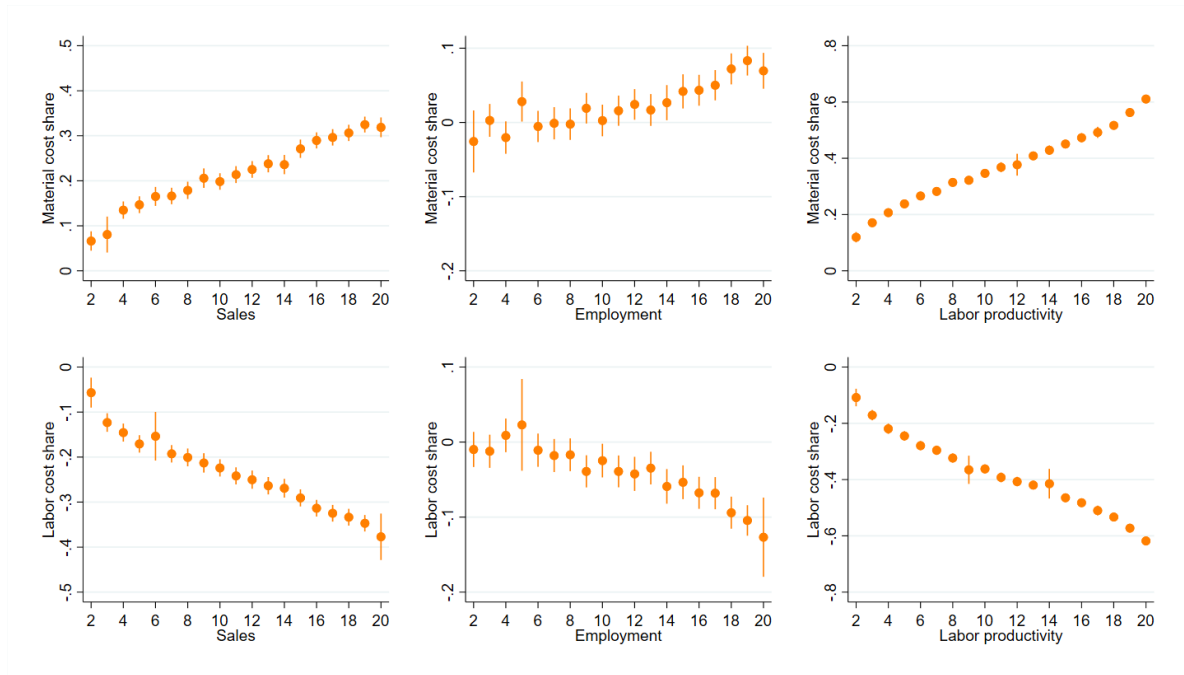


Figure 9: Material and labor cost shares across firms.

This figure shows how material and labor cost shares vary across quantiles of firm size (sales or employment) and labor productivity, conditional on 2-digit sector $\times$ year fixed effects. Confidence intervals at the 95% level are plotted.

I first examine whether output elasticities are likely to vary systematically across firms by looking at input cost shares. In my model in Section 2, the material and labor cost

shares can be written as:

$$CS_{m,jt} = \frac{\alpha_{m,jt}}{\alpha_{k,jt} + \alpha_{h,jt} + \Lambda_{jt}\alpha_{h,jt}} \quad \text{and} \quad CS_{h,jt} = \frac{\Lambda_{jt}\alpha_{h,jt}}{\alpha_{k,jt} + \alpha_{h,jt} + \Lambda_{jt}\alpha_{h,jt}}$$

where the cost shares are measured as the expenditure on an input divided by total costs. Total costs are measured as the sum of material, labor, and capital expenditure, where capital expenditure is measured as expenditure on capital investments. These cost shares map directly into firm-specific output elasticities ($CS_{m,jt} = \alpha_{m,jt}$, $CS_{h,jt} = \alpha_{h,jt}$) if the following assumptions hold: constant returns to scale and perfectly competitive input markets. In my model, labor markets are imperfectly competitive, implying that observed input cost shares do not directly correspond to output elasticities—disentangling output elasticities from labor wedges requires estimating production functions. Nevertheless, output elasticities are key components of input cost shares. Therefore, the evidence on cost shares presented in this section is suggestive of how actual output elasticities vary across firms.

Figure 9 plots the material and labor cost shares against firm size (measured as sales and employment) and labor productivity (measured as sales per hour). Material cost shares increase systematically with firm size and labor productivity, while labor cost shares decrease with firm size and labor productivity. Moreover, Table 16 shows that there is substantial dispersion in cost shares within 2-digit sectors. These patterns suggest that material and labor output elasticities vary systematically across firms.

Table 16: The distribution of material and labor cost shares.

Cost shares	Mean	Median	10 th Pct	25 th Pct	75 th Pct	90 th Pct	Variance
Material	0.62	0.65	0.35	0.50	0.77	0.85	0.04
Labor	0.35	0.32	0.13	0.21	0.47	0.62	0.03

This table reports the summary statistics for the material and labor cost shares, conditional on 2-digit sector×year fixed effects.

To the extent that the cross-sectional correlations between cost shares and firm size are driven by output elasticities, assuming Cobb-Douglas production functions will lead to systematic biases in the measurement of markups and labor wedges. To illustrate this, I write the relationship between Cobb-Douglas markups μ_{jt}^{CD} and “true” markups μ_{jt} (markups obtained without assuming homogenous output elasticities) as:

$$\mu_{jt}^{CD} = \mu_{jt} \frac{\alpha_m^{CD}}{\alpha_{m,jt}}$$

where $\mu_{jt}^{CD} \equiv \alpha_m^{CD} \frac{P_{jt}Y_{jt}}{P_{m,t}M_{jt}}$ and $\mu_{jt} = \alpha_{m,jt} \frac{P_{jt}Y_{jt}}{P_{m,t}M_{jt}}$. The covariance between Cobb-Douglas markups and firm size can then be written as:

$$CV(\log \mu_{jt}^{CD}, \log size_{jt}) = CV(\log \mu_{jt}, \log size_{jt}) - CV(\log \alpha_{m,jt}, \log size_{jt}) \quad (11)$$

Therefore, the covariance between Cobb-Douglas markups and firm size will be *downward-biased* if material output elasticities ($\alpha_{m,jt}$) increase with firm size.

Similarly, the relationship between Cobb-Douglas labor wedges and “true” labor wedges is:

$$\Lambda_{jt}^{CD} = \Lambda_{jt} \left(\frac{\alpha_m^{CD}}{\alpha_{m,jt}} \right) \left(\frac{\alpha_{h,jt}}{\alpha_h^{CD}} \right)$$

where $\Lambda_{jt}^{CD} \equiv \frac{\alpha_m^{CD}}{\alpha_h^{CD}} \frac{\Phi_{jt} H_{jt}}{P_{m,t} M_{jt}}$ and $\Lambda_{jt} = \frac{\alpha_{m,jt}}{\alpha_{h,jt}} \frac{\Phi_{jt} H_{jt}}{P_{m,t} M_{jt}}$. The covariance between Cobb-Douglas labor wedges and firm size can be written as:

$$CV(\log \Lambda_{jt}^{CD}, \log size_{jt}) = CV(\log \Lambda_{jt}, \log size_{jt}) - CV(\log \alpha_{m,jt}, \log size_{jt}) + CV(\log \alpha_{h,jt}, \log size_{jt}) \quad (12)$$

Therefore, covariance between Cobb-Douglas labor wedges and firm size will also be *downward-biased* if material output elasticities increase with firm size, or if labor output elasticities decrease with firm size.

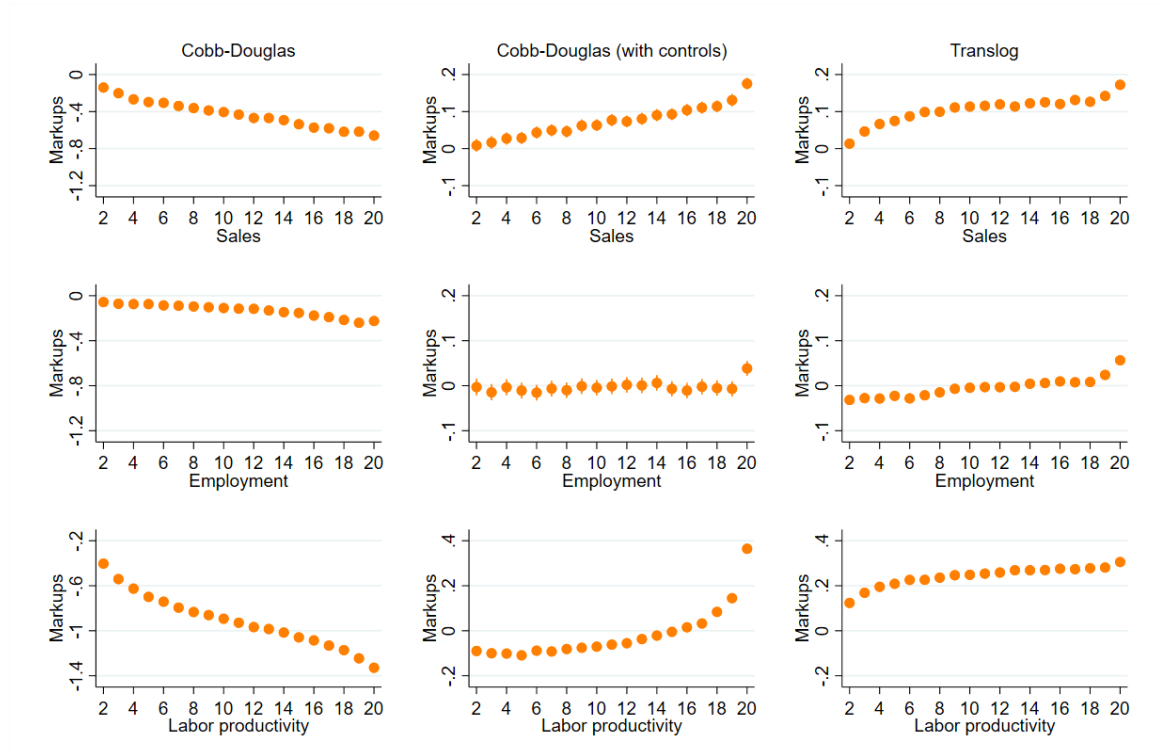


Figure 10: Cobb-Douglas and Translog (log) markup measures across firms.

This figure shows how different log markup measures vary by firm size (sales or employment) and labor productivity quantiles, conditional on 2-digit sector×year fixed effects. Confidence intervals at the 95% level are plotted.

Figure 10 shows how different markup measures vary by firm size and labor productivity. The first column shows that Cobb-Douglas markups decline steeply with firm size and labor productivity, which is inconsistent with existing evidence on lower passthrough of cost shocks to prices among large firms (see, e.g., [Amity et al. \(2014\)](#)). As shown above, this negative correlation may be due to Cobb-Douglas markups capturing a positive cor-

relation between material output elasticities and firm size.

Without estimating production functions, I now attempt to assess the extent to which Cobb-Douglas markup measures are likely to understate the relationship between actual markups and firm size. Suppose that material output elasticities are positively correlated with firm size. Then, equation (11) shows that controlling material output elasticities will increase the correlation between Cobb-Douglas markups and firm size. Although material output elasticities are not observed, under the assumption that production functions exhibit constant returns to scale, they depend on the capital-material and labor-material ratios, $\alpha_{m,jt} = \alpha_m(\frac{K_{jt}}{M_{jt}}, \frac{H_{jt}}{M_{jt}})$.⁴⁷ This suggests that the correlation between Cobb-Douglas markups and firm size would increase if one controls for these input ratios.

The second column of Figure 10 shows that the slope between Cobb-Douglas markups and firm size becomes positive upon controlling for capital-material and labor-material ratios. The third column shows that markups implied by the estimated translog production functions are also increasing with firm size (without controlling for input ratios). Figure 11 shows that a similar pattern holds for labor wedges: the Cobb-Douglas labor wedges decline with firm size more steeply than the translog counterpart.

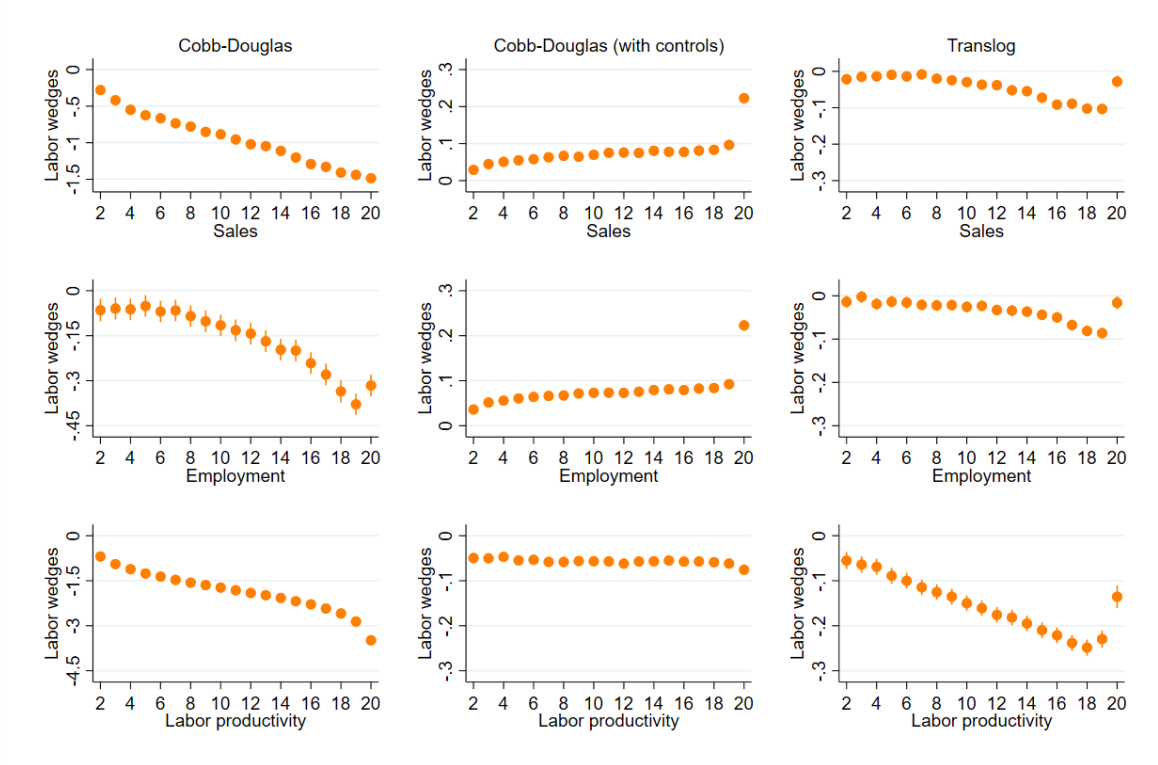


Figure 11: Cobb-Douglas and Translog (log) labor wedge measures across firms.

This figure shows how different log labor wedge measures vary by firm size (sales or employment) and labor productivity quantiles, conditional on 2-digit sector×year fixed effects. Confidence intervals at the 95% level are plotted.

⁴⁷Similarly, the labor output elasticity can be written as $\alpha_{h,jt} = \alpha_h(\frac{K_{jt}}{H_{jt}}, \frac{M_{jt}}{H_{jt}})$. Suppose labor output elasticities decrease with firm size, then equation (12) suggests that the correlation between Cobb-Douglas labor wedges and firm size will increase if one controls for the relevant input ratios.

Finally, I show that the estimated translog output elasticities vary across firms in a similar way compared to the input cost shares, and that they are stable over time. Table 17 shows that the translog material output elasticities are on average lower than their cost shares, while labor output elasticities are higher than their cost shares. This is what one would expect if labor wedges are below 1. Consistent with the fact that material cost shares increase with firm size depicted in Figure 9, Figure 12 shows that translog material output elasticities are also increasing with firm size, while these patterns are inverted for translog labor output elasticities. Further, Figure 13 shows that both the distribution of translog material and labor output elasticities are stable over time. This implies that the translog markup and labor wedge measures are also stable over time (see Figure 16).

Table 17: The distribution of output elasticities.

Elasticities	Mean	Median	10 th Pct	25 th Pct	75 th Pct	90 th Pct	Variance
Material	0.48	0.49	0.26	0.38	0.60	0.70	0.04
Labor	0.46	0.45	0.25	0.35	0.55	0.66	0.03
Capital	0.06	0.06	0.02	0.04	0.08	0.11	0.00

This table reports the summary statistics for the output elasticities, conditional on 2-digit sector×year fixed effects.

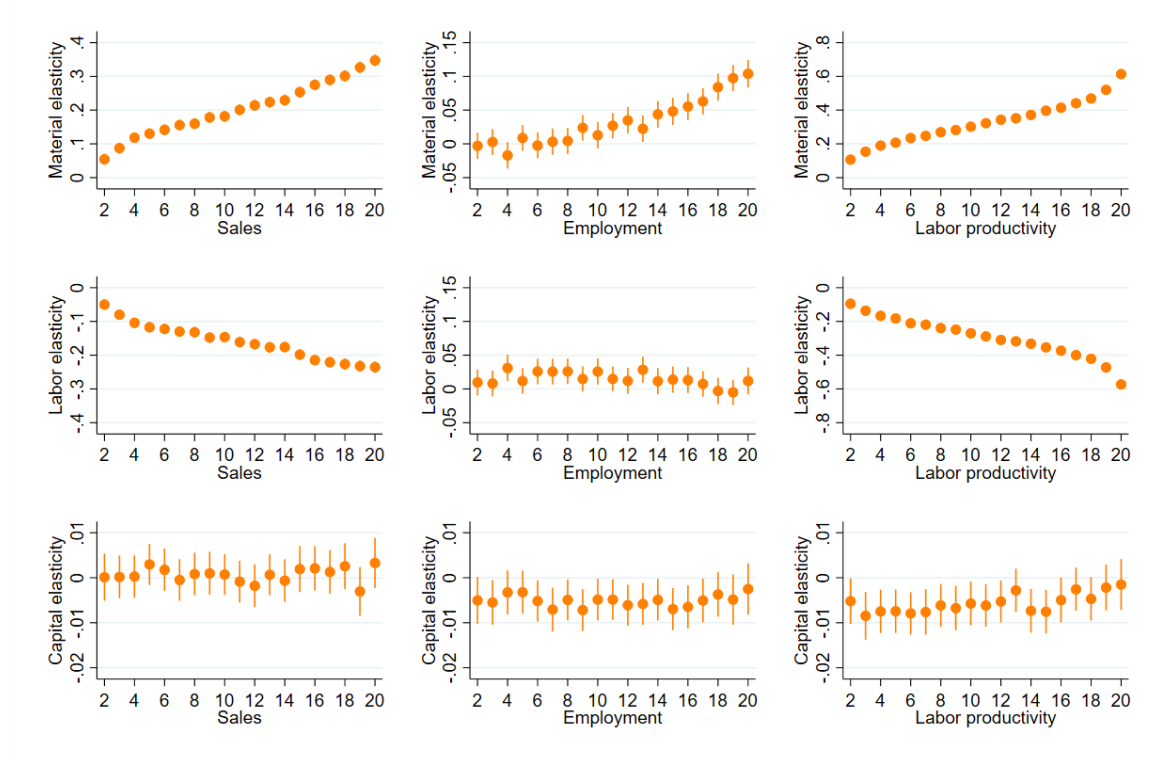


Figure 12: Translog output elasticities across firms.

This figure shows how output elasticities vary across quantiles of firm size (sales or employment) and labor productivity, conditional on 2-digit sector×year fixed effects. Confidence intervals at the 95% level are plotted.

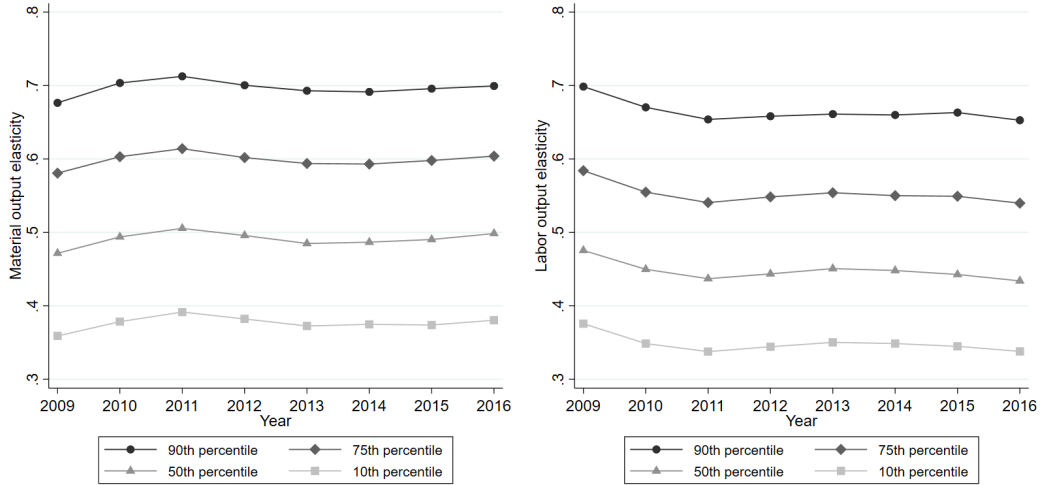


Figure 13: Translog output elasticities over time.

B.4 Monopsony in the market for materials

In Section 3, I measure markups and labor wedges under the assumption that firms have no monopsony power in material inputs. If firms have monopsony power in material markets, then even if one has data on actual material and labor output elasticities, the material and labor first-order conditions do not deliver the correct measures of markups and labor wedges; they will be contaminated by a material monopsony wedge. These first-order conditions become:

$$\mu_{jt} = \lambda_{m,jt} \alpha_{m,jt} \frac{P_{jt} Y_{jt}}{P_{m,jt} M_{jt}} \quad \text{and} \quad \Lambda_{jt} = \lambda_{m,jt} \frac{\alpha_{m,jt}}{\alpha_{h,jt}} \frac{\Phi_{jt} H_{jt}}{P_{m,jt} M_{jt}}$$

where $\lambda_{m,jt} = \frac{\epsilon_{m,jt}}{1+\epsilon_{m,jt}}$ is the monopsony markdown for materials, which depends on the material supply elasticity $\epsilon_{m,jt}$.

I now show that, in the presence of material monopsony power, it is still possible to obtain correct measures of markups and labor wedges—not contaminated by material markdowns—if production functions are estimated using *material expenditure* instead of quantities. Without material price data, one recovers the output elasticity with respect to material expenditure when estimating production functions. In what follows, I derive this material expenditure output elasticity and show that (a) it is equal to the composite elasticity $\tilde{\alpha}_{m,jt} \equiv \lambda_{m,jt} \alpha_{m,jt}$ when firms have material monopsony power, and (b) it is equal to the material quantity output elasticity $\alpha_{m,jt}$ when firms are pricetakers in material markets. Therefore, when firms have material monopsony power, the correct markup and labor wedge measures can still be obtained, although one cannot disentangle material monopsony power from material (quantity) output elasticities.

Consider the following upward-sloping material supply curve: $M_{jt} = \mathcal{V}(P_{m,jt})$. The inverse material supply curve, expressed as a function of material expenditure, is: $P_{m,jt} = \tilde{\mathcal{V}}(\tilde{M}_{m,jt})$, where $\tilde{M}_{m,jt} \equiv P_{m,jt} M_{jt}$. Variables written in lowercase letters are in logs. The

production function takes the form:

$$\begin{aligned} y_{jt} &= f(k_{jt}, h_{jt}, m_{jt}) + \omega_{jt} \\ &= f(k_{jt}, h_{jt}, \tilde{m}_{jt} - \tilde{v}(\tilde{m}_{jt})) + \omega_{jt} \end{aligned}$$

where the second equality uses the inverse material supply curve. The output elasticity with respect to material expenditure is:

$$\tilde{\alpha}_{m,jt} \equiv \frac{\partial y_{jt}}{\partial \tilde{m}_{jt}} = \alpha_{m,jt} \left(1 - \frac{\partial \tilde{v}(\tilde{m}_{jt})}{\partial \tilde{m}_{jt}} \right)$$

What remains to be shown is that the term $\left(1 - \frac{\partial \tilde{v}(\tilde{m}_{jt})}{\partial \tilde{m}_{jt}} \right)$ is equal to the material mark-down. From the inverse material supply curve, the elasticity of material prices with respect to material expenditure is:

$$\frac{\partial p_{m,jt}}{\partial \tilde{m}_{jt}} = \frac{\partial p_{m,jt}}{\partial m_{jt}} \left(\frac{\partial \tilde{M}_{jt}}{\partial M_{jt}} \right)^{-1} P_{m,jt} \quad (B6(a))$$

and the partial derivative of material expenditure with respect to material quantities is:

$$\begin{aligned} \frac{\partial \tilde{M}_{jt}}{\partial M_{jt}} &= \left(1 + \underbrace{\left(\frac{\partial m_{jt}}{\partial p_{m,jt}} \right)}_{\equiv \epsilon_{m,jt}} \right)^{-1} P_{m,jt} \end{aligned} \quad (B6(b))$$

Combining equations (B6(a)) and (B6(b)) gives $\lambda_{m,jt} \equiv \frac{\epsilon_{m,jt}}{1+\epsilon_{m,jt}} = \left(1 - \frac{\partial \tilde{v}(\tilde{m}_{jt})}{\partial \tilde{m}_{jt}} \right)$. Therefore, the output elasticity with respect to material expenditure is equal to the composite elasticity $\tilde{\alpha}_{m,jt} \equiv \lambda_{m,jt} \alpha_{m,jt}$. When firms have no material monopsony power, so that $\lambda_{m,jt} = 1$ and $P_{m,jt} = P_{m,t}$, the output elasticity with respect to material expenditure or quantities are the same. This result is analogous to one by [Bond et al. \(2021\)](#), who show that estimating production functions using revenues instead of output recovers revenue elasticities instead of output elasticities, complicating the measurement of markups. Similarly, my result shows that estimating production functions based on material expenditures rather than quantities recovers the output elasticity with respect to material expenditure, which hinders the measurement of material markdowns, but not markups or labor wedges.

The presence of material markdowns has implications for the use of material inputs as a proxy variable when estimating production functions. Given the material supply equation faced by the firm, the material input FOC can be written as:

$$\tilde{M}_{jt} = \left(1 - \frac{\partial \tilde{v}(\tilde{m}_{jt})}{\partial \tilde{m}_{jt}} \right) P_{jt} \mu_{jt}^{-1} \alpha_{m,jt} \Omega_{jt} F \left(K_{jt}, H_{jt}, \frac{\tilde{M}_{jt}}{\tilde{\mathcal{V}}(\tilde{M}_{m,jt})} \right)$$

Thus, the control function for productivity (Ω_{jt}) is formed on material expenditure rather than quantities. The presence of material markdowns do not violate the scalar unobservable assumption if firm size (expenditure on materials) is a sufficient statistic for material

markdowns, analogous to the implication that firm size in terms of wage-bill shares are sufficient statistics for labor market monopsony markdowns in recent models such as [Berger et al. \(2022\)](#).

In deriving these results, I abstract from material price differences driven by quality heterogeneity. In the presence of quality heterogeneity, there will be material price dispersion across firms even when firms do not have material monopsony power. The lack of material price data will cause an input price bias, as discussed in Section 3. [De Loecker et al. \(2016\)](#) develop an approach to addressing input price bias using output price data, and I implement their approach to address this concern.

B.5 Alternative methods for production function estimation

The production approach to markup measurement requires estimating the output elasticity of a flexible input (De Loecker and Warzynski, 2012) (DLW henceforth). In Section 3, I estimate a translog gross output production function to obtain the flexible input’s (materials) output elasticity. As Bond and Söderbom (2005) and Gandhi et al. (2020) show, this output elasticity is challenging to identify, particularly in the context of gross output production functions and perfectly competitive goods markets. In such cases, persistence in material prices is the only source of persistence in material input choices, which is crucial for the identification of the material output elasticity. However, De Ridder et al. (2021) show that with imperfectly competitive product markets—a key feature of my model—variation in output prices and productivity shocks to firms’ competitors generate additional variation in firms’ material input demand, which aids the estimation of the material output elasticity.

To more directly address the identification concern raised in Gandhi et al. (2020), I compare my baseline estimates of output elasticities to those estimated using two alternative methods that aim to address this identification challenge: (i) control function method with constant returns to scale production functions (Flynn et al., 2019), and (ii) dynamic panel method (Blundell and Bond, 1998).

Table 18: Output elasticities under alternative production function estimation methods.

	DLW			FGT			BB		
	Mean	Median	Var	Mean	Median	Var	Mean	Median	Var
Material	0.48	0.49	0.04	0.47	0.46	0.04	0.35	0.35	0.02
Labor	0.46	0.45	0.03	0.45	0.47	0.05	0.25	0.24	0.02
Capital	0.06	0.06	0.00	0.08	0.07	0.00	0.09	0.09	0.00
Returns to scale	1.00	0.99	0.00	1.00	1.00	0.00	0.69	0.68	0.01

This table reports the estimated output elasticities under alternative production function approaches. My baseline approach follows De Loecker and Warzynski (2012), referred to as DLW. This table also presents estimates from implementing constant returns to scale (Flynn et al. (2019); FGT) and dynamic panel methods (Blundell and Bond (1998); BB). All output elasticities are trimmed by 1% on either side of their distribution.

In the context of imperfectly competitive output markets, Flynn et al. (2019) (FGT henceforth) show that the challenge of identifying the material output elasticity can be recast as a challenge of separately identifying returns to scale from the level of markups. They further show that under the restriction of constant returns to scale, the material output elasticity is identified. Table 18 compares my baseline estimates of the output elasticities to those obtained under constant returns to scale. Comparing the first three columns with the middle three columns shows that my baseline estimates (based on DLW) and those using FGT are similar. The FGT material output elasticities are slightly lower than those estimated following DLW, while the labor output elasticities are slightly

higher. Figure 14 shows that material output elasticities estimated with either DLW or FGT are increasing with firm size (as defined by sales). The figure also shows that the implied markups and labor wedges are also increasing with firm size. Figures 15 and 16 further show that the estimated material output elasticities, markups, and labor wedges are stable over time.

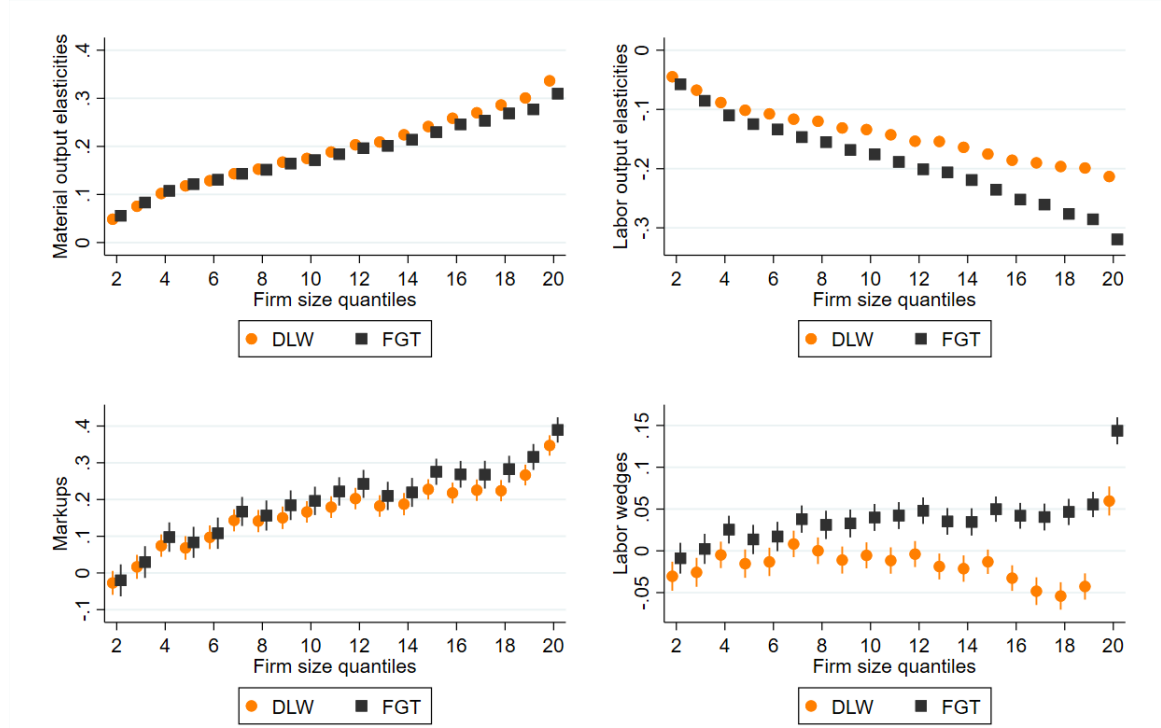


Figure 14: Output elasticities, markups, and labor wedges by firm size under DLW and FGT.

This figure compares the estimated output elasticities, markups, and labor wedges using two methods: (i) my baseline approach following [De Loecker and Warzynski \(2012\)](#), and (ii) the [Flynn et al. \(2019\)](#) approach which imposes a constant returns to scale restriction. Firm size quantiles are defined based on sales. The plots control for 2-digit sector $\times$ year fixed effects. Vertical bars are 95% confidence intervals.

As an alternative to the control function method, the dynamic panel data method of [Blundell and Bond \(1998\)](#) (BB henceforth) circumvents the need to select a proxy variable, instead relying on a log-linear AR(1) process for Hicks-neutral productivity to identify production function parameters. In practice, the estimated output elasticities and returns to scale are much lower compared to my baseline estimates, as shown in Table 18. This leads to markup measures that fall below 1 for most firms in my sample. Because the BB method requires first-differencing input and output data (and their lags), it significantly reduces variation in the data that can be used for identification since cross-sectional variation in factor inputs are differenced away. Further, attenuation bias due to measurement error is exacerbated in dynamic panel methods. In a Monte Carlo simulation comparing control function methods and dynamic panel methods when the true production function has constant returns to scale, [Yeh et al. \(2022\)](#) also find that dynamic panel methods tend to estimate far lower returns to scale. The practical difficulty

of obtaining reasonable estimates of production function parameters using dynamic panel methods is also discussed in [De Loecker and Syverson \(2021\)](#).

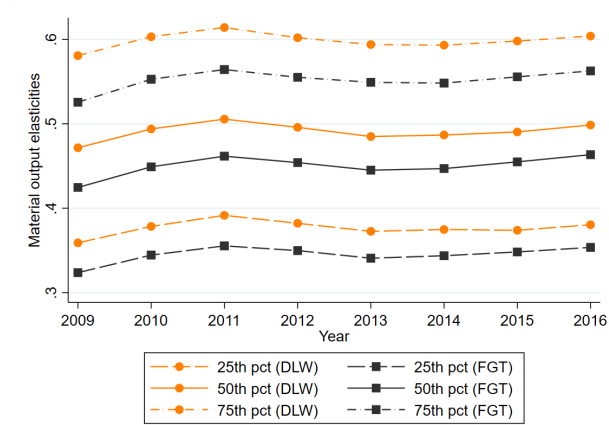


Figure 15: Time-series of material output elasticities under DLW and FGT.

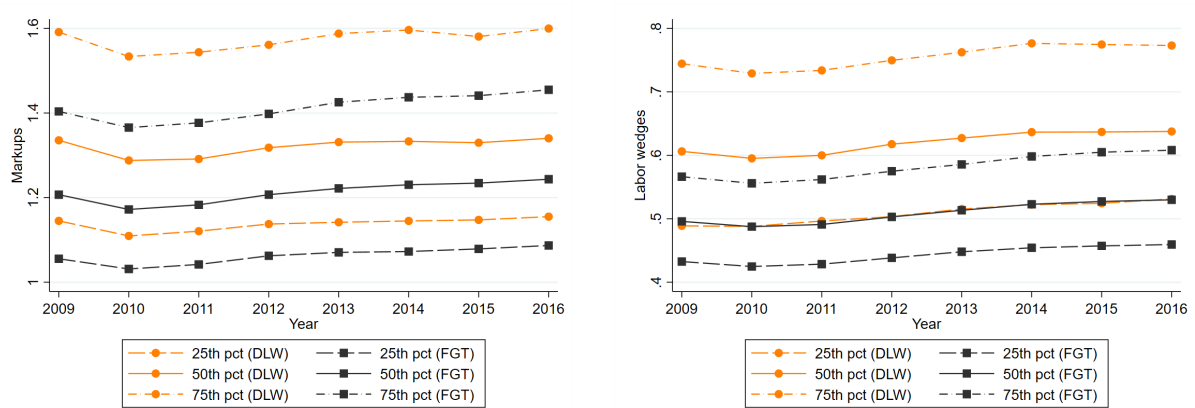


Figure 16: Time-series of markups and labor wedges under DLW and FGT.

B.6 On the role of data on hours and prices

The French matched employer-employee data (DADS) and production survey of manufacturing firms (EAP) offer three key pieces of information: worker-level hours worked, worker mobility between employers, and output prices. Data on hours worked allows me to measure labor inputs as total hours. Data on workers' employer switches allows me to measure the total effective hours at a firm as the difference between the firm-level average hourly wage and firm wage premium (see Section 2). Data on output prices allows me to measure output in quantities instead of revenues, to address unobserved input price heterogeneity stemming from input quality differences (De Loecker et al., 2016), as well as to include controls for markups and output prices in the productivity control function (an issue that arises when price data is not available (Bond et al., 2021)). I compare the labor wedges and markups that I obtain under different specifications or measurements of labor inputs and output and arrive at two main findings.

1. Role of hours data: I find that measuring labor as total hours (or effective hours) instead of employment results in a lower *level* and smaller *dispersion* of labor wedges and markups. This finding may be useful to keep in mind when interpreting production-based labor wedge and markup measures, particularly in light of an important insight of recent macroeconomic models of labor and product market power: the level of labor wedges or markups act as a uniform proportional tax on firms, while the dispersion leads to resource misallocation across firms (Berger et al., 2022; Edmond et al., 2023).

Rows (1), (2), and (3) of Table 19 present the results when I estimate a basic specification that *does not* account for output price bias, input price bias, or markups in the productivity control function—only the measure of labor inputs differs. The mean and median labor wedge is lower when labor inputs are measured as total hours or total effective hours. The interquartile range and 90-10 difference for both labor wedges and markups are also significantly smaller when hours data is used. Rows (7), (8), and (9) show that the same findings emerge in my baseline specification, which addresses output price bias, input price bias, and the presence of markups and output prices in the productivity control function.

To see how failing to account for variation in labor hours across firms could bias labor wedge and markup estimates, consider the following example. For brevity, I drop all firm and time subscripts. Let l , h , and $\bar{h}$ denote employment, total hours, and average hours in logs at a given firm. Suppose the production function consists of only labor and materials (m):

$$y = f(m, h) + \omega = f(m, l + \bar{h}) + \omega$$

I now discuss two sources of bias when data on hours is not available.

OMITTED VARIABLE BIAS. This equation shows that measuring labor inputs as employment instead of total hours leads to average hours ($\bar{h}$) being an omitted variable.

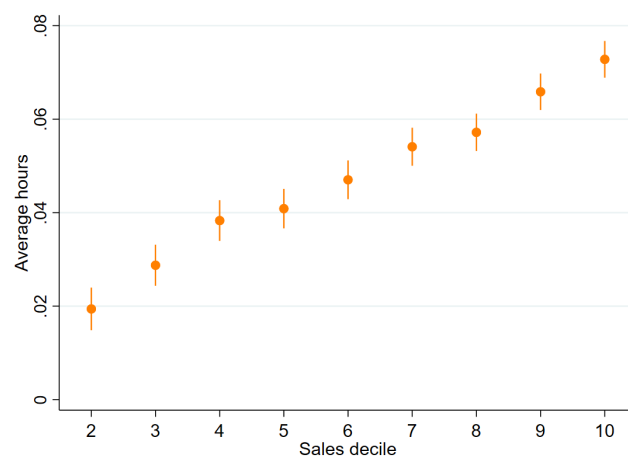


Figure 17: Average hours (in logs) by firm size.

Note: This figure presents how log average hours vary across firm size deciles, controlling for 2-digit sector×year fixed effects. Vertical bars are 95% confidence intervals.

If workers at larger firms work longer hours, then the production function parameter estimates (and returns to scale) will be biased upwards. Figure 17 shows that, within two-digit manufacturing sectors, average hours are higher at larger firms. In this case, material output elasticities will be upward-biased, leading to upward-biased markups. However, labor output elasticities will also be upward-biased, so it is ambiguous in which direction measured labor wedges will be biased. When I use employment instead of total effective hours, I find a 0.02 increase in the median returns to scale.

MISMEASUREMENT. When output elasticities depend on labor inputs, mismeasurement of labor inputs will also bias the output elasticities, even if production function parameters are known. The bias in measured output elasticities then lead to biased labor wedges and markups. Consider the following more specific example. Let $f(m, h)$ be translog with constant returns to scale:

$$y = \beta m + (1 - \beta)h - \frac{\sigma_{mh}}{2}(m - h)^2 + \omega$$

where σ_{mh} is the elasticity of substitution between materials and labor. Materials and labor are substitutes when $\sigma_{mh} < 0$, and complements when $\sigma_{mh} > 0$. Suppose the production function parameters (β, σ_{mh}) , TFPQ (ω), materials (m), and employment (l) are known, but not average hours $\bar{h}$. Let measured output elasticities be denoted with a hat. Then, the relationship between actual and measured output elasticities are:

$$\begin{aligned} \text{Materials: } \hat{\alpha}_m &= \alpha_m + \sigma_{mh}\bar{h} \\ \text{Labor: } \hat{\alpha}_h &= \alpha_h - \sigma_{mh}\bar{h} \end{aligned}$$

where $\sigma_{mh}\bar{h}$ is the source of the mismeasurement. In this case, the returns to scale is not affected. However, the sign of the bias on materials and labor output elasticities go in opposite directions and depends on the elasticity of substitution between materials and

labor. Suppose materials and labor are complements ($\sigma_{mh} > 0$). Then, measured material output elasticities will be upward-biased, while measured labor output elasticities will be downward-biased. In turn, this implies that both measured markups ($\hat{\mu}$) and labor wedges ($\hat{\Lambda}$) will be biased upwards—recall that $\hat{\mu} = \hat{\alpha}_m \frac{PY}{P_m M}$ and $\hat{\Lambda} = \frac{\hat{\alpha}_m}{\hat{\alpha}_h} \frac{WL}{P_m M}$. These biases are consistent with the results from comparing rows (1), (2), and (3), as well as the comparison of rows (7), (8), and (9).

2. Role of output price data: I find that output and input price biases appear to roughly offset each other. Specifications that address neither of the biases that stem from unobserved output nor input prices deliver similar labor wedges and markups as specifications that address both. However, addressing only output price bias without addressing input price bias leads to biased labor wedges and markups.

I start with the comparison between row (3) and row (9). In row (3), I implement a “basic” version of production function estimation that *does not* address potential output and input price bias, or the presence of markups and output prices in the productivity control function. In contrast, the baseline implementation in row (9) addresses all three potential sources of bias. In both rows (3) and (9), I measure labor inputs as total effective hours. The measured labor wedges and markups in both implementations are similar in levels and dispersion.

However, comparing row (3) with row (4), where the only difference is that row (4) measures output in quantity terms, leads to different levels and dispersion of markups and labor wedges. Correcting for output price bias appears to lead to lower and more dispersed markups, and higher and more dispersed labor wedges. In particular, correcting only for output price bias increases the median labor wedge from 0.62 to 0.73. Comparing row (4) with row (5) shows that controlling for output prices and markup proxies (market shares, firm age, and export status) in the productivity control function does not materially change the measured labor wedges. However, comparing row (4) with row (6) shows that controlling for unobserved input price variation using output prices, following [De Loecker et al. \(2016\)](#), returns similar labor wedges and markups as the “basic” and baseline implementations in rows (3) and (9).

Why might addressing the output price bias only, but not the input price bias, lead to lower markups and higher labor wedges? In what follows, I show that this could be the case when these two sources of bias offset each other, a point discussed in detail in [De Loecker and Goldberg \(2014\)](#) and [De Loecker and Syverson \(2021\)](#). Consider again the production function:

$$\begin{aligned} y &= f(m, h) + \omega \\ \Rightarrow \tilde{y} &= f(\tilde{m} - p_m, h) + \omega + p \end{aligned}$$

where $\tilde{y}$ and $\tilde{m}$ represent log sales and log material expenditure. Let both output prices $p = p(d)$ and material prices $p_m = p_m(d)$ be a function of output quality d .

OUTPUT PRICE BIAS. Suppose productivity, material quantities, and labor are observed, but not output prices p . Then, the production function estimates are *upward-biased* if output quality d is not perfectly correlated with productivity ω . This is because, all else equal, firms with higher output quality attract higher demand, charge higher prices, and demand more inputs, resulting in a positive correlation between input choices and unobserved output prices.

INPUT PRICE BIAS. Suppose productivity, output quantities, and labor are observed, but not material prices p_m . Then, the production function estimates are *downward-biased* if output quality d is not perfectly correlated with productivity ω . This is because, all else equal, firms with higher output quality attract higher demand and purchase higher-quality material inputs. This generates a positive correlation between material choices and unobserved material prices.

Indeed, in the basic specification that does not address any of the potential sources of bias (row (3)), I find a median returns to scale of 1.01, which reduces to 0.96 when I address only the output price bias (row (4)). This suggests the presence of input price bias. Although there is no reason ex-ante to expect the output and input price biases to offset each other, the comparison between rows (3), (4), (5), (6), and (9) appear to support this interpretation.

Table 19: Measures of markups and labor wedges with and without data on hours and prices.

Specification	h measure	y measure	Markups				Labor wedges			
			Mean	Median	75-25	90-10	Mean	Median	75-25	90-10
(1) Basic	Emp.	Sales	1.45	1.37	0.48	0.99	0.74	0.71	0.34	0.68
(2) Basic	Hours	Sales	1.41	1.34	0.46	0.93	0.66	0.64	0.26	0.54
(3) Basic	E. hours	Sales	1.37	1.31	0.44	0.88	0.63	0.62	0.23	0.48
(4) Basic	E. hours	Quantity	1.37	1.29	0.51	1.13	0.79	0.73	0.46	1.01
(5) Markup+price in control function	E. hours	Quantity	1.35	1.28	0.51	1.11	0.77	0.71	0.47	1.04
(6) Input price controls only	E. hours	Quantity	1.36	1.32	0.43	0.89	0.62	0.62	0.25	0.50
(7) Baseline	Emp.	Quantity	1.44	1.37	0.50	1.01	0.73	0.71	0.35	0.72
(8) Baseline	Hours	Quantity	1.41	1.35	0.47	0.96	0.64	0.62	0.28	0.55
(9) Baseline	E. hours	Quantity	1.37	1.32	0.43	0.89	0.63	0.62	0.25	0.50

This table reports the measured markups and labor wedges when production functions are estimated with and without data on hours and/or output prices. All markup and labor wedge measures are trimmed by 1% on either side of their distributions. The ‘baseline’ specification refers to the specification estimated in Section 3, which includes output prices and controls for markups in the control function, and uses output prices to address unobserved quality-driven input price variation (De Loecker et al., 2016). The ‘basic’ specification neither controls for markups and output prices in the control function, nor addresses unobserved input quality differences. In row (5), the specification ‘Markup+price in control function’ includes output prices and controls for markups (i.e., market shares, firm age, export status) in the control function, but does not address input price bias. In row (6), the specification ‘Input price controls only’ addresses both input and output price biases, but does not address the presence of markups in the control function, leading to violations of the scalar unobservable assumption. The columns ‘ h measure’ and ‘ y measure’ show the measure of labor and output used in the estimated production function. ‘E. hours’ means effective hours.

B.7 Multiplication bias

The key estimating equation (3') for worker bargaining power requires first measuring labor wedges (Λ) and product market rents ($\tilde{\mu}$). Measuring them requires data on material expenditure as well as estimates of material and labor output elasticities (α_m, α_h). Because these variables appear on both sides of the estimating equation multiplicatively, any measurement error or estimation bias in these variables will be positively correlated and, therefore, lead to a mechanical upward bias of the bargaining power estimate $\hat{\kappa}$. I refer to this bias on $\hat{\kappa}$ as *multiplication bias*.

There are two reasons why the output elasticities could be measured with error, conditional on the production function specification. First, material and labor output elasticities are functions of factor inputs, and these inputs could be measured with error. Second, the parameters governing the material and labor output elasticities could be estimated with a bias due to the difficulties of identifying the flexible input's output elasticity (Gandhi et al., 2020).

To see how biased or mismeasured output elasticities can affect the $\hat{\kappa}$, I return to the example in Section 5.4. As a reminder, I denote measured or estimated parameters or variables with a hat. Let the true markdown be constant and homogenous, $\lambda_{jt} = \lambda$. Let ϵ_{jt}^{bias} be the source of bias (such as measurement error or estimation biases), such that $\frac{\hat{\alpha}_{m,jt}}{\hat{\alpha}_{h,jt}} = \frac{\alpha_{m,jt}}{\alpha_{h,jt}} \epsilon_{jt}^{bias}$. Let $\tilde{\epsilon}_{jt}$ be a random error uncorrelated with $\hat{\mu}_{jt}$. The estimating equation becomes:

$$\underbrace{\frac{\Phi_{jt} H_{jt}}{P_{m,t} M_{jt}} \frac{\hat{\alpha}_{jt}^m}{\hat{\alpha}_{jt}^h}}_{\hat{\Lambda}_{jt}} = \underbrace{\kappa \left(\frac{P_{jt} Y_{jt} - P_{m,t} M_{jt} - P_{k,t} K_{jt}}{P_{m,t} M_{jt}} \right)}_{\hat{\mu}_{jt}} \frac{\hat{\alpha}_{jt}^m}{\hat{\alpha}_{jt}^h} + (1 - \kappa) \lambda + \tilde{\epsilon}_{jt}$$

Then, an OLS regression of $\hat{\Lambda}_{jt}$ on $\hat{\mu}_{jt}$ will give an upward-biased estimate of κ :

$$\hat{\kappa}^{OLS} = \kappa + (1 - \kappa) \frac{CV(\lambda \epsilon_{jt}^{bias}, \tilde{\mu}_{jt} \epsilon_{jt}^{bias})}{V(\tilde{\mu}_{jt} \epsilon_{jt}^{bias})} \geq \kappa, \text{ where } \kappa \in [0, 1]$$

Therefore, the upward bias is larger when the true κ is smaller, and one may estimate positive worker bargaining power even when $\kappa = 0$.⁴⁸

In the rest of this section, I discuss several attempts to address this concern, depending on the source of the bias (ϵ^{bias}). First, under a classical measurement error assumption, I obtain an upper bound estimate of the measurement error in $(\hat{\alpha}_m/\hat{\alpha}_h)$ and simulate the extent of bias on $\hat{\kappa}$. Second, I follow a well-established tradition of using instrumental

⁴⁸Here, I have assumed that $\lambda_j = \lambda$, so there is only one source of bias—the multiplication bias, which biases $\hat{\kappa}$ upwards. If λ_j is endogenous and depends on firm size and amenities, then monopsony markdowns bias $\hat{\kappa}$ downwards when firms with high product market rents (higher $\tilde{\mu}_j$) also have more monopsony power (lower λ_j). While the two sources of bias work in opposite directions, there is no reason *a priori* to expect them to cancel out.

variables to address measurement error. In the same simulation, I show that instrumenting product market rents using their lags ($\hat{\mu}_{jt-1}$) recovers the true κ . I therefore also report estimates when product market rents are instrumented using their lags. However, this lags-as-IV strategy does not address the multiplication bias when ϵ^{bias} are serially correlated. This is the case when the parameters governing the output elasticities are estimated with bias. This is a concern since the production function estimation literature has shown that identifying the output elasticities of flexible inputs is not trivial (see, for example, [Bond and Söderbom \(2005\)](#) and [Gandhi et al. \(2020\)](#)). To address this concern, I propose an alternative estimating equation that avoids having to first estimate output elasticities. However, as I explain later in this section, this alternative requires approximating a more complex, potentially highly nonlinear function.

Measurement error and instrumental variables. Assuming that the source of bias on $\hat{\kappa}$ is classical measurement error in the output elasticities, I now use the panel structure of the data to infer the size of the measurement error. I then use this variance to simulate the likely extent of bias for a given distribution of $\tilde{\mu}_{jt}$, and known λ and κ .

Let the log of the ratio of material-to-labor output elasticities be $x_{jt} \equiv \log \frac{\alpha_{m,jt}}{\alpha_{h,jt}}$, and let it be measured with error: $\hat{x}_{jt} = x_{jt} + \epsilon_{jt}^{me}$, where $\hat{x}_{jt}$ is the observed measure and $\epsilon_{jt}^{me} = \log \epsilon_{jt}^{me}$ is classical measurement error. Following [Krueger and Summers \(1988\)](#), I compute the within-firm variance as $V_j(\hat{x}_{jt}) = V_j(x_{jt}) + V(\epsilon_{jt}^{me})$. Suppose that x_{jt} is highly persistent (or fixed) over time, so that $V_j(x_{jt}) \approx 0$. Then, taking the average of the firm-level variance gives $E[V_j(\hat{x}_{jt})] = V(\epsilon_{jt}^{me})$. This provides an upper bound estimate of the extent of measurement error, because any within-firm variation in x_{jt} will be attributed to $V(\epsilon_{jt}^{me})$.

Given a measure of the size of the measurement error, I assess the potential size of the multiplication bias. I simulate the relationship between labor wedges and product market rents for a known κ and estimate κ with and without the presence of measurement error. For this purpose, I continue to assume that monopsony markdowns are constant and homogenous across firms, and calibrate it to $\lambda = 0.5$. I then take the estimated distribution product market rents as the true distribution and compute the implied Λ_{jt} under different values of κ . Next, I draw a sequence of ϵ_{jt}^{me} from $\mathcal{N}(0, V(\epsilon_{jt}^{me}))$, where $V(\epsilon_{jt}^{me})$ is estimated from the previous step, and multiply $\tilde{\mu}_{jt}$ and Λ_{jt} by ϵ_{jt}^{me} . I then compare the difference between the true κ and those estimated by OLS (with and without the presence of measurement error). Without measurement error, the OLS estimator for κ is unbiased under the assumption of a constant and homogenous λ .

Table 20 compares the true κ and the OLS estimate of κ for different sizes of measurement error. The estimate of the size of measurement error following [Krueger and Summers \(1988\)](#) is $V(\epsilon^{me}) = 0.084$. For measurement error of this size, panel (a) shows that the OLS estimate of κ is close to the true κ , with an upward bias of around 0.005. Panels (b) and (c) show that for much larger measurement error the bias is also much

larger, although the size of the bias declining in the level of κ .

Table 20: Effect of measurement error on estimated worker bargaining power.

	True κ						
	0.05	0.10	0.15	0.20	0.25	0.30	0.35
Panel (a)	$V(\epsilon^{me})=0.084$						
OLS	0.055	0.105	0.154	0.204	0.254	0.304	0.353
IV	0.050	0.100	0.150	0.200	0.250	0.300	0.350
Panel (b)	$V(\epsilon^{me})=0.20$						
OLS	0.075	0.124	0.173	0.221	0.270	0.319	0.367
IV	0.049	0.100	0.150	0.200	0.250	0.300	0.350
Panel (c)	$V(\epsilon^{me})=0.30$						
OLS	0.099	0.146	0.194	0.242	0.287	0.336	0.383
IV	0.051	0.100	0.149	0.200	0.250	0.300	0.350

This table compares the true and estimated workers' bargaining power parameter in the presence of measurement error in material expenditures or output elasticities.

A well-established approach to addressing measurement error in the explanatory variable is to use instrumental variables (IV). The instrument needs to be correlated with the explanatory variable and be orthogonal to the measurement error. Under a classical measurement error assumption, product market rents can be instrumented using their lags. Table 20 compares the OLS and IV estimates of κ , where the IV is lagged product market rent. In all three panels, the instrument successfully recovers the true κ .

An alternative estimating equation for worker bargaining power. The strategy above of using lagged product market rent as an IV to address the multiplication bias induced by measurement error relies on the assumption of classical measurement error. This is a strong assumption; if the production function parameters are not identified, then the bias in the estimated output elasticities will be serially correlated. This causes $\hat{\kappa}$ to be upward-biased and lagged product market rent is no longer a valid instrument. To avoid having to first estimate the output elasticities (α), I propose using an alternative estimating equation to (3'). I rearrange the baseline estimating equation (3') to obtain:

$$\begin{aligned} \frac{\Phi_{jt}H_{jt}}{P_{m,t}M_{jt}} = & \kappa \left(\frac{P_{jt}Y_{jt} - P_{m,t}M_{jt} - P_{k,t}K_{jt}}{P_{m,t}M_{jt}} \right) \\ & + (1 - \kappa)\lambda(H_{jt}, \mathcal{H}^{-1}(H_{jt}, \Phi_{jt}H_{jt})) \frac{\alpha_h(K_{jt}, M_{jt}, H_{jt})}{\alpha_m(K_{jt}, M_{jt}, H_{jt})} \end{aligned} \quad (13)$$

Compared to equation (3'), estimating equation (13) does not require estimates of output elasticities. However, equation (13) includes a more complicated term to approximate $(\lambda(\cdot)\alpha_h(\cdot)/\alpha_m(\cdot))$.⁴⁹ In practice, I approximate $\lambda(\cdot)\alpha_h(\cdot)/\alpha_m(\cdot)$ with a fourth-order poly-

⁴⁹Note that, if the production function were Cobb-Douglas, then we need only approximate $\lambda(\cdot)$, which would make equation (13) more attractive than equation (3').

nomial. Similar to the baseline equation (3'), this alternative equation is also not immune to measurement-error-induced multiplication bias because material expenditure and wage bills appear on both sides of the equation and these may be measured with error. Therefore, when estimating κ using equation (13), I also instrument all right-hand-side variables using their lags under a classical measurement error assumption.

Table 21 presents the bargaining power estimates from implementing equation (13). Columns (1) through (3) show that controlling for the term $\lambda(\cdot)\frac{\alpha_h(\cdot)}{\alpha_m(\cdot)}$ and firm fixed effects are important, as they substantially reduce the estimated bargaining power. Indeed, column (4) shows that a specification that controls for both $\lambda(\cdot)\frac{\alpha_h(\cdot)}{\alpha_m(\cdot)}$ and firm fixed effects gives an estimated κ of 0.134, close to my baseline estimate of 0.124 in column (4) of Table (4). Column (6) shows that additionally using lagged left-hand-side variables to address multiplication bias due to classical measurement error further reduces the estimated κ to 0.111.

Table 21: Estimated worker bargaining power using an alternative equation.

	$\frac{\Phi_{jt}H_{jt}}{P_{m,t}M_{jt}}$					
	(1)	(2)	(3)	(4)	(5)	(6)
$\frac{P_{jt}Y_{jt}-P_{m,t}M_{jt}-P_{k,t}K_{jt}}{P_{m,t}M_{jt}}$	0.344	0.266	0.196	0.134	0.221	0.111
	(0.003)	(0.005)	(0.004)	(0.006)	(0.001)	(0.014)
Sector×year fixed effects	✓	✓	✓	✓	✓	✓
Firm fixed effects		✓		✓		✓
Control for $\lambda(\cdot)\frac{\alpha_h(\cdot)}{\alpha_m(\cdot)}$			✓	✓	✓	✓
IV for measurement error					✓	✓
Observations	126,836	122,239	126,836	122,239	99,238	95,666

This table reports the estimated worker bargaining power parameter using the alternative estimating equation (13). When controlling for the term $\lambda(\cdot)\frac{\alpha_h(\cdot)}{\alpha_m(\cdot)}$, I use a second-order polynomial of its arguments: capital, materials, effective labor, and wage bills. Bootstrapped standard errors are reported in parentheses. In columns (5) and (6), all left-hand-side variables are instrumented with their lags.

Taken together, these findings suggest that measurement error in factor inputs and estimation biases due to challenges in identifying production function parameters do not result in significant upward bias in my estimates of worker bargaining power. Nevertheless, it is useful to compare how measured monopsony markdowns depend on the value of inferred worker bargaining power.

Measured monopsony markdowns under different values of worker bargaining power. To compare measured markdowns under different values of κ , I start by deriving an upper and lower bound for κ . Given the estimated $\tilde{\mu}_{jt}$ and Λ_{jt} , we can obtain upper and lower bounds for κ by noting that $\lambda_{jt} \in [0, \min\{1, \Lambda_{jt}\}]$. The labor wedge equation

(3) can be rearranged as follows:

$$\kappa = \frac{\Lambda_{jt} - \lambda_{jt}}{\tilde{\mu}_{jt} - \lambda_{jt}}$$

In this expression, κ is decreasing in λ . Since the maximum value λ_{jt} can take is Λ_{jt} , the lower bound for κ is 0. Since the minimum value λ_{jt} can take is 0, the upper bound for κ is $\frac{\Lambda_{jt}}{\tilde{\mu}_{jt}}$. Because firms are assumed to have the same bargaining parameter, I set the upper bound as the level of κ such that $\lambda_{jt} = 0$ for at most 10% for firms. The bounds on worker bargaining power are then $\kappa \in [0, 0.27]$.

Table 22 shows the summary statistics of measured markdowns for different values of κ . The table shows that the mean and median markdowns decrease steeply with κ . Therefore, the larger the estimated κ is, the greater the labor market power firms are inferred to have.

Table 22: Measured monopsony markdowns under different κ .

κ	Measured λ				
	0.05	0.10	0.15	0.20	0.25
Mean	0.57	0.49	0.41	0.31	0.20
Median	0.57	0.50	0.42	0.33	0.24
25 th percentile	0.49	0.41	0.31	0.21	0.08
75 th percentile	0.66	0.59	0.52	0.44	0.36
75-25 difference	0.17	0.18	0.21	0.23	0.28

This table compares the distribution of measured monopsony markdowns under different values of worker bargaining power κ .

How does the value of κ affect the correlation between measured markdowns and firm characteristics? The first panel of Figure 18 shows that the correlation between markdowns and firm wage premia is relatively flat across different values of κ , although this correlation flips from positive to negative as κ increases. The second panel of Figure 18 shows that measured markdowns decline with firm size for each value of κ , although the relationship between monopsony markdowns and firm size is also relatively flat.

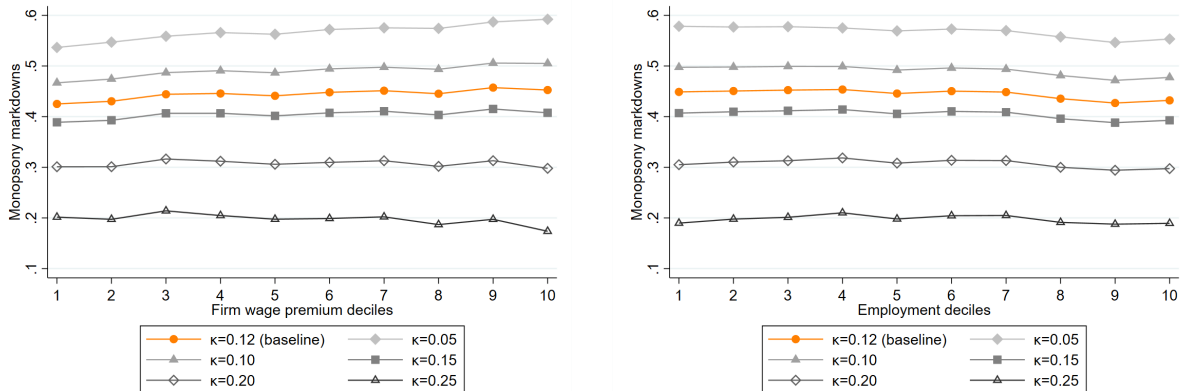


Figure 18: Implied monopsony markdowns by firm wage premia and firm size.

B.8 Labor adjustment costs

The estimated labor wedge $\hat{\Lambda}_{jt}$ could capture labor adjustment costs. I now derive labor wedges in the presence of labor adjustment costs and discuss how labor adjustment costs could bias the estimated bargaining power $\hat{\kappa}$ and the inferred monopsony markdown $\hat{\lambda}_{jt}$. I then attempt to assess the magnitude of this source of bias and incorporate labor adjustment costs in my estimation of κ . In this section, I will use the term ‘employment’ to mean effective labor (H_{jt}).

Specification of labor adjustment costs. There are two alternatives—convex and non-convex cost functions—and they have different implications for the observed distribution of employment growth. The former implies that employment growth rates should be smooth and centered around zero. The latter implies that employment growth rates should be multimodal—most firms do not adjust employment, but when they do, they tend to make lumpy adjustments. In the French data, employment growth rates are smoothly distributed around zero (see Figure 19), hence I specify convex labor adjustment costs. Specifically, I work with quadratic adjustment costs in the derivations of labor wedges below.

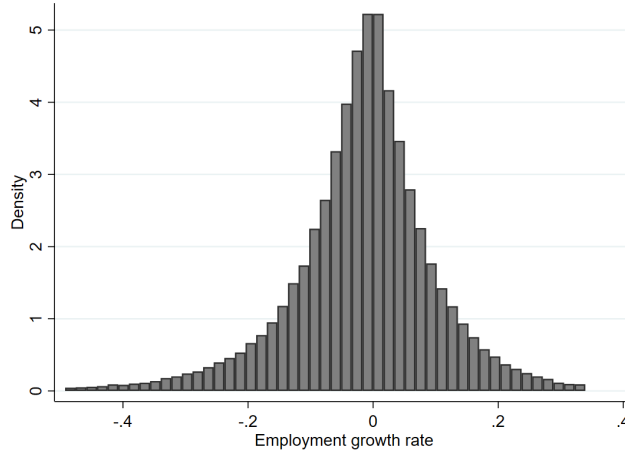


Figure 19: Distribution of employment growth rates.

A deterministic model of firm growth with labor adjustment costs. I now use a stylized model to show that labor adjustment costs: (i) imply that labor wedges depend on firms’ current and future employment growth rates; (ii) imply that one will find $\Lambda_j \in [0, 1]$ even if product and labor markets were perfectly competitive; (iii) disappear as firms grow towards their optimal sizes—that is, Λ_j grows as firms grow.

For simplicity, suppose that firms face a constant elasticity goods demand function $P_{jt} = Y_{jt}^{-1/\sigma}$, a constant elasticity labor supply curve $\Phi_{jt} = H_{jt}^{1/\eta}$, and operate a production technology that uses only labor inputs $Y_{jt} = \Omega_j H_{jt}$. Further, assume that firm productivity Ω_j is heterogeneous across firms but constant over time. Consider the following wage bargaining problem:

$$\max_{H_{jt}} \left(\Phi_{jt} H_{jt} \right)^\kappa \left(V(\Omega_j, H_{jt-1}) - V^o(\Omega_j, H_{jt-1}) \right)^{1-\kappa} \quad (14)$$

subject to the labor supply curve and production technology. The firm's surplus is $V(\Omega_j, H_{jt-1}) - V^o(\Omega_j, H_{jt-1})$. If workers and firms reach an agreement on the terms of employment, the value function of the firm is:

$$V(\Omega_j, H_{jt-1}) = P_{jt} Y_{jt} - \Phi_{jt} H_{jt} - C(H_{jt}, H_{jt-1}, \Phi_{jt}) + \beta V(\Omega_j, H_{jt})$$

where β is the discount factor and the quadratic adjustment costs are $C(H_{jt}, H_{jt-1}, \Phi_{jt}) = \frac{\gamma}{2} \left(\frac{H_{jt}}{H_{jt-1}} - 1 \right)^2 \Phi_{jt} H_{jt}$. If workers and firms do not arrive at an agreement, the firm's value function becomes:

$$V^o(\Omega_j, H_{jt-1}) = \beta V(\Omega_j, H_{jt}^o)$$

where H_{jt}^o are the workers that firms carry over from time t to $t+1$ if no agreement was reached at t . I assume that $H_{jt}^o = H_{jt-1}$, the stock of workers available to the firm at $t+1$ if workers and firms do not agree at t is equal to employment at $t-1$, so no new hires are added at t . Workers supply no labor at t if an agreement is not reached at t , and so there is no production, no wage expenditure, and no adjustment cost in that time. The FOC with respect to labor combined with the envelope theorem gives:

$$\Phi_{jt} = \frac{\mu^{-1} \Omega_j^{\frac{\sigma-1}{\sigma}} H_{jt}^{-\frac{1}{\sigma}} + \beta \frac{V_{jt} - V_{jt}^o}{H_{jt}}}{\left[\left(1 + \frac{\gamma}{2} g_{h,jt}^2 \right) + (1 - \kappa) \lambda \left(\gamma g_{h,jt} (1 + g_{h,jt}) - \beta \gamma g_{h,jt+1} (1 + g_{h,jt+1})^2 (1 + g_{\Phi,jt+1}) \right) \right]}$$

where the firm's surplus can be written as:

$$V_{jt} - V_{jt}^o = P_{jt} Y_{jt} - \Phi_{jt} H_{jt} - C(H_{jt}, H_{jt-1}, \Phi_{jt}) + \beta [V(\Omega_j, H_{jt}) - V(\Omega_j, H_{jt}^o)]$$

To simplify the FOC, I approximate the last term in the firm's surplus as follows:

$$V(\Omega_j, H_{jt}) - V(\Omega_j, H_{jt}^o) \approx \frac{\partial V(\Omega_j, H_{jt}^o)}{\partial H_{jt}^o} (H_{jt} - H_{jt}^o) = \gamma g_{h,jt+1} (1 + g_{h,jt+1})^2 \Phi_{jt+1} (H_{jt} - H_{jt-1})$$

where the last equality uses the envelope condition and the assumption that $H_{jt}^o = H_{jt-1}$. We can now define the labor wedge as the ratio of firm-specific wages over their marginal revenue product of labor:

$$\Lambda_{jt} \equiv \frac{\Phi_{jt}}{MRPH_{jt}} = \frac{\kappa \mu + (1 - \kappa) \lambda}{\Gamma_{jt}} \quad (15)$$

where the wedge induced by labor adjustment costs (Γ_{jt}) can be written as:

$$\Gamma_{jt} \equiv \left(1 + \frac{\gamma}{2}g_{h,jt}^2\right) + (1 - \kappa)\lambda \left(\gamma g_{h,jt}(1 + g_{h,jt}) - \beta \gamma g_{h,jt+1}(1 + g_{h,jt+1})^2(1 + g_{\Phi,jt+1}) \right) - \kappa \gamma \beta g_{h,jt+1}(1 + g_{h,jt+1})^2(1 + g_{\Phi,jt+1})g_{h,jt}$$

The presence of adjustment costs imply that the labor wedge depends on current and future employment growth rates.

How will the labor wedge evolve over the firm's lifecycle? Suppose all firms begin their life with one employee. Due to convex adjustment costs, firms cannot instantaneously achieve their optimal size, but must grow towards it gradually. I now solve the firm's problem backwards from time $\bar{T}_j$, defined as the time at which firm j achieves its optimal steady state size. With this understanding, I now drop the subscript j for brevity. At time $t = \bar{T}$, firm size is such that $H_{t+1} = H_t = H_t^o$; firms no longer grow, and the labor wedge becomes $\bar{\Lambda} = \kappa\mu + (1 - \kappa)\lambda$. At time $\bar{T} - 1$, just before the firm achieves its optimal size, the labor wedge is:

$$\Lambda_{\bar{T}-1} = \frac{\Phi_{\bar{T}-1}}{\mu^{-1}\Omega^{\frac{\sigma-1}{\sigma}}H_{\bar{T}-1}^{-\frac{1}{\sigma}}}$$

Comparing time $\bar{T}$ with $\bar{T} - 1$, we have that:

$$\frac{\Lambda_{\bar{T}}}{\Lambda_{\bar{T}-1}} = \frac{\Phi_{\bar{T}}H_{\bar{T}}^{\frac{1}{\sigma}}}{\Phi_{\bar{T}-1}H_{\bar{T}-1}^{\frac{1}{\sigma}}} = \left(\frac{H_{\bar{T}}}{H_{\bar{T}-1}}\right)^{\frac{1}{\eta} + \frac{1}{\sigma}} \geq 1 \Rightarrow \Lambda_{\bar{T}} \geq \Lambda_{\bar{T}-1}, \text{ since } H_{\bar{T}} \geq H_{\bar{T}-1}.$$

where the second equality uses the labor supply curve. Therefore, the labor wedge at steady state is larger than that one period before the firm achieves its steady state size. Extending this logic further back in time implies that the firm's labor wedge grows towards $\bar{\Lambda} = \kappa + (1 - \kappa)\lambda$ over time as firms grow towards their steady state size.

Main implications. This stylized model provides some guidance on how labor adjustment costs might bias the bargaining power and markdown estimates, and how one might assess the extent to which labor wedges reflect labor adjustment costs:

- (1.) All else equal, older firms have a higher labor wedge than younger firms, because they are closer to their optimal size. As firms grow closer to their optimal size, Λ converges to $\kappa\mu + (1 - \kappa)\lambda$ from below. The extent to which firms' labor wedges rise as they age, conditional on their latent type, therefore contains information about the magnitude of labor adjustment costs in measured labor wedges.
- (2.) Similarly, all else equal, growing firms have a lower labor wedge than non-growing firms (Λ_{jt} further below 1), because they are further from their optimal size.
- (3.) The above stylized example abstracts from variable markups, although my estimation of worker bargaining power relies on the presence of variable markups. To the

extent that firms charge higher markups as they grow larger (towards their optimal size), there will be a positive correlation between markups and measured labor wedges, implying that $CV(\tilde{\mu}_{jt}, (1/\Gamma_{jt})) > 0$. Therefore, not accounting for labor adjustment costs could lead to an *upward bias* in the estimated κ .

- (4.) The presence of labor adjustment costs implies that one would *overstate* firm monopsony power λ for a given labor wedge Λ , markup μ , and bargaining power κ .

Incorporating labor adjustment costs into my baseline model. The previous model is stylized. It assumes that markups (μ) and markdowns (λ) are constant and homogenous, and that labor is the only factor input. I now incorporate labor adjustment costs into my structural framework in Section 2 for guidance on: (i) how to assess the extent to which labor wedges reflect labor adjustment costs; (ii) how to account for labor adjustment costs when estimating bargaining power. I use the same quadratic labor adjustment cost function as the one above. The firm's state variables are now $Z_{jt} = \{\Omega_{jt}, D_{jt}, A_{jt}, H_{jt-1}, K_{jt}\}$, where capital K_{jt} is pre-determined.

Labor wedge. With labor adjustment costs, the labor wedge in my model (see equation (3)) becomes:

$$\Lambda_{jt} = \frac{\kappa \tilde{\mu}_{jt} + (1 - \kappa) \lambda_{jt}}{\Gamma_{jt}} \quad (16)$$

where the wedge induced by labor adjustment costs is:

$$\Gamma_{jt} \equiv \left(1 + \frac{\gamma}{2} g_{h,jt}^2\right) + (1 - \kappa) \lambda \left(\gamma g_{h,jt} (1 + g_{h,jt}) - \beta \gamma E_t \left[g_{h,jt+1} (1 + g_{h,jt+1})^2 (1 + g_{\Phi,jt+1}) \right] \right) \\ - \kappa \gamma \beta E_t \left[g_{h,jt+1} (1 + g_{h,jt+1})^2 (1 + g_{\Phi,jt+1}) \right] g_{h,jt}$$

and $\tilde{\mu}_{jt} \equiv \left(1 - \frac{P_{m,t} M_{jt} + P_{k,t} K_{jt}}{P_{jt} Y_{jt}}\right) \frac{\mu_{jt}}{\alpha_{h,jt}}$ are product market rents.

Estimating equation for bargaining power. To assess whether labor adjustment costs might significantly bias my bargaining power estimates downwards, I derive the estimating equation for worker bargaining power that accounts for labor adjustment costs:

$$1 + \log \Lambda_{jt} \approx \kappa \tilde{\mu}_{jt} + (1 - \kappa) \lambda_{jt} - \log \Gamma_{jt} \quad (17)$$

Compared to the original estimating equation (3'), this new equation includes the "adjustment cost wedge" Γ_{jt} . The derivation of equation (17) relies on the following first-order approximations:

$$\log [\kappa \tilde{\mu}_{jt} + (1 - \kappa) \lambda_{jt}] \approx \kappa \tilde{\mu}_{jt} + (1 - \kappa) \lambda_{jt} - 1, \text{ and}$$

$$\log \Gamma_{jt} \approx \frac{\gamma}{2} g_{h,jt}^2 + (1 - \kappa) \lambda \left(\gamma g_{h,jt} (1 + g_{h,jt}) - \beta \gamma E \left[g_{h,jt+1} (1 + g_{h,jt+1})^2 (1 + g_{\Phi,jt+1}) \middle| Z_{jt} \right] \right) - \kappa \gamma \beta E \left[g_{h,jt+1} (1 + g_{h,jt+1})^2 (1 + g_{\Phi,jt+1}) \middle| Z_{jt} \right] g_{h,jt}$$

I discuss next how I use equation (17) to assess the magnitude of labor adjustment costs and estimate worker bargaining power.

Assessing the extent to which the labor wedge (Λ_{jt}) reflects labor adjustment costs (Γ_{jt}).

In light of the discussion above on how labor wedges are expected to differ by firm age and employment growth rates (see main implications (1.) and (2.) of the deterministic model), I now compare the estimated labor wedges $\log \hat{\Lambda}_{jt}$ along these two observed firm characteristics. Motivated by equation (17), I run the following regression:

$$\log \hat{\Lambda}_{jt} = \text{Category}_{jt} + \mathcal{X}' \mathcal{B} + \varepsilon_{jt}$$

where Category_{jt} refers categories of firm age and employment growth rates to which firm j belongs at time t . Firm age is defined by ten groups of 5-year increments, starting at the age group between 0 to 5 years old. Employment growth rate categories are defined in terms of deciles. The vector of controls $\mathcal{X}$ includes product market rents $\hat{\mu}_{jt}$, firm size as measured by employment and wage bills (implied by $\lambda_{jt} = \lambda(H_{jt}, \mathcal{H}^{-1}(H_{jt}, \Phi_{jt} H_{jt}))$), firm fixed effects, and sector $\times$ year fixed effects. I restrict the sample to a balanced panel of firms that appeared in every year between 2009 and 2016, and on average experienced a positive growth rate in employment in this time frame. I present the findings in Figures 20 and 21. Each figure contains four panels. In panel (a), $\mathcal{X}$ includes firm size controls and sector $\times$ year fixed effects. In panel (b), $\mathcal{X}$ additionally includes product market rents. In panel (c), $\mathcal{X}$ additionally includes firm fixed effects. In panel (d), $\mathcal{X}$ additionally includes both product market rents and firm fixed effects.

Figure 20 presents how labor wedges vary depending on the age of the firm. Panel (a) shows that labor wedges increase with firm age. Panels (b), (c), and (d) show that controlling for product market rents or looking at within-firm variation significantly flattens the profile of labor wedges across firm age. A similar pattern holds when looking at employment growth rates, as Figure 21 shows. When I control for product market rents or look at within-firm variation (or both), the profile of labor wedges along firm employment growth rates largely flattens. Overall, these findings suggest that while labor adjustment costs may be present, they seem unlikely to be the main determinant of labor wedges or to significantly affect the bargaining power estimates. Nevertheless, I now attempt to account for labor adjustment costs when estimating worker bargaining power.

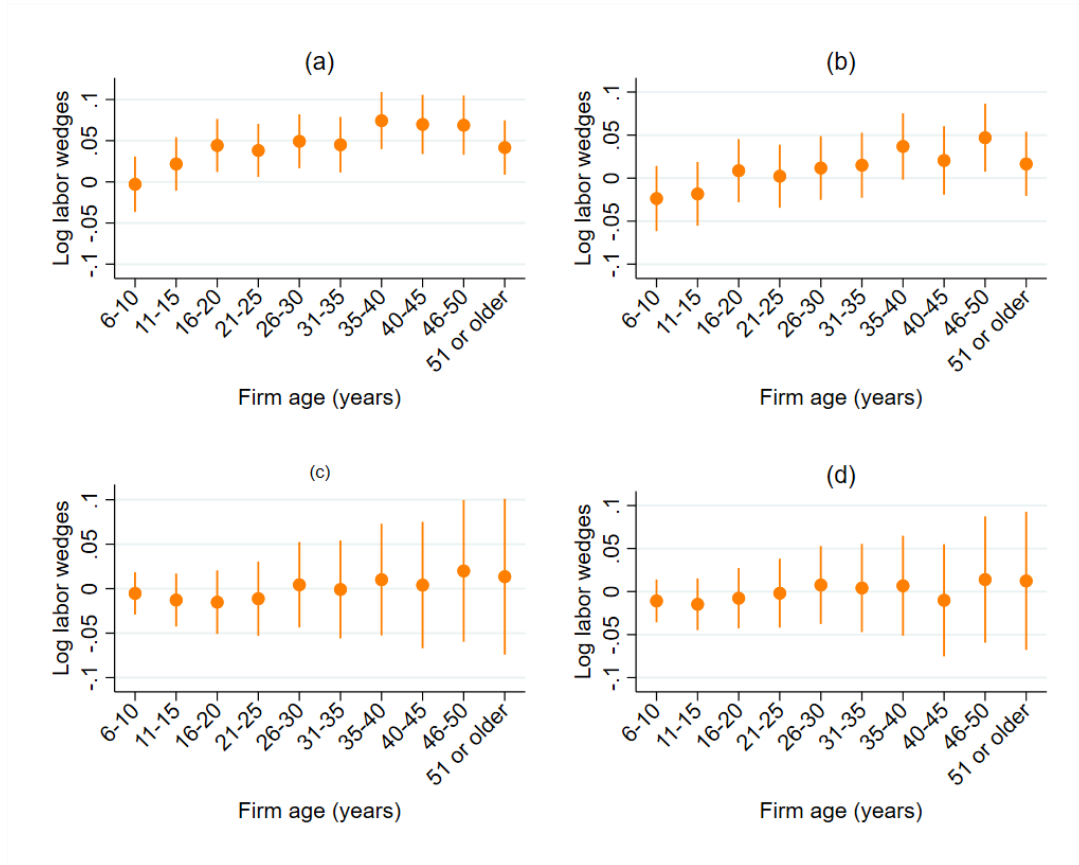


Figure 20: Labor wedges by firm age.

Note: Panel (a) includes firm size controls and sector×year fixed effects. Panel (b) additionally includes product market rents. Panel (c) additionally includes firm fixed effects, but not product market rents. Panel (d) additionally includes both product market rents and firm fixed effects. Vertical bars are 95% confidence intervals.

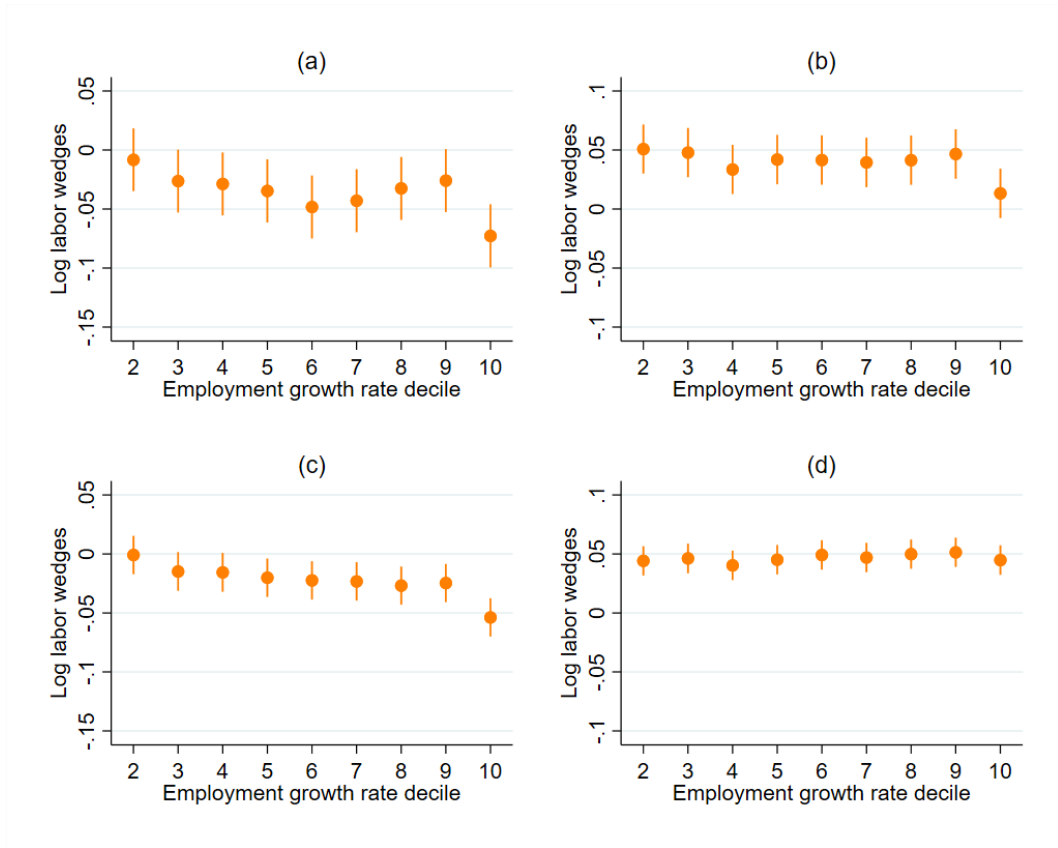


Figure 21: Labor wedges by employment growth rate.

Note: Panel (a) includes firm size controls and sector×year fixed effects. Panel (b) additionally includes product market rents. Panel (c) additionally includes firm fixed effects, but not product market rents. Panel (d) additionally includes both product market rents and firm fixed effects. Vertical bars are 95% confidence intervals.

Accounting for labor adjustment costs when estimating worker bargaining power (κ).

I now adapt my approach (proposed in Section 3) to estimating bargaining power to account for labor adjustment costs using equation (17). The control function approach taken to address unobserved amenities when estimating κ is unchanged—the monopsony markdowns can still be written as a function of firm size in employment and wage bills $\lambda(H_{jt}, \mathcal{H}^{-1}(H_{jt}, \Phi_{jt}H_{jt}))$. However, the presence of the labor adjustment wedge (Γ_{jt}) introduces the following component into equation (17):

$$E \left[\underbrace{g_{h,jt+1}(1 + g_{h,jt+1})^2(1 + g_{\Phi,jt+1})}_{\equiv g_{j,t+1}^{adj}} \middle| Z_{jt} \right]$$

Therefore, the estimation procedure is now complicated by the presence of *expected* future employment and wage growth rates ($g_{h,jt+1}$ and $g_{\Phi,jt+1}$), which are unobserved.

To address unobserved future employment and wage growth rates, I replace the $t + 1$ growth rates expected at time t with the actual growth rates at $t + 1$ in the expectation term above. That is, I compute the realized version of the term in expectations above, $g_{j,t+1}^{adj} \equiv g_{h,jt+1}(1 + g_{h,jt+1})^2(1 + g_{\Phi,jt+1})$. I then note that, under the assumption of rational expectations, expected growth rates ($E[g_{j,t+1}^{adj} | Z_{jt}]$) are on average equal to realized growth rates ($g_{j,t+1}^{adj}$). Under this assumption, deviations of actual growth rates from their expectations can be thought of as random expectation errors, or “measurement error”. This gives a *lower bound* estimate for κ due to attenuation bias.

In an alternative approach to addressing the unobserved expected future growth rates, I project realized growth rates at time $t + 1$, $g_{j,t+1}^{adj}$, on the firm’s state variables Z_{jt} to compute the expected future growth rates at time t , $E[g_{j,t+1}^{adj} | Z_{jt}]$. That is, I use the predicted future growth rates at time t ($\hat{g}_{j,t+1}^{adj}$) in place of the expected growth rates ($E[g_{j,t+1}^{adj} | Z_{jt}]$) in equation (17). However, among the state variables, quality D_{jt} and amenities A_{jt} are unobserved. When computing the predicted future growth rates $\hat{g}_{j,t+1}^{adj}$, I can only project $g_{j,t+1}^{adj}$ on $\hat{\Omega}_{jt+1}$, H_{jt} , and K_{jt+1} . Therefore, to the extent that a higher current quality or amenity raises expected future growth rates, these will be omitted variables that upward bias the estimated κ . This gives an *upper bound* estimate for κ .

Table 23 shows the estimates of worker bargaining power under regression specifications that use realized future growth rates (columns 1 and 2), and those that use predicted future growth rates (columns 3 and 4). Column 2 shows that the point estimate for κ of 0.124 turns out to be the same as my baseline estimate in column 4 of Table 4 of 0.124. Column 4 in Table 23 shows that the estimates are similar when I use predicted future growth rates in place of realized growth rates.

Table 23: Estimates of κ , controlling for quadratic labor adjustment costs.

	Labor wedges (Λ_{jt})			
	(1)	(2)	(3)	(4)
Product market rents ($\tilde{\mu}_{jt}$)	0.121 (0.007)	0.124 (0.022)	0.123 (0.007)	0.125 (0.006)
Controls for $E[g_{j,t+1}^{adj} Z_{jt}]$	Realized	Realized	Predicted	Predicted
Sector $\times$ Year FE	✓	✓	✓	✓
Firm FE		✓		✓
Observations	68,169	65,599	68,169	65,599

This table presents estimates of worker bargaining power (κ) that accounts for quadratic labor adjustment costs. I control for expected future growth rates, $E[g_{j,t+1}^{adj}|Z_{jt}]$, induced by the presence of labor adjustment costs, in two ways, using realized $g_{j,t+1}^{adj}$ (denoted as “Realized”) and using predicted $g_{j,t+1}^{adj}$ (denoted as “Predicted”). Bootstrapped standard errors in parentheses. All regressions in this table include fourth-order polynomials of employment and wage bills to address unobserved amenities, following the control function approach for monopsony markdowns detailed in Section 3.3.

B.9 Estimating labor wedges using hiring wages only

A caveat for the results presented in Section 5 is that firm wage premia and labor wedges are estimated for all workers; both new hires and incumbent workers. As discussed in Section 3, it may be important to allow the wages of incumbent workers to be determined separately from those of new hires. In this section, I estimate labor wedges using hiring wages only, following Di Addario et al. (2020). I do not take a stance on the wage-setting protocol for incumbent workers.

To repeat the main estimation exercise of this paper, I first k-means cluster firms into groups using only hiring wages (W^n) and estimate firm wage premia (ϕ^n). To estimate production functions taking into account differences in worker ability across firms, I then compute the average worker ability at each firm using the following relationship: $W_{jt}^n = \bar{E}_{jt}^n \Phi_{jt}^n$. The rest of the estimation routine is as described in Section 3.

Once production functions and price-cost markups are estimated, labor wedges are measured as follows:

$$\Lambda_{jt}^n = \frac{W_{jt}^n L_{jt}}{P_{jt} Y_{jt}} \cdot \mu_{jt} \cdot \alpha_{h,jt}^{-1}$$

Λ^n represents the labor wedges for new hires.

Table 24 below shows that the estimated price-cost markups are similar to those estimated using all workers in Table 3 in Section 5. However, the estimated labor wedges are lower in this case: new hires are paid a lower share of their marginal revenue product than incumbent workers. This is consistent with the findings of Kline et al. (2019), who show that patent-induced labor productivity shocks pass through to incumbent workers’ wages, but not the wages of new hires.

Table 24: Summary statistics for estimated firm wage premia and labor wedges in 2016 (using hiring wages only).

	Mean	Median	25 th Pct	75 th Pct	Var	Var (i)	Var (ii)
ϕ	2.58	2.58	2.52	2.63	0.01	0.01	0.01
Λ	0.45	0.40	0.30	0.53	0.20	0.05	0.06
μ	1.49	1.38	1.18	1.68	0.07	0.04	0.02
# firms	12,826						

This table reports the summary statistics for estimated firm wage premia and labor wedges using hiring wages only (Di Addario et al., 2020). Variances are reported for the logarithms of those variables. The column Var (i) reports the variances corrected for measurement error following Krueger and Summers (1988) and Kline et al. (2020), while the column Var (ii) reports the variances for firm-groups. Markups and labor wedges are winsorized by 2%.

B.10 Bootstrapped standard errors

In Section 3.3, I use the labor wedge equation (3) to estimate worker bargaining power (κ). Implementing this equation requires measuring labor wedges (λ_j) and product market rents ($\tilde{\mu}_j$), which requires estimating production functions. Tables (4), (21), and (23) therefore report bootstrapped standard errors for the κ estimates. I now describe the bootstrapping procedure I implement to compute those standard errors:

- i. For each two-digit French manufacturing sector, I randomly sample (with replacement) 90% of firms. For each sampled firm, I extract their entire panel in the time span that I observe them (2009-2016).
- ii. I estimate the production function for each two-digit sector using the sampled firms and compute the labor wedges (λ_j) and product market rents ($\tilde{\mu}_j$).
- iii. I then implement the labor wedge equation (3) on only the sampled firms to estimate κ , then compute their λ_j and $\tilde{\mu}_j$.

I repeat steps (i.) to (iii.) 500 times and compute the bootstrapped standard errors.

C Appendix: Theory

C.1 Firm wage premia with outside wages

As before, workers bargain collectively with their employer j and bargaining is efficient: workers and firms jointly choose wages, prices, materials, and capital to maximize total rents, taking into account the product demand curve and labor supply curve. Firms have an outside option of zero profits. The firm-specific labor supply curve is now $H_j = \mathcal{H}(\Phi_j - \Phi^o, A_j)$; workers do not supply labor unless firms pay at least the exogenous outside wage Φ^o . Workers and firms maximize the following Nash product:

$$\max_{\Phi_j, P_j, M_j, K_j} \left((\Phi_j - \Phi^o) H_j \right)^\kappa \left(\Pi_j \right)^{1-\kappa}$$

subject to $H_j = \mathcal{H}(\Phi_j - \Phi^o, A_j)$, $Y_j = \mathcal{G}(P_j, D_j)$, and $Y_j = \Omega_j F(K_j, M_j, H_j)$. The firm's profit is $\Pi_j = P_j Y_j - \Phi_j H_j - P_m M_j - P_k K_j$. The firm-specific wage premium is:

$$\Phi_j = \Phi^o + \underbrace{\kappa \left(\frac{P_j Y_j - P_m M_j - P_k K_j}{H_j} - \Phi^o \right)}_{\text{Total rents per effective labor}} + (1 - \kappa) \underbrace{\left[\lambda_j (MRPH_j - \Phi^o) \right]}_{\text{Monopsony wages}} \quad (18)$$

Equation (18) shows that the firm wage premium is again a weighted average of a pure bargaining outcome and a pure monopsony outcome. When $\kappa = 0$, workers receive a mark down of monopsony rents in addition to the outside wage. The markdown λ_j is determined by the firm-specific labor supply elasticity. In this case, workers do not receive rents from the firm's product market power. When $\kappa = 1$, workers receive the total amount of rents generated by firms' labor and product market power.

Firm wage premia can be written in exactly the same form as in equation (2). The labor wedge Λ_j in this case is:

$$\Lambda_j = \frac{1}{1 - (1 - \kappa)(1 - \lambda_j) \frac{\Phi^o}{\Phi_j}} \left[\kappa \left(1 - \frac{\alpha_{m,j} + \alpha_{k,j}}{\mu_j} \right) \frac{\mu_j}{\alpha_{h,j}} + (1 - \kappa) \lambda_j \right] \quad (19)$$

Labor wages are therefore higher when workers have outside wages, all else equal.

Estimating workers' bargaining power. In Section 3, I proposed a control function approach to address unobserved variation in amenities when estimating bargaining power. The estimating equation assumed that outside wages are zero. I now adjust the estimating equation to account for positive outside wages. The adjusted estimating equation is:

$$\tilde{\Lambda}_{jst} = \kappa \mathcal{M}_{jst} + (1 - \kappa) \lambda (H_{jst}, \mathcal{A}(H_{jst}, \Phi_{jst} H_{jst})) + \tilde{\epsilon}_{jst}$$

where $\tilde{\Lambda}_{jst} \equiv \frac{\Phi_{jst} - \Phi_{jst}^o}{MRPH_{jst} - \Phi_{jst}^o}$, $\mathcal{M}_{jst} \equiv \frac{\left(1 - \frac{\alpha_{m,j} + \alpha_{k,j}}{\mu_j} \right) \frac{\mu_j}{\alpha_{h,j}} MRPH_{jst} - \Phi_{jst}^o}{MRPH_{jst} - \Phi_{jst}^o}$, and $\tilde{\epsilon}$ is a residual

capturing measurement error. As in Section 3, I approximate the markdown function $\lambda(\cdot)$ with a 4th-order polynomial in employment and wage bills. The markdowns are then computed using the estimated bargaining parameter: $\lambda_{jst} = \frac{1}{1-\hat{\kappa}} (\tilde{\Lambda}_{jst} - \hat{\kappa} \mathcal{M}_{jst} - \hat{\epsilon}_{jst})$.

Implementing the estimation procedure requires measuring the outside wage. One option is to set Φ^o equal to the level of unemployment benefits. However, Jäger et al. (2020) show that even large reforms to unemployment benefit levels in Austria did not materially affect wages. Similarly, Le Barbanchon et al. (2019) show that extending the potential unemployment benefit duration in France did not affect reservation wages reported by jobseekers. Because the model implies that firms paying below the outside wage will not hire any workers, I set the outside wage to the minimum observed wage.

The estimated workers' bargaining power using this measure of outside wages is reported in Table 25. These estimates are smaller than the ones that assume zero outside wages in Table 4. I then compute the implied monopsony wage markdown, taking the estimated bargaining power in column (4) of Table 25. The wage markdowns at the average firm is 0.45 under $\Phi^o = 0$ in Table 5. The markdowns at the average firm computed with the current measure of Φ^o is 0.31.

Table 25: Estimated worker bargaining power (when workers have outside wages).

	Labor wedge (Λ_{jt})					
	(1)	(2)	(3)	(4)	(5)	(6)
Product market rents ($\tilde{\mu}_{jt}$)	0.052 (0.006)	0.061 (0.005)	0.048 (0.007)	0.062 (0.006)	0.046 (0.007)	0.090 (0.009)
Sector×year fixed effects	✓	✓	✓	✓	✓	✓
Firm fixed effects		✓		✓		✓
Control for $\lambda(\cdot)$			✓	✓	✓	✓
IV for measurement error					✓	✓
Observations	126,836	122,239	126,836	122,239	99,238	95,666

This table reports the estimated workers' bargaining power parameter when workers have outside wages. Even columns controls for firm fixed effects. Columns (3) through (6) controls for differences in monopsony markdowns reflecting differences in amenities. Columns (5) and (6) instruments measured product market rent $\tilde{\mu}_{jt}$ with its lag to address classical measurement error. Bootstrapped standard errors are reported in parentheses.

Table 26: The distribution of monopsony markdowns (when workers have outside wages).

Summary statistics	Mean	Median	25 th Pct	75 th Pct	Var	Var (i)	Var (ii)
Markdowns (λ_{jt})	0.31	0.28	0.19	0.39	0.03	0.02	0.02

This table reports the summary statistics in 2016 for the estimated monopsony markdowns when workers have outside wages. The column Var (i) reports the variances corrected for measurement error following Krueger and Summers (1988) and Kline et al. (2020), while the column Var (ii) reports the variances for firm-groups. Each variable is winsorized by 2%.

C.2 Firm wage premia with Stole & Zwiebel (1996) bargaining

Consider a setting in which firms bargain individually with their employees. Workers supply one unit of labor inelastically. Firms produce goods with only labor inputs. The labor market is characterized by search frictions. Firms post vacancies V_j subject to a convex vacancy posting cost $c(V_j)$. The product market is imperfectly competitive; firms face the following product demand curve $Y_j = P_j^{-\sigma}$. The production function is $Y_j = \Omega_j H_j^{\alpha_h}$. There are no idiosyncratic shocks to firms. Firms steady state size is $H_j = \frac{q}{\delta} V_j$, where q is the job-filling rate and δ is the separation rate. Firms post vacancies to maximize profits:

$$\max_{V_j} P_j Y_j - \Phi_j H_j - c(V_j)$$

subject to the product demand curve and the firm size constraint. The solution to the Stole and Zwiebel (1996) bargaining problem is then:

$$(1 - \kappa)(\Phi_j - \Phi_j^o) = \kappa(MRPH_j - \frac{\partial \Phi_j}{\partial H_j} H_j - \Phi_j)$$

Solving this differential equation yields the following firm wage premium equation:

$$\Phi_j = \left(\frac{\kappa \mu \alpha_h^{-1}}{(1 - \kappa) \mu \alpha_h^{-1} + \kappa} \right) MRPH_j + (1 - \kappa) \Phi^o \quad (20)$$

where markups are constant $\mu = \frac{\sigma}{\sigma-1}$ and the marginal revenue product of labor is $MRPH_j = \mu^{-1} \alpha_h \frac{P_j Y_j}{H_j}$.

Comparison with collective (efficient) bargaining. When $\alpha_k = \alpha_m = 0$ and labor supply is completely inelastic (as is the case when deriving equation (20) under Stole-Zwiebel bargaining), the firm wage premium equation (18) becomes:

$$\Phi_j = \kappa \mu \alpha_h^{-1} MRPH_j + (1 - \kappa) \Phi^o \quad (21)$$

Comparing equation (21) with equation (20) shows that, under collective (efficient) bargaining, workers are able to extract a higher share of $MRPH$ as wages than under individual bargaining, since $\frac{1}{(1-\kappa)\mu\alpha_h^{-1}+\kappa} < 1$.

C.3 Market power and misallocation: subsidizing firm size

In Section 6, I assess the importance of variable markups, markdowns, and worker bargaining power for wages and welfare. To implement either the social planner's equilibrium or an efficient allocation of factor inputs across firms, I derive a set of subsidies that accomplish this goal, following Edmond et al. (2023). These subsidies are meant as a tool to quantify the aggregate welfare effects of markup or markdown dispersion, and to quantify how far raising worker bargaining power can go in raising aggregate welfare. My model

does not speak to the distributional welfare effects of these subsidies as I do not model household heterogeneity. In what follows, I start by deriving the output and labor subsidies that offset firm market power when workers have no bargaining power ($\kappa = 0$). I then show how the subsidies are affected when workers have some bargaining power. I assume that these subsidies are financed by a lump sum tax on the household.

Case 1: No worker bargaining power. Suppose workers have no bargaining power ($\kappa = 0$). For a given set of size-based subsidies $T_y(Y_{jg})$ and $T_h(H_{js})$, the firm now maximizes:

$$\max_{\Phi_{js}, Y_{jg}, K_j, M_j} \Pi_j^{gross} = P_{jg}Y_{jg} - \Phi_{js}H_{js} - P_kK_j - P_{m,j}M_j + T_y(Y_{jg}) + T_h(H_{js})$$

subject to the production function $Y_j = \Omega_j K_j^{\alpha_k} M_j^{\alpha_m} H_j^{\alpha_h}$, labor supply curve (8), and product demand curve (9).

Equalizing or removing markups. The first-order conditions for capital and materials are:

$$\begin{aligned} \text{Capital FOC: } & \left(P_{jg} + Y_{jg} \frac{\partial P_{jg}}{\partial Y_{jg}} + T'_y(Y_{jg}) \right) \alpha_k \frac{Y_{jg}}{K_j} = P_k \\ \text{Materials FOC: } & \left(P_{jg} + Y_{jg} \frac{\partial P_{jg}}{\partial Y_{jg}} + T'_y(Y_{jg}) \right) \alpha_m \frac{Y_{jg}}{M_j} = P_{m,j} \end{aligned}$$

The term $Y_{jg} \frac{\partial P_{jg}}{\partial Y_{jg}}$ captures the distortion induced by the firm's price-setting power. As in Section 6, let ρ_{jg} be the price elasticity of demand. A subsidy schedule that satisfies:

$$T'_y(Y_{jg}) = -Y_{jg} \frac{\partial P_{jg}}{\partial Y_{jg}} = \frac{1}{\rho_{jg}} P_{jg}$$

offsets the firm's price-setting power, inducing the firm to behave as a pricetaker in the product market. Alternatively, a subsidy schedule that satisfies:

$$T'_y(Y_{jg}) = \left(\frac{1}{\rho_{jg}} - \frac{1}{\bar{\sigma}} \right) P_{jg} \quad (22)$$

induces firms to charge a constant markup $\bar{\mu} = \frac{\bar{\sigma}}{\bar{\sigma}-1}$. Any level of constant markups can be implemented by an appropriate choice of the policy parameter $\bar{\sigma}$. For example, $\bar{\sigma} \rightarrow \infty$ induces firms to behave as pricetakers. The output subsidy schedule is then $T_y(Y_{jg}) = \int T'_y(Y_{jg}) dY_{jg}$. These results have been shown by [Edmond et al. \(2023\)](#).

Equalizing or removing monopsony markdowns. I now write the first-order condition for labor to show that an appropriately chosen labor subsidy schedule can offset firm's labor market power:

$$\text{Labor FOC: } \left(P_{jg} + Y_{jg} \frac{\partial P_{jg}}{\partial Y_{jg}} + T'_y(Y_{jg}) \right) \alpha_h \frac{Y_{jg}}{H_{js}} = \Phi_{js} + H_{js} \left(\frac{\partial H_{js}}{\partial \Phi_{js}} \right)^{-1} - T'_h(H_{js})$$

The term $H_{js} \left(\frac{\partial H_{js}}{\partial \Phi_{js}} \right)^{-1}$ captures firms' wage-setting power and distorts labor allocations away from the social planner's allocation. A labor subsidy schedule that satisfies:

$$T'_h(H_{js}) = H_{js} \left(\frac{\partial H_{js}}{\partial \Phi_{js}} \right)^{-1} = \frac{1}{\xi_{js}} \Phi_{js}$$

offsets firms labor market power, where ξ_{js} is the labor supply elasticity. Alternatively, a labor subsidy schedule that satisfies:

$$T'_h(H_{js}) = \left(\frac{1}{\xi_{js}} - \frac{1}{\bar{\eta}} \right) \Phi_{js} \quad (23)$$

induces firms to mark wages down by a constant $\bar{\lambda} = \frac{\bar{\eta}}{1+\bar{\eta}}$. That is, $\bar{\eta}$ is a policy parameter than can be chosen to implement a specific level of the constant markdown $\bar{\lambda}$. The labor subsidy schedule is then $T_h(H_{js}) = \int T'_h(H_{js}) dH_{js}$. These results extend the results in [Edmond et al. \(2023\)](#), whose model does not feature imperfect labor market competition.

Case 2: Positive worker bargaining power. I now derive the output and labor subsidies that address firm product and labor market power when workers have some bargaining power $\kappa > 0$. For convenience, I restate the bargaining problem between workers and their employer:

$$\max_{\Phi_{js}, Y_{jg}, K_j, M_j} \left(\Phi_{js} H_{js} \right)^\kappa \left(\Pi_j^{gross} \right)^{1-\kappa}$$

subject to the production function $Y_j = \Omega_j K_j^{\alpha_k} M_j^{\alpha_m} H_j^{\alpha_h}$, labor supply curve (8), and product demand curve (9). The gross profits are $\Pi_j^{gross} = P_{jg} Y_{jg} - \Phi_{js} H_{js} - P_k K_j - P_{m,j} M_j + T_y(Y_{jg}) + T_h(H_{js})$.

Equalizing or removing markups. To equalize or remove *only* markup-induced distortions, one can use the same output subsidy schedule as in Case 1 (where $\kappa = 0$), because the first-order conditions for capital and materials are the same. However, when workers have bargaining power, workers will appropriate some of the profits due to the output subsidy. That is, workers benefit directly from the output subsidy, and not only through the effects of the subsidy on the firm's pricing behavior. Therefore, to isolate the welfare effects solely from equalizing markups, an appropriate set of labor *taxes* on the firm needs to be designed. To see this, consider the wage equation under output subsidies, but without any labor taxes:

$$\Phi_{js} = \kappa Q_{js}^{rent} + (1 - \kappa) \left[\left(\frac{\xi_{js}}{1 + \xi_{js}} \right) \left(1 - \frac{1}{\rho_{jg}} + \frac{T'_y(Y_{jg})}{P_{jg}} \right) \alpha_h \frac{P_{jg} Y_{jg}}{H_{js}} \right] + \kappa \frac{T_y(Y_{jg})}{H_{js}} \quad (24)$$

where $Q_{js}^{rent} \equiv \frac{P_{jg} Y_{jg} - P_k K_j - P_{m,j} M_j}{H_{js}}$. Equation (24) is almost identical to the original wage equation (1) where there are no output subsidies; the main differences are that (i) markups are now offset by the output subsidy ($T'_y(Y_{jg})$), and (ii) workers capture some of the output

subsidies to the firm because they have bargaining power ($\kappa \frac{T_y(Y_{jg})}{H_{js}}$). With labor taxes, this equation becomes:

$$\begin{aligned} \Phi_{js} = & \kappa Q_{js}^{rent} + (1 - \kappa) \left[\left(\frac{\xi_{js}}{1 + \xi_{js}} \right) \left(1 - \frac{1}{\rho_{jg}} + \frac{T'_y(Y_{jg})}{P_{jg}} \right) \alpha_h \frac{P_{jg} Y_{jg}}{H_{js}} \right] \\ & + \kappa \left(\frac{T_y(Y_{jg})}{H_{js}} + \frac{T_h(H_{js})}{H_{js}} \right) + (1 - \kappa) T'_h(H_{js}) \end{aligned} \quad (25)$$

Equation (25) shows that to isolate the welfare effects of equalizing markups, without capturing the direct effect of the output subsidy through worker bargaining power—that is, without capturing $\kappa \frac{T_y(Y_{jg})}{H_{js}}$ —the appropriate labor tax must satisfy:

$$\kappa \left(\frac{T_y(Y_{jg})}{H_{js}} + \frac{T_h(H_{js})}{H_{js}} \right) + (1 - \kappa) T'_h(H_{js}) = 0 \quad (26)$$

Equalizing or removing monopsony markdowns. Suppose that a policymaker wishes only to equalize or remove markdowns and is not interested in markups ($T_y(Y_{js}) = 0$). Then, given equation (24), the labor subsidy schedule that achieves the policymaker's goal of implementing a monopsony markdown of $\bar{\lambda}$ must satisfy:

$$(1 - \kappa) T'_h(H_{js}) = (1 - \kappa) \left(\frac{1}{\xi_{js}} - \frac{1}{\bar{\eta}} \right) \Phi_{js} - \kappa \left(1 + \frac{1}{\xi_{js}} \right) \frac{T_h(H_{js})}{H_{js}} \quad (27)$$

The first term on the right-hand side of the equality addresses firm monopsony power. When $\bar{\eta} \rightarrow \infty$, the subsidy removes monopsony markdowns entirely, although it does not remove the labor wedge Λ_{js} because workers are still able to appropriate profits from markups. The second term takes into account that workers are able to capture a part of the labor subsidy given to the firm. If workers have no bargaining power ($\kappa = 0$), then the labor subsidy reduces to the one in Case 1: $T'_h(H_{js}) = \left(\frac{1}{\xi_{js}} - \frac{1}{\bar{\eta}} \right) \Phi_{js}$.

Finally, if the policymaker also chooses output subsidies to equalize markups, then the labor subsidy that implements constant monopsony markdowns must satisfy:

$$(1 - \kappa) T'_h(H_{js}) = (1 - \kappa) \left(\frac{1}{\xi_{js}} - \frac{1}{\bar{\eta}} \right) \Phi_{js} - \kappa \lambda_{js}^{-1} \frac{T_y(Y_{jg}) + T_h(H_{js})}{H_{js}} \quad (28)$$

Welfare effects of the output and labor subsidies. The output and labor subsidies only affect the household's budget constraint by affecting firms' input demand decisions—they do not otherwise directly affect the household's budget constraint. There are two reasons for this. First, because the output and labor subsidies are financed by a lump sum tax on the representative household, the subsidies do not distort household consumption and labor supply decisions. Second, because firm profits (inclusive of the subsidies) are rebated to the household in full, the total amount of taxes paid by the household is fully compensated by the subsidies that return to the household in the form of firm profits. To

see this, define Π as net profits (which does not include the subsidies), and gross profits as net profits plus the output and labor subsidies:

$$\Pi^{gross} = \underbrace{\int_g \sum_{j \in g} (P_{jg} Y_{jg} - P_k K_j - P_{m,j} M_j) dg - \int_s \sum_{j \in s} \Phi_{js} H_{js} ds}_{\equiv \Pi} + \int_g \sum_{j \in g} T_y(Y_{jg}) dg + \int_s \sum_{j \in s} T_h(H_{js}) ds$$

Then, the household's budget constraint under the subsidies is:

$$\begin{aligned} C &= \Phi H + \Pi^{gross} - T_y^{total} - T_h^{total} \\ &= \Phi H + \Pi \end{aligned}$$

where $T_y^{total} = \int_g \sum_{j \in g} T_y(Y_{jg}) dg$ and $T_h^{total} = \int_s \sum_{j \in s} T_h(H_{js}) ds$ are the lump sum taxes on the household. The subsidies therefore only affect the household budget constraint through firms' input demand conditions. These subsidies are meant as a tool to assess the aggregate welfare effects of firm market power. In a richer setting that considers household heterogeneity, the implementation of such taxes and subsidies can have direct distributional implications that my model does not address.

C.4 Measuring product and labor market shares

The nested-CES model presented in Section 6 imposes strong restrictions on markups and markdowns. First, markups and markdowns are a function of sales market shares and wage-bill market shares. Second, their relationship with market shares follow:

$$\begin{aligned} \text{Markups:} \quad & \frac{\mu_{jg} - 1}{\mu_{jg}} = \frac{1}{\sigma} + \left(\frac{1}{\theta} - \frac{1}{\sigma} \right) \frac{P_{jg} Y_{jg}}{\sum_{j'}^{n_g} P_{j'g} Y_{j'g}} \\ \text{Markdowns:} \quad & \frac{\lambda_{js}}{1 - \lambda_{js}} = \eta + (\nu - \eta) \frac{\Phi_{js} H_{js}}{\sum_{j'}^{n_s} \Phi_{j's} H_{j's}} \end{aligned}$$

In Section 6.3, I describe the implementation of these equations to calibrate the preference elasticities $(\theta, \sigma, \nu, \eta)$, and assess how well the variable markups implied by these equations fit the relationship between my empirical markup estimates and measured market shares. In this section, I discuss the measurement issues regarding market shares and how they may affect the inferred preference elasticities and welfare calculations. My discussion below makes the following assumptions: (i) product demand and labor supply preferences are nested-CES, and (ii) product and labor market structures are oligopolistic and oligopsonistic.

Market share measurement. The implementation of the markup equation implied by the nested-CES model requires measuring market shares for a given definition of product markets. Defining a product market is not straightforward, as researchers do not generally observe the direct competitors of a firm. I define a product market as a 5-digit manufacturing sector, as is standard in the literature (De Ridder et al., 2021; Amiti and Heise, 2024). There are 211 5-digit sectors. Under this definition, I use firm-level total revenues to compute market shares in my baseline implementation in Section 6.3.

There are two main challenges to measuring market shares in a world with international trade. First, if firms export across national borders, their exports do not compete directly with the output of domestic firms within the same market. Not accounting for exports may therefore overstate the market share of large firms. To address this concern, in Table 27, I report the values of the preference parameters (θ, σ) that I obtain when I measure market shares using domestic revenues. This is possible because the FARE firm balance sheet data breaks down firm-level total sales into domestic sales and exports, although the data does not contain any additional information about exports (e.g. destination). Overall, I find similar parameter values for (θ, σ) compared to my baseline calibration.

Second, market share measures that do not account for import competition will overstate the market share of domestic firms.⁵⁰ I now show that, in the presence of import competition, my calibration approach may understate degree of passthrough of market

⁵⁰See, for example, Amiti and Heise (2024), who show that accounting for foreign competition flattens the trend of rising market concentration in the US economy.

shares to markups $(\frac{1}{\theta} - \frac{1}{\sigma})$, and overstate the between-market elasticity of substitution (θ). To see this, let the true market share of a firm j in sector g be $s_{jgt}^y = \frac{P_{jgt}Y_{jgt}}{\sum_{j'}^{ng} P_{j'gt}Y_{j'gt} + P_{Fgt}Y_{Fgt}}$, where $P_{Fgt}Y_{Fgt}$ are competing imports from foreign firms, and the measured market share be $\hat{s}_{jgt}^y = \frac{P_{jgt}Y_{jgt}}{\sum_{j'}^{ng} P_{j'gt}Y_{j'gt}}$. The relationship between the true market share and the measured market share is $s_{jgt}^y = \hat{\gamma}_{gt}\hat{s}_{jgt}^y$, where $\hat{\gamma}_{gt} \equiv \frac{\sum_{j'}^{ng} P_{j'gt}Y_{j'gt}}{\sum_{j'}^{ng} P_{j'gt}Y_{j'gt} + P_{Fgt}Y_{Fgt}} \in [0, 1]$. The relationship between markups and market shares can then be written as:

$$\frac{\hat{\mu}_{jg} - 1}{\hat{\mu}_{jg}} = \frac{1}{\sigma} + \left(\frac{1}{\theta} - \frac{1}{\sigma}\right)\hat{\gamma}_{gt}\hat{s}_{jgt}^y$$

Therefore, the estimated market share passthrough becomes $\widehat{(\frac{1}{\theta} - \frac{1}{\sigma})}\hat{\gamma}_{gt}$, which is less than $(\frac{1}{\theta} - \frac{1}{\sigma})$. Understating the market share passthrough does not necessarily affect the inferred σ , because σ is measured using $\frac{1}{\sigma} = \frac{\hat{\mu}_{jg}-1}{\hat{\mu}_{jg}} - \widehat{(\frac{1}{\theta} - \frac{1}{\sigma})}\hat{\gamma}_{gt}\hat{s}_{jgt}^y$. However, it would lead to an upward-bias in the inferred θ , because this parameter is measured as the inverse of $\widehat{(\frac{1}{\theta} - \frac{1}{\sigma})}\hat{\gamma}_{gt} + \frac{1}{\sigma}$.

How might the biases induced by unobserved import competition affect the welfare calculations in Section 7.4? Section 6.2 shows that the dispersion of markups reduce welfare by misallocating factor inputs across firms. In the nested-CES model, the lower bound for markups is $\mu_{min} = \frac{\sigma}{\sigma-1}$ and the upper bound is $\mu_{max} = \frac{\theta}{\theta-1}$. The responsiveness coefficient $\frac{1}{\theta} - \frac{1}{\sigma}$ governs whether a firm with a given market share is closer to μ_{min} or μ_{max} . Therefore, the downward bias in the responsiveness of markups to market shares $(\frac{1}{\theta} - \frac{1}{\sigma})$ and the resulting upward bias in the between-market elasticity of substitution (θ) reduces the possible extent of markup dispersion. In turn, this will understate the welfare cost of markups operating through the misallocation channel. In this case, one can read the welfare costs of markups in Section 7.4 as a lower bound.

On the labor market side, the measurement of wage-bill market shares for local labor markets are less susceptible to issues concerning international trade. In my baseline calibration, I define a local labor market as a 5-digit sector and commuting zone pair. There are 211 5-digit sectors and 281 commuting zones in my sample, leading to 11,531 local labor markets. This approach follows Berger et al. (2022), who define local labor markets by 3-digit NAICS and commuting zone pairs. In Table 27, I also present the calibrated labor market preference parameters (ν, η) when I define local labor markets at the more aggregated 2-digit sector $\times$ commuting-zone level.

Table 27: Elasticities of substitution between goods (σ) and between jobs (η) by 2-digit French manufacturing sectors.

Within-market elasticities of substitution	σ		η	
	(1)	(2)	(3)	(4)
Manufacturing sectors				
Textile	3.22	3.20	1.55	1.33
Apparel	3.95	3.95	1.15	1.12
Leather	3.50	3.49	1.49	1.55
Wood products (excluding furniture)	-19.38	27.78	0.57	0.67
Paper and publishing	3.50	3.29	0.50	0.63
Recorded media	2.31	2.31	1.82	1.77
Chemicals	4.33	4.34	0.47	0.54
Pharmaceutical	-1.49	-1.48	-0.15	-0.09
Rubber & plastics	4.01	3.99	0.90	0.82
Non-metallic minerals	4.17	4.11	0.46	0.65
Basic metals	4.64	4.62	0.80	0.61
Fabricated metals (excluding machinery)	3.35	3.35	0.95	0.89
Computers, electronic, & optical	3.96	3.94	0.89	0.83
Electrical equipment	4.52	4.37	0.84	0.79
Machinery & equipment	7.84	7.84	0.62	0.55
Motor vehicles	7.16	7.12	0.72	0.74
Other transport equipment	4.96	5.12	0.77	0.52
Furniture	3.36	3.47	1.14	1.24
Other manufacturing	3.26	3.25	0.60	0.79
Repair & installation of machinery	3.76	3.74	1.01	1.02
Economy-wide	4.53	4.58	0.87	0.88
Between-market elasticities of substitution	θ		ν	
	1.53	1.73	0.74	0.80

This table reports calibrated values of the elasticities of substitution between goods (σ) and between jobs (η) for 2-digit sectors (2009-2016). The 2-digit sector-specific parameters are obtained by implementing the method described in Section 6.3 for each 2-digit sector. My baseline measures of (σ, η) are reported columns (1) and (3) in the row ‘Economy-wide’. In columns (1) and (2), a product market is defined as a 5-digit sector. Market shares are measured using firm-level total sales in column (1), and as domestic sales in column (2). In column (3), a local labor market is defined as a 5-digit sector×commuting-zone pair. In column (4) a local labor market is a 2-digit sector×commuting-zone pair. Labor market shares are measured as a firm’s wage-bill share in columns (3) and (4).

C.5 Measuring quality and amenities under variable elasticity of substitution

In Section 6.3, I measure firm heterogeneity in product quality and non-wage amenities under the assumption of CES preferences. An important concern is whether the measured degree of heterogeneity depends heavily on the assumed preference structure. In this section, I measure these sources of heterogeneity using other well-known preference structures that nest the CES as a special case. Specifically, I compare the CES-implied measures with two, more general, classes of variable-elasticity-of-substitution (VES) product demand and labor supply systems: (i) Pollak (1971) additively separable preferences and (ii) Kimball (1995) preferences.⁵¹ My main finding is that the implied firm heterogeneity under these two alternative preference structures are closely correlated with my baseline measures based on the CES structure.

In what follows, I maintain the assumption that the product demand aggregator over product markets, and the labor supply aggregator over labor markets, are CES. That is, at the aggregate level, the model structure is as in Section 6. However, at the product and labor market-level, the aggregator over firms are now VES instead of CES. The VES preference structure implies that markups and markdowns are firm-specific and depends on firm size, even if market structures are monopolistic and monopsonistic.⁵² As such, in this section I assume that firms behave monopolistically in product markets and monopsonistically in labor markets.

C.5.1 Pollak's additive preferences

Let the aggregator over firms (product varieties) in market g be:

$$Y_g = \sum_{j \in g} \tilde{D}_{jg}^{\frac{1}{\sigma}} \left(\frac{\sigma}{\sigma - 1} \right) (Y_{jg} + \delta^y)^{\frac{\sigma-1}{\sigma}}$$

where $\sum_{j \in g} \tilde{D}_{jg} = 1$. The parameter $\delta^y \geq 0$ governs the extent of the departure from CES, with $\delta^y = 0$ giving the CES demand system. Solving for the expression for market shares gives:

$$\frac{P_{jg} Y_{jg}}{\sum_{j' \in g} P_{j'g} Y_{j'g}} = \frac{\tilde{D}_{jg}^{\frac{1}{\sigma}} (Y_{jg} + \delta^y)^{-\frac{1}{\sigma}} Y_{jg}}{\sum_{j' \in g} \tilde{D}_{j'g}^{\frac{1}{\sigma}} (Y_{j'g} + \delta^y)^{-\frac{1}{\sigma}} Y_{j'g}} \quad (29)$$

and the price elasticity of demand is:

$$\rho_{jg}^{Pollak} = \sigma \left(1 + \frac{\delta^y}{Y_{jg}} \right) \quad (30)$$

⁵¹These preferences have been used extensively in research in international trade and macroeconomics. See, for example, Morlacco and Arkolakis (2017), Arkolakis, Costinot, Donaldson, and Rodrigues-Clare (2019), and Edmond et al. (2023).

⁵²See Morlacco and Arkolakis (2017) for a detailed exposition of these demand systems.

As $\delta^y \rightarrow 0$, the price elasticity of demand and markups converge to the CES case: $\rho_{jg}^{Pollak} \rightarrow \sigma$ and $\mu_{jg}^{Pollak} = \frac{\rho_{jg}^{Pollak}}{\rho_{jg}^{Pollak}-1} \rightarrow \frac{\sigma}{\sigma-1}$. When $\delta^y > 0$, the price elasticity of demand decreases with firm size, while markups increase with firm size.

One feature of Pollak preferences is that price elasticities of demand (ρ_{jg}^{Pollak}) are infinitely large, and so markups ($\mu_{jg} - 1$) are approximately zero for small firms (see equation (30)).⁵³ That is, small firms are as good as pricetakers. This implication appears to be inconsistent with my baseline markup estimates in Table 3, as most firms charge markups significantly above zero. However, the nested-CES product demand system in Section 6 avoids this inconsistency, as even small firms can charge non-negligible markups in that set up.

Analogously, let the labor supply aggregator over firms in labor market s be:

$$H_s = \sum_{j \in s} \tilde{A}_{js}^{-\frac{1}{\eta}} \left(\frac{\eta}{\eta - 1} \right) (H_{js} + \delta^h)^{\frac{1+\eta}{\eta}}$$

where $\sum_{j \in s} \tilde{A}_{js} = 1$. The parameter $\delta^h \geq 0$ governs the extent of the departure from CES. Solving for the labor market shares gives:

$$\frac{\Phi_{js} H_{js}}{\sum_{j' \in s} \Phi_{j's} H_{j's}} = \frac{\tilde{A}_{js}^{-\frac{1}{\eta}} (H_{js} + \delta^h)^{\frac{1}{\eta}} H_{js}}{\sum_{j' \in s} \tilde{A}_{j's}^{-\frac{1}{\eta}} (H_{j's} + \delta^h)^{\frac{1}{\eta}} H_{j's}} \quad (31)$$

and the labor supply elasticity:

$$\xi_{js}^{Pollak} = \eta \left(1 + \frac{\delta^h}{H_{js}} \right) \quad (32)$$

When $\delta^h > 0$, larger firms face lower labor supply elasticities and are able to mark down wages below the marginal revenue product of labor further. Just as for markups, the Pollak labor supply system implies that small firms face infinitely large labor supply elasticities, and therefore, pay wages that are approximately the full marginal revenue product of labor ($\lambda_{js}^{Pollak} = 1$). This implication is also inconsistent with my baseline markdown estimates in Table 5, which shows that most firms would have non-negligible monopsony power in the absence of worker bargaining power. By contrast, the nested-CES labor supply system in Section 6 avoids this inconsistency.

Calibration and comparison. To compare my quality and amenity measures based on the CES structure to those obtained under the Pollak VES structure, I must first calibrate the parameters $(\delta^y, \delta^h, \sigma, \eta)$. To do so, I use moments from my baseline markup and markdown estimates, $\hat{\mu}_{jg}$ and $\hat{\lambda}_{js}$. Symbols with a hat are estimates. I maintain the same definition of a product market (5-digit sector) and labor market (5-digit sector $\times$ commuting-zone) as in Section 6. Because the Pollak product demand and labor supply systems nest

⁵³Other popular VES preferences also share this feature, for example, the Kimball demand system.

the CES system I use in Section 6, I maintain the same calibrated values of σ and η as in my baseline calibration.

To calibrate the extent of the departure from CES in the Pollak product demand system, δ^y , I use equation (30). I write this equation in relative terms, comparing a firm j to the median firm in the markup distribution:

$$\frac{\hat{\rho}_{jg}^{Pollak}}{\hat{\rho}_{\mu,pct50}^{Pollak}} = \left(1 + \frac{\delta^y}{Y_{jg}}\right) \bigg/ \left(1 + \frac{\delta^y}{Y_{\mu,pct50}}\right) \quad (33)$$

where $\hat{\rho}_{jg}^{Pollak} \equiv \frac{\hat{\mu}_{jg}}{\hat{\mu}_{jg}-1}$. The parameter δ^y is then calibrated to match the covariance between firm size Y_{jg} and ratio of price demand elasticities $\frac{\hat{\rho}_{jg}^{Pollak}}{\hat{\rho}_{\mu,pct50}^{Pollak}}$. This moment is informative about the degree of departure from CES in the Pollak demand system, because CES (in combination with monopolistic competition) implies that price demand elasticities should be independent of firm size.

Similarly, I use equation (32) to calibrate the extent of the departure from CES in the labor supply system. I express this equation in relative terms, comparing a firm j to the median firm in the markdown distribution:

$$\frac{\hat{\xi}_{js}^{Pollak}}{\hat{\xi}_{\lambda,pct50}^{Pollak}} = \left(1 + \frac{\delta^h}{H_{js}}\right) \bigg/ \left(1 + \frac{\delta^h}{H_{\lambda,pct50}}\right) \quad (34)$$

where $\hat{\xi}_{js}^{Pollak} \equiv \frac{\hat{\lambda}_{js}}{1-\hat{\lambda}_{js}}$. I then calibrate the parameter δ^h to match the covariance between firm size H_{js} and ratio of labor supply elasticities $\frac{\hat{\xi}_{js}^{Pollak}}{\hat{\xi}_{\lambda,pct50}^{Pollak}}$.

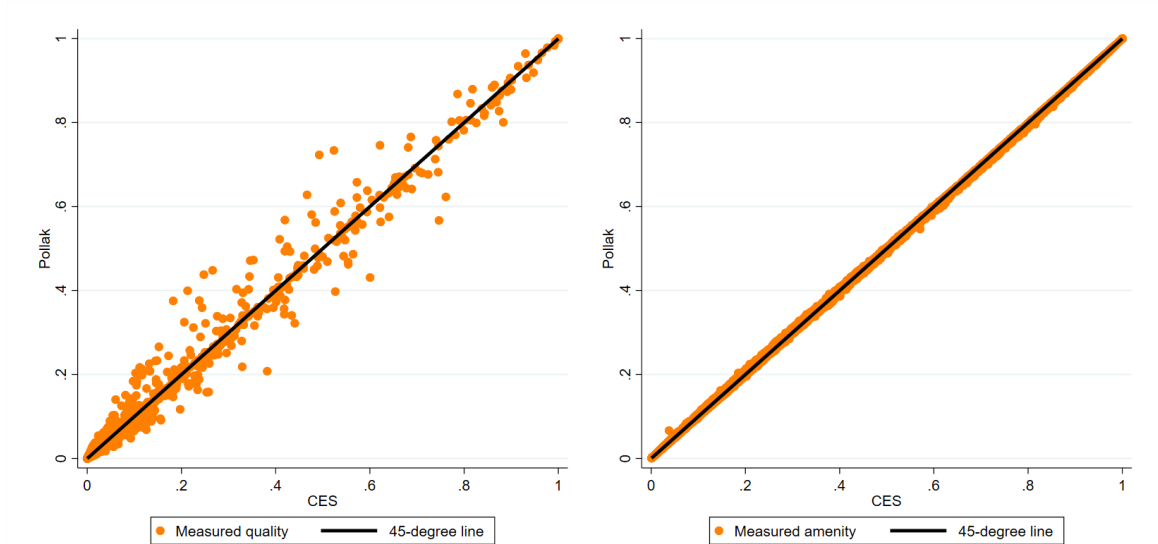


Figure 22: Measured product quality and non-wage amenity under nested-CES and Pollak's additively separable (VES) preferences in 2016.

Note: Each data point is a firm.

Finally, given the calibrated parameters $(\delta^y, \delta^h, \sigma, \eta)$, I measure firm heterogeneity

in product quality using equation (29), and heterogeneity in non-wage amenities using equation (31). Figure 22 shows how the CES-based and VES-based (Pollak) measures of firm heterogeneity in quality and amenities compare. The left-hand-side panel of the figure shows that CES-based and Pollak-based quality measures correlate strongly with each other—they are centered around the 45-degree line. Similarly, CES-based and Pollak-based amenity measures correlate even more strongly with each other. The larger CES-Pollak disparity for quality measures compared to amenity measures arises because my markup estimates increase more sharply with firm size, while markdown estimates decrease at a slower rate with firm size (see Figure 14).

C.5.2 Kimball preferences

Let the aggregator over firms (product varieties) in market g be:

$$1 = \sum_{j \in g} \tilde{D}_{jg} \Upsilon \left(\frac{Y_{jg}}{\tilde{D}_{jg} Y_g} \right)$$

where $\sum_{j \in g} \tilde{D}_{jg} = 1$. The function $\Upsilon(\cdot)$ is homothetic, satisfies $\Upsilon(1) = 1$, $\Upsilon' > 0$, $\Upsilon'' < 0$, and nests CES preferences as a special case. Solving for the expression for market shares gives:

$$\frac{P_{jg} Y_{jg}}{\sum_{j' \in g} P_{j'g} Y_{j'g}} = \Upsilon' \left(\frac{Y_{jg}}{\tilde{D}_{jg} Y_g} \right) \frac{Y_{jg}}{Y_g} \mathcal{D}_g^{-1} \quad (35)$$

where $\mathcal{D}_g \equiv \sum_{j \in g} \Upsilon' \left(\frac{Y_{jg}}{\tilde{D}_{jg} Y_g} \right) \frac{Y_{jg}}{Y_g}$ is an endogenous demand shifter that is determined in general equilibrium.

Operationalizing the Kimball preferences requires making an assumption on the functional form of $\Upsilon(\cdot)$. I follow the common approach of assuming that $\Upsilon(\cdot)$ is an upper incomplete gamma function (Klenow and Willis, 2016). That is:

$$\Upsilon(x) = 1 + (\sigma - 1) \exp \left(\frac{1}{\delta^y} \right) (\delta^y)^{\frac{\sigma}{\delta^y} - 1} \left[\text{Gamma} \left(\frac{\sigma}{\delta^y}, \frac{1}{\delta^y} \right) - \text{Gamma} \left(\frac{\sigma}{\delta^y}, \frac{x^{\frac{\delta^y}{\sigma}}}{\delta^y} \right) \right]$$

where, with a slight abuse of notation, the parameter $\delta^y \geq 0$ now governs the degree of departure of Kimball preferences from the CES; $\delta^y \rightarrow 0$ takes the preferences to the CES limit. With this functional form, one can write the a firm's product market share as:

$$\frac{P_{jg} Y_{jg}}{\sum_{j' \in g} P_{j'g} Y_{j'g}} = \tilde{D}_{jg} \exp \left[\frac{1}{\delta^y} - \frac{1}{\delta^y} \left(\frac{Y_{jg}}{\tilde{D}_{jg} Y_g} \right)^{\frac{\delta^y}{\sigma}} \right] \frac{Y_{jg}}{\tilde{D}_{jg} Y_g} \mathcal{D}_g^{-1} \quad (36)$$

and the price elasticity of demand as:

$$\rho_{jg}^{Kimball} = \frac{\sigma}{1 - \delta^y \log \left(\frac{P_{jg}}{P_g} \mathcal{D}_g \right)} \quad (37)$$

As $\delta^y \rightarrow 0$, the price elasticity of demand and markups converge to the CES case: $\rho_{jg}^{Kimball} \rightarrow \sigma$ and $\mu_{jg}^{Kimball} = \frac{\rho_{jg}^{Kimball}}{\rho_{jg}^{Kimball}-1} \rightarrow \frac{\sigma}{\sigma-1}$. When $\delta^y > 0$, the price elasticity of demand decreases with firm size, while markups increase with firm size.

Similarly, one can model the aggregator over firms in each labor market s as Kimball:

$$1 = \sum_{j \in s} \tilde{A}_{js} \mathcal{L} \left(\frac{H_{js}}{\tilde{A}_{js} H_s} \right)$$

where $\sum_{j \in s} \tilde{A}_{js} = 1$. The function $\mathcal{L}(\cdot)$ is homothetic, satisfies $\mathcal{L}(1) = 1$, $\mathcal{L}' > 0$, $\mathcal{L}'' > 0$, and nests CES preferences as a special case. However, operationalizing the implied equations for labor supply (elasticities) requires assuming a functional form that satisfies these conditions; one that is symmetric to the upper incomplete Gamma function of [Klenow and Willis \(2016\)](#). This is a challenging task, and to the best of my knowledge, no analogous function for labor supply has yet been developed. As such, I focus on comparing the Kimball-implied measures product quality heterogeneity to those implied by CES below.

Calibration and comparison. As with the calibration of the Pollak preference parameters, I maintain the same calibrated value of σ as in Section 6.3, as well as the same definition of product and labor markets. To calibrate the extent of the departure from CES in the Kimball product demand system, δ^y , I rearrange the equation (37) for the price elasticity of demand as:

$$\frac{1}{\hat{\rho}_{jg}^{Kimball}} = \frac{1}{\sigma} - \frac{\delta^y}{\sigma} \log P_{jg} + \frac{\delta^y}{\sigma} \log \left(\frac{\mathcal{D}_g}{P_g} \right) \quad (38)$$

where $\hat{\rho}_{jg}^{Kimball} \equiv \frac{\hat{\mu}_{jg}}{\hat{\mu}_{jg}-1}$. To calibrate the parameter δ^y , I first estimate the ratio $\frac{\delta^y}{\sigma}$ by running an OLS regression of the inverse of the price elasticity of demand $\frac{1}{\hat{\rho}_{jg}^{Kimball}}$ on output prices $\log P_{jg}$, treating the last term in the equation, $\frac{\delta^y}{\sigma} \log \left(\frac{\mathcal{D}_g}{P_g} \right)$, as a sector $\times$ year fixed effect. Then, given the estimate $\widehat{\delta^y/\sigma}$, I measure δ^y as $\sigma(\widehat{\delta^y/\sigma})$. Finally, given these calibrated preference parameters, I measure firm heterogeneity in product quality using equation (35). This requires simultaneously solving for the general equilibrium quantities P_g , Y_g , and $\mathcal{D}_g$.

Figure 23 shows that CES-based and Kimball-based quality measures are closely aligned with each other, as they are approximately centered around the 45-degree line. However, it is worth noting that the Kimball demand system implies a higher quality measure than the CES for high-quality firms, but a lower quality measure than the CES for low-quality firms. Thus, the Kimball preference system implies a larger dispersion in product quality than the CES.

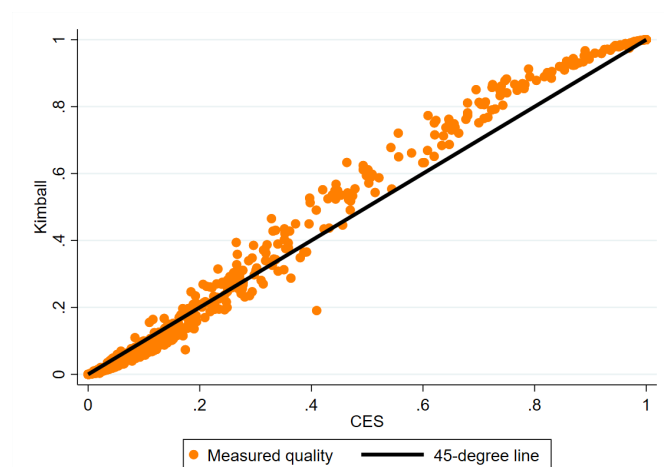


Figure 23: Measured product quality under nested-CES and Kimball (VES) preferences in 2016.

D Appendix: Additional Tables and Figures

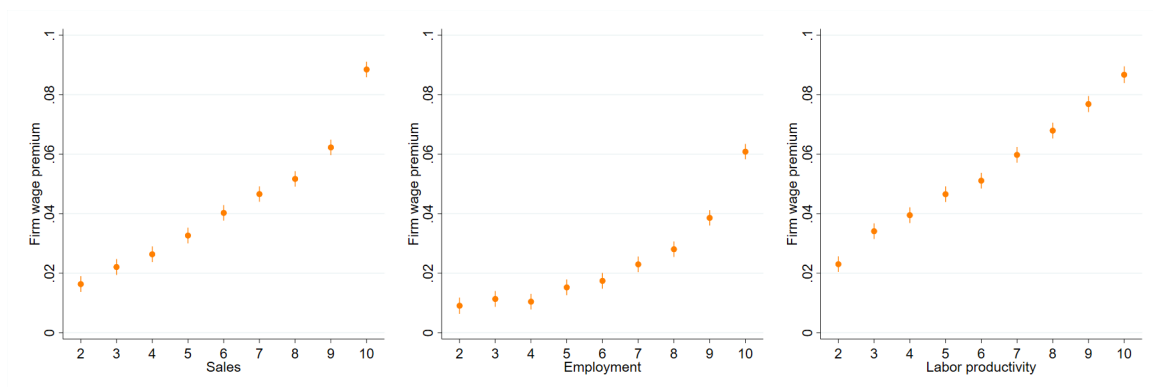


Figure 24: Firm wage premia by deciles of firm size and labor productivity.

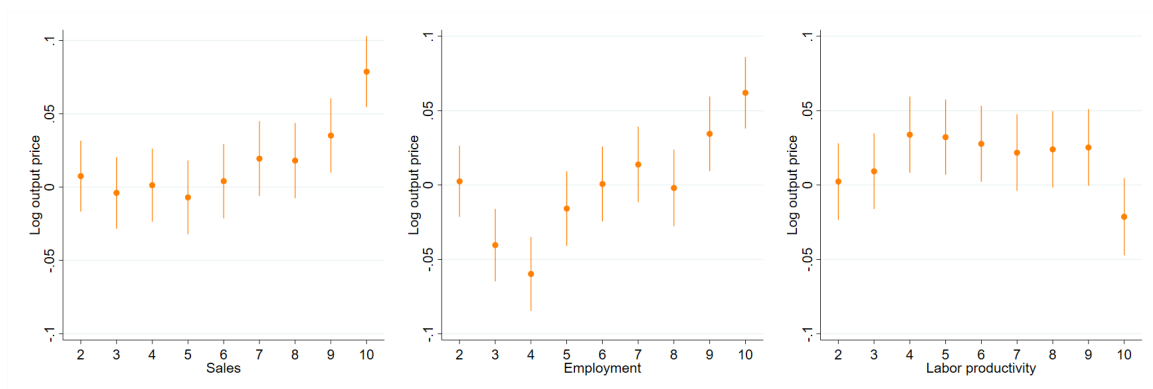


Figure 25: Output prices by deciles of firm size and labor productivity.

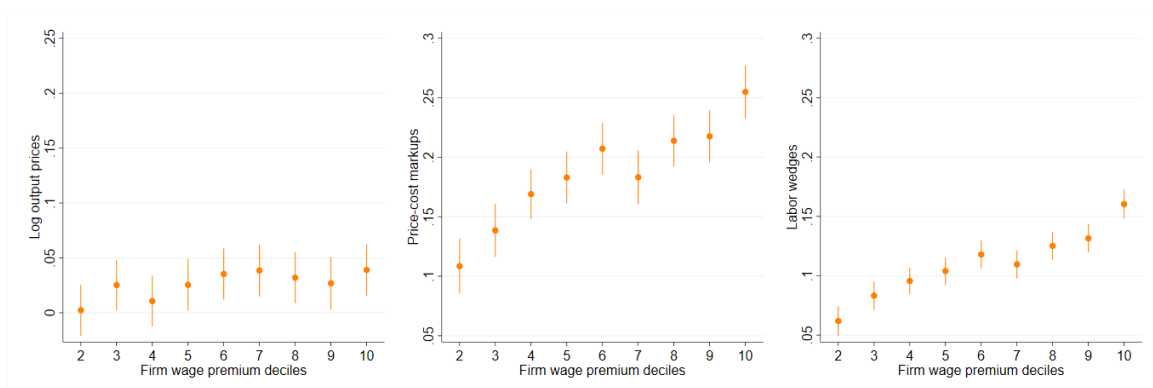


Figure 26: Output prices, markups, and labor wedges by high-wage status.

Notes: This figure shows how output prices, price-cost markups, and labor wedges vary by deciles of firm wage premia relative to firms in the first decile (low-wage firms). These are unconditional correlations. Decile 10 represents high-wage firms. Confidence intervals at the 95% level are plotted.

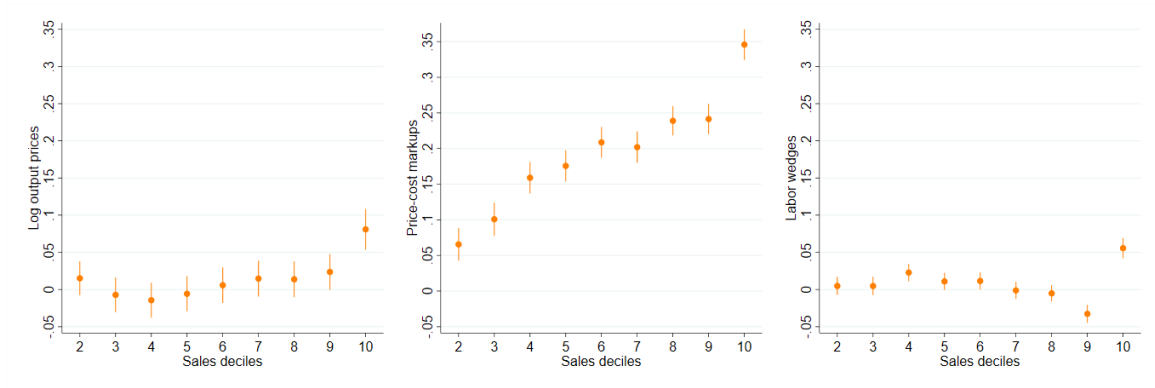


Figure 27: Output prices, markups, and labor wedges by deciles of firm size.

Notes: This figure shows how output prices, price-cost markups, and labor wedges vary by deciles of firm size (measured as sales) relative to firms in the first decile. These are unconditional correlations. Decile 10 represents large firms. Confidence intervals at the 95% level are plotted.

Table 28: Estimates of worker bargaining power and monopsony markdowns by 2-digit French manufacturing sectors.

Manufacturing sectors	κ	λ
Apparel	0.194	0.290
Leather	0.171	0.470
Repair & installation of machinery	0.168	0.363
Other manufacturing	0.166	0.442
Pharmaceutical	0.165	0.095
Computers, electronic, & optical	0.153	0.476
Motor vehicles	0.153	0.352
Basic metals	0.151	0.304
Furniture	0.149	0.492
Electrical equipment	0.131	0.433
Fabricated metals (excluding machinery)	0.125	0.412
Chemicals	0.121	0.301
Machinery & equipment	0.119	0.324
Wood products (excluding furniture)	0.110	0.412
Non-metallic minerals	0.104	0.369
Textile	0.101	0.495
Recorded media	0.101	0.606
Other transport equipment	0.088	0.345
Rubber & plastics	0.073	0.483
Paper and publishing	0.069	0.493

This table reports estimates of worker bargaining power and median monopsony markdowns for 2-digit sectors (2009-2016).

Table 29: Competition and the direct component of shock passthrough.

Shocks	$\sigma = 1.1$	$\sigma = 5$	$\sigma = 10$	$\eta = 1$	$\eta = 5$	$\eta = 10$
Δ TFPQ	0.05	1.11	1.55	0.98	0.46	0.28
Δ product quality	0.54	0.28	0.17	0.28	0.13	0.08
Δ amenity	-0.54	-0.28	-0.17	-0.28	-0.08	-0.08

This table reports the size of the *direct* component of shock passthrough under different degrees substitutability of product varieties and jobs. The passthrough measures are with respect to a positive 1% shock. When varying σ in the first three columns, η is kept at the baseline calibrated value of 0.87. When varying η in the last three columns, σ is kept at the baseline calibrated value of 4.53.

Table 30: Correlations between firm wage premia and the sources of firm heterogeneity.

	Wage premium	TFPQ	Quality	Amenity
Wage premium (ϕ_{js})	1.00			
TFPQ (ω_{js})	0.05	1.00		
Product quality (d_{js})	0.08	-0.63	1.00	
Non-wage amenity (a_{js})	-0.04	-0.04	0.27	1.00

This table reports the Pearson correlation coefficients between estimated firm wage premia, estimated TFPQ, measured product quality, and measured non-wage amenities in 2016. Variables are in logs.

E Appendix: Extension with Two Skill Types

One limitation of the current analysis is that the model does not feature skill heterogeneity. Allowing for heterogeneous skill may shed light on the extent of differences in bargaining power and monopsony power by skill. Moreover, if skill compositions differ across firms, then my baseline model may mismeasure labor wedges, subsequently affecting the bargaining power and markdown estimates. In this section, I extend the analysis in Section 5 to a setting with two skill types—high-skill and low-skill occupations. This extension allows the composition of skills to differ across firms. My findings suggest that my baseline estimates in Section 5, which do not allow for skill heterogeneity, masks interesting differences in bargaining power (higher for high-skill workers) and monopsony markdowns (lower for high-skill workers) between skills, but captures the firm-level average markdown and labor wedge well.

E.1 Model environment

Skill-specific labor supply. Let the subscript $o = \{0, 1\}$ denote high-skill ($o = 1$) and low-skill ($o = 0$) labor, and let Φ_{oj} be the firm-specific wage premium for skill o . A worker i with efficiency E_i and skill o obtains a wage $W_{ioj} = E_i \Phi_{oj}$. Efficiency units of labor of skill o in the firm is $H_{o,j} = \bar{E}_{o,j} L_{o,j}$, where $\bar{E}_{o,j}$ denotes average efficiency and $L_{o,j}$ denotes amount of labor. The upward-sloping labor supply curve facing each firm is $H_{o,j} = \mathcal{H}_o(\Phi_{o,j}, A_{o,j})$. The skill and firm-specific labor supply shifter $A_{o,j}$ represents the non-wage amenities at a firm j for skill o and is taken as given by the firm.

Product demand. As in the baseline model, the downward-sloping demand curve for firm j 's output is $Y_j = \mathcal{G}(P_j, D_j)$, where P_j is the price and D_j is a demand shifter (product quality), which is taken as given by the firm.

Production technology. Firms operate a general production function with diminishing marginal returns to each input $Y_j = \Omega_j F(K_j, M_j, H_{0j}, H_{1j})$, where the main change compared to my baseline model is that the production function now features two skill types. This production function allows skill compositions to differ across firms due to (i) skill-specific complementarities or substitutabilities with non-labor inputs, and (ii) non-homotheticities in skills.

As before, Ω_j is the Hicks-neutral productivity term, K_j are physical capital, and M_j are material inputs. I maintain the assumption that the markets for capital and material inputs are perfectly competitive with prices P_k and P_m .

Wage determination. Workers of each skill type bargain collectively with their employer j . I maintain the assumption that bargaining is efficient—workers and firms jointly decide on wages, prices, materials, and capital to maximize total profits, taking into account the product demand curve and labor supply curves.

Denote high-skill workers' bargaining power as κ_1 and low-skill workers' bargaining power as κ_0 . The employer's bargaining power is then $1 - \kappa_0 - \kappa_1$. As before, in the event that firms and their employees do not arrive at an agreement, I assume that firms do not produce and have an outside option of zero profits; workers do not supply labor and have an outside option of zero wages. The worker-firm bargaining problem is:

$$\max_{\Phi_{0j}, \Phi_{1j}, P_j, M_j, K_j} \left(\Phi_{1j} H_{1j} \right)^{\kappa_1} \left(\Phi_{0j} H_{0j} \right)^{\kappa_0} \left(\Pi_j \right)^{1-\kappa_0-\kappa_1}$$

subject to $H_{oj} = \mathcal{H}_o(\Phi_{oj}, A_{oj})$, $Y_j = \mathcal{G}(P_j, D_j)$, and $Y_j = \Omega_j F(K_j, M_j, H_{0j}, H_{1j})$. The firm's profit is $\Pi_j = P_j Y_j - \Phi_{0j} H_{0j} - \Phi_{1j} H_{1j} - P_m M_j - P_k K_j$.

Skill-specific labor wedges. Solving the worker-firm bargaining problem gives the following skill-specific labor wedge equation at firm j :

$$\begin{aligned} \Lambda_{o,j} &= \kappa_o \left(1 - \frac{P_k K_j + P_m M_j + \Phi_{o',j} H_{o',j}}{P_j Y_j} \right) \frac{\mu_j}{\alpha_{o,j}} + (1 - \kappa_0 - \kappa_1) \lambda_{o,j} \\ &= \kappa_o \left(1 - \frac{\alpha_{k,j} + \alpha_{m,j} + \alpha_{o',j} \Lambda_{o',j}}{\mu_j} \right) \frac{\mu_j}{\alpha_{o,j}} + (1 - \kappa_0 - \kappa_1) \lambda_{o,j}, \text{ s.t. } o \neq o' \end{aligned} \quad (39)$$

where the α 's are output elasticities, λ 's are monopsony markdowns, and Λ 's are labor wedges. This equation is analogous to the one in the baseline model (see equation (3)). As the bargaining power of workers with skill o tend to zero, their labor wedge converges to the monopsony markdown. If, instead, workers of type o have some bargaining power, then they are able to capture some of the profits generated by markups, as well as profits from the employer's monopsony power over workers of the other skill type o' .

E.2 Estimating firm market power and worker bargaining power

Defining high-skill and low-skill occupations. I use the one-digit occupation classifications in the DADS matched employer-employee data set to define low-skill occupations as blue-collar occupations (e.g. maintenance workers and welders) and administrative support occupations (e.g. clerical workers and secretaries); I define high-skill occupations as senior staff in top management positions (e.g. head of logistics or human resources), employees in supervisory roles (e.g. accounting and sales managers), and technical workers (e.g. IT and quality control technicians).

Estimating firm wage premia. Let the subscript $o = \{h, l\}$ denote high-skill and low-skill labor. The estimation procedure is as described in Section 3.1. However, firm-group effects are now occupation-specific:

$$\ln W_{it} = \iota_i + \phi_{g(j(i,t)),t} + \text{Occ}_{o(i,t)} \times \phi_{g(j(i,t)),t} + \chi'_{it} \beta + \nu_{it}$$

where i denotes the individual, j denotes the firm, $g(j)$ denotes the group of firm j at time t , $o(i, t)$ denotes worker i 's occupational group at time t , ι_i are worker fixed effects, $\phi_{g(j(i,t))}$ are firm-group fixed effects, and χ_{it} is a vector of time-varying worker characteristics.

Estimating skill-specific labor wedges. Under the assumption that material inputs are flexible and that firms are pricetakers in material markets, the labor wedges can be measured as:

$$\Lambda_{o,j} = \left(\frac{\Phi_{o,j} H_{o,j}}{P_m M_j} \right) \left(\frac{\alpha_{m,j}}{\alpha_{o,j}} \right) \quad (40)$$

The key challenge to identifying the skill-specific labor wedge is to separately estimate them from skill-specific output elasticities.

To estimate the skill-specific output elasticities, I estimate a production function of the form:

$$y_j = \bar{\alpha}_0 f(k_j, m_j, h_{0,j}) + \bar{\alpha}_1 h_{1,j} + \omega_j$$

where, as a reminder, variables lowercase letters are in logs. This functional form assumes that high-skill labor are complements to the bundle of inputs comprised of capital, materials, and low-skill labor. This assumption is motivated by a large literature in macroeconomics on capital-skill complementarities (Krusell, Ohanian, Rios-Rull, and Violante, 2000), and on the complementarity between imported inputs and skills in international trade (Verhoogen, 2008). This Cobb-Douglas functional form in high-skill labor and the bundle of other inputs also mitigates the dimensionality problem associated with adding more factor inputs into the production function; with a flexible functional form such as the translog, the number of parameters to be estimated increases exponentially with the number of factor inputs. I approximate $f(\cdot)$ with a translog functional form. This accommodates the possibility that low-skill labor are substitutes with capital and materials.

Estimating this production function gives the high-skill output elasticity $\bar{\alpha}_1$ and low-skill output elasticity $\bar{\alpha}_0 \frac{\partial f}{\partial h_{0,j}}$. With these in hand, I back out the skill-specific labor wedges using equation (40).

Estimating skill-specific bargaining power and monopsony markdowns. My approach to estimating skill-specific worker bargaining power, which is motivated by the need to avoid the unobserved endogenous monopsony wages, is as described in Section 3.3. The estimating equation is equation (39), restated below for convenience:

$$\Lambda_{o,j,t} = \kappa_o \underbrace{\left(1 - \frac{P_{k,t} K_{j,t} + P_{m,t} M_{j,t} + \Phi_{o',j,t} H_{o',j,t}}{P_{j,t} Y_{j,t}} \right)}_{\equiv \tilde{\mu}_{o,j}} \frac{\mu_{j,t}}{\alpha_{o,j,t}} + (1 - \kappa_0 - \kappa_1) \lambda(H_{o,j,t}, A_{o,j,t}), \text{ s.t.}$$

where $o \neq o'$ and $\tilde{\mu}_{o,j}$ are profits that can be appropriated by workers of skill o (as a share of revenues). As before, the main challenge to disentangling worker bargaining

power (κ_o) from monopsony markdowns ($\lambda_{o,j,t}$) is the presence of unobserved amenities ($A_{o,j,t}$). To address the unobserved amenities, I apply the same insight from Section 3.3 that firm size in terms of employment and wage-bills jointly controls for unobserved amenities, $\lambda(H_{o,j,t}, A_{o,j,t}) = \lambda(H_{o,j,t}, \mathcal{H}_o^{-1}(H_{o,j,t}, \Phi_{o,j,t} H_{o,j,t}))$.⁵⁴ As before, I approximate the function $\lambda(\cdot)$ with a fourth-order polynomial. Given the estimated bargaining power parameters, monopsony markdowns are measured as $\lambda_{o,j,t} = \frac{\Lambda_{o,j,t} - \hat{\kappa}_o \hat{\mu}_{o,j,t}}{1 - \hat{\kappa}_0 - \hat{\kappa}_1}$.

Main results. Table 31 reports the estimated firm wage premia, labor wedges, and monopsony markdowns by skill type. Comparing the first and second rows shows that high-skill workers receive higher wages, but have similar wage dispersion. Comparing the third and fourth rows shows that the average level of labor wedges are similar for high-skill and low-skill workers. However, there is much more dispersion in labor wedges among the high-skill. The labor wedges of the high-skill are higher than those of the low-skill at the 75th percentile, but lower than those of the low-skill at the 25th percentile. The fifth row shows the firm-level weighted average labor wedge (Λ_f), where the weights are the employment share of a skill type within the firm. The level of the average firm-level labor wedges are relatively close to my baseline estimates, suggesting that my baseline estimates approximate a firm-level average labor wedge.

Table 31: Summary statistics for estimated firm wage premia, labor wedges, and monopsony markdowns by skill type.

Panel A	Mean	Median	25 th Pct	75 th Pct	Var
ϕ_1	2.87	2.88	2.81	2.92	0.01
ϕ_0	2.26	2.26	2.21	2.32	0.01
Λ_1	0.67	0.55	0.32	0.90	0.23
Λ_0	0.65	0.63	0.52	0.78	0.06
Λ_f	0.73	0.66	0.51	0.89	0.31
λ_1	0.16	0.15	0.06	0.27	0.04
λ_0	0.47	0.48	0.38	0.56	0.02
λ_f	0.37	0.37	0.30	0.44	0.01

This table reports the summary statistics for estimated firm wage premia, labor wedges, and wage markdowns by skill group. Monopsony markdowns are computed using the estimated bargaining power parameters in specifications (2) and (4) in Table 32 as the baseline.

Table 32 reports the estimates of worker bargaining power by skill type. Comparing columns (2) and (4), I find that high-skill workers obtain about 11% of the firm profits, while low-skill workers obtain about half of that. The sixth and seventh rows of Table 31 report the distribution of monopsony markdowns received by each skill group. The level

⁵⁴Note that, even when amenities vary across firms, they do not necessarily affect monopsony markdowns directly. This is the case when amenities are multiplicatively separable from wages in workers' labor supply functions. Under multiplicative separability, the control function for amenities is not needed. See the discussion in Section 3.3.

of monopsony markdowns for high-skill workers are lower than that for low-skill workers, although the labor wedge for high-skill workers is higher.

Finally, I compare how labor wedges and monopsony markdowns vary across deciles of firm wage premia. Firm wage premia are computed as within-firm weighted averages of the skill-specific firm wage premium at a given firm, with weights being the employment share of a given skill-group in the firm. Figure 28 presents the findings. Consistent with the finding that high-skill workers have a higher bargaining power, the labor wedges of high-skill workers are increasing in wage premia more steeply than for low-skill workers. As such, the firm-level (weighted) average labor wedge is also increasing with firm wage premia, consistent with my baseline estimates in Figure 1. Figure 28 also shows that high-wage firms tend to have slightly more monopsony power than low-wage firms—monopsony markdowns mildly decrease with firm wage premia. The opposite appears to be the case for high-skill workers, as monopsony markdowns rise with firm wage premia. The firm-level (weighted) average monopsony markdown has a flat profile across firm wage premia, in line my baseline estimates in Figure 2.

Table 32: Estimated workers' bargaining power.

Labor wedge	High-skill		Low-skill	
	(1)	(2)	(3)	(4)
$\tilde{\mu}_{o,j}$	0.119 (0.000)	0.114 (0.001)	0.052 (0.001)	0.056 (0.001)
Sector×year fixed effects	✓	✓	✓	✓
Firm fixed effects		✓		✓
Number of firms	114,950			

This table reports the estimated workers' bargaining power by skill group. Sector fixed effects are at the 5-digit level. Block bootstrapped standard errors are reported in parentheses.

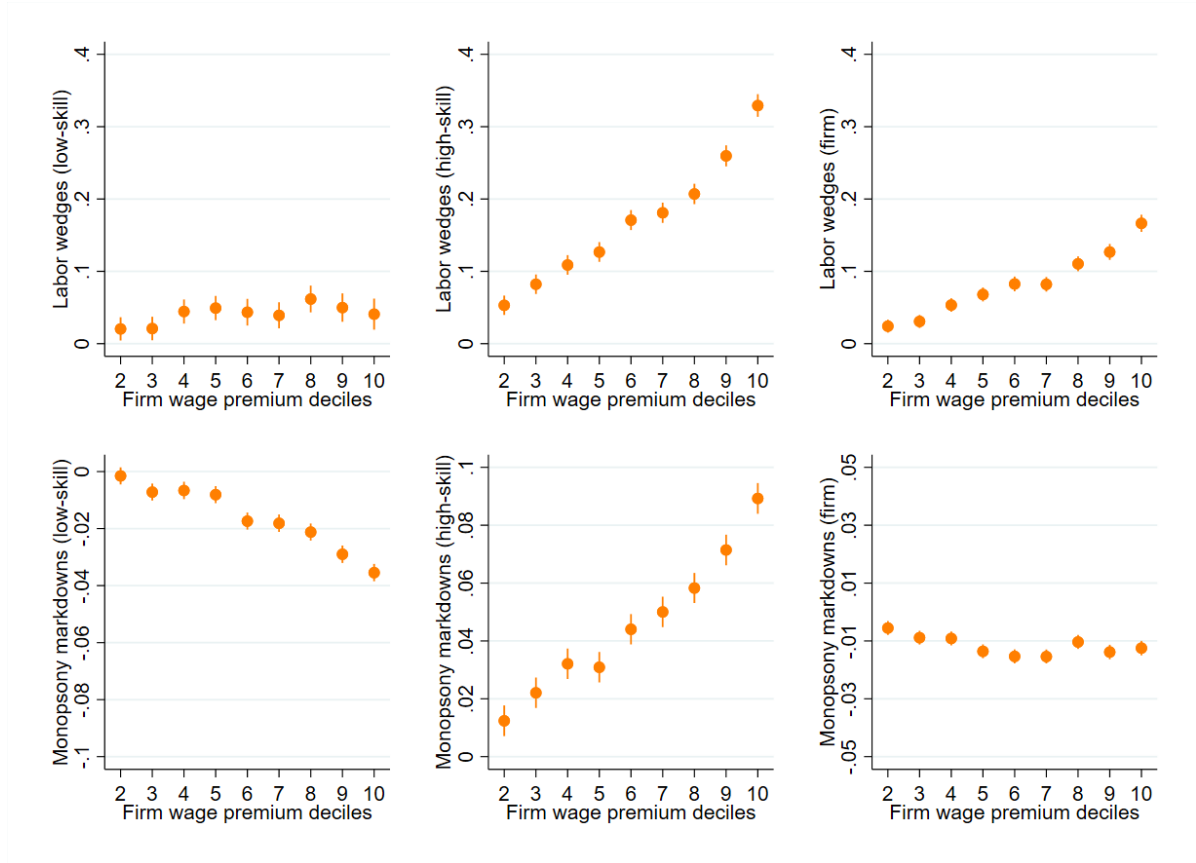


Figure 28: Skill-specific labor wedges and monopsony markdowns high-wage status.

Notes: This figure shows how skill-specific labor wedges and monopsony markdowns vary by deciles of firm wage premia relative to firms in the first decile, controlling for 5-digit sector $\times$ year fixed effects. Firm wage premia are constructed at the firm level by computing weighted average of skill-specific wage premia within firms, with the weights being the employment share of each skill-group within the firm. Decile 10 represents high-wage firms. Vertical bars are 95% confidence intervals.